

ALGEBRA 2

Unit 3

Topic: _____ Name: _____ Date: _____

Main Ideas/Questions Notes/Examples

PIECEWISE FUNCTIONS

EVALUATING Piecewise Functions

1. Given $f(x) = \begin{cases} x^2 - 1 & \text{if } x < 0 \\ 5x + 3 & \text{if } x \geq 0 \end{cases}$
a) $f(-5)$

2. Given $g(x) = \frac{1}{2}x + 3$

Topic: _____ Name: _____ Date: _____

Main Ideas/Questions Notes/Examples

ABSOLUTE VALUE Functions

- The absolute value function is written as $f(x) = |x|$.
- Complete the table below to graph and range.

x	x
-3	
-2	
-1	
0	
1	
2	
3	

Topic: _____ Name: _____ Date: _____

Main Ideas/Questions Notes/Examples

FUNCTION FAMILY

PARENT

A parent function is the simplest function of the family. Give the parent function for each function family. Then graph the function.

NEAR: _____ ABSOLUTE VALUE: _____

Topic: _____ Name: _____ Date: _____

Main Ideas/Questions Notes/Examples

VERTEX FORM of a Quadratic Function

- Vertex Form of a Quadratic Equation: $f(x) = a(x - h)^2 + k$
- _____ is the vertex; _____ is the direction of the parabola.
- _____ determines the width and direction of the parabola.

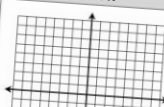
WHY USE Vertex form shows the _____

Topic: _____ Name: _____ Date: _____

Main Ideas/Questions Notes/Examples

PARENT FUNCTIONS

A parent function is the simplest function of the family. A family of graphs displays similar characteristics.

PARENT FUNCTION	GRAPH	CHARACTERISTICS
LINEAR		- D: _____; R: _____ - x-intercept: _____; y-intercept: _____ - End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

Examples: $f(x) = x^2 + 2x - 5$

Parts of a PARABOLA

- The vertical line that divides the axis of symmetry.
- Axis of Symmetry Formula: $x = -\frac{b}{2a}$
- The turning point of the parabola. If it's the lowest point, it's called the vertex. If it's the highest point, it's called the maximum.

PARENT FUNCTIONS & TRANSFORMATIONS

NOTES • HOMEWORK • QUIZZES • TEST

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Unit 3 – Parent Functions & Transformations: Sample Unit Outline

	TOPIC	HOMEWORK
DAY 1	Piecewise Functions	HW #1
DAY 2	Absolute Value Functions & Inequalities (Graphing by Table)	HW #2
DAY 3	Quiz 3-1	None
DAY 4	Parent Functions & Transformations	HW #3
DAY 5	Vertex Form of an Absolute Value Function	HW #4
DAY 6	Quiz 3-2	None
DAY 7	Standard Form of Quadratic Functions & Inequalities	HW #5
DAY 8	Vertex Form of Quadratic Functions & Inequalities	HW #6
DAY 9	Converting Standard Form to Vertex Form: Completing the Square	HW #7
DAY 10	Quiz 3-3	None
DAY 11	End Behavior & Increasing/Decreasing Intervals (Linear, Absolute Value, and Quadratic Functions)	HW #8
DAY 12	Functions Review: Linear, Absolute Value, & Quadratic Identify Parent Function, Domain, Range, Roots, End Behavior, etc.	HW #9
DAY 13	*Greatest Integer Function	HW #10
DAY 14	Unit 3 Review	Study for Test
DAY 15	Unit 3 Test	None

Notes:

- Piecewise Functions (Day 1) includes linear functions only. Piecewise functions that include absolute and quadratic functions are integrated throughout the later lessons.
- A Parent Functions Chart is included on Day 12 that includes linear, absolute value, quadratic, cubic, square root, cube root, reciprocal, exponential, and logarithmic. This chart was meant to be introduced in this unit with linear, absolute value, and quadratic, then used throughout future units.
- Greatest Integer Function (Day 13) is included as a bonus lesson and does not appear on the quizzes, study guide, or test.

Name:

Date:

Topic:

Class:

Main Ideas/Questions

Notes/Examples

PIECEWISE FUNCTIONS

EVALUATING

*Piecewise
Functions*

1. Given $f(x) = \begin{cases} x^2 - 1 & \text{if } x < -2 \\ 5x + 3 & \text{if } x \geq -2 \end{cases}$, find each value.

a) $f(-5)$ b) $f(-2)$ c) $f(7)$

2. Given $g(x) = \begin{cases} \frac{1}{2}x + 3 & \text{if } x \leq -4 \\ -x - 1 & \text{if } -4 < x < 1 \\ 2x^3 + 9 & \text{if } x \geq 1 \end{cases}$, find each value.

a) $g(2)$ b) $g(-1)$ c) $g(-6)$

GRAPHING

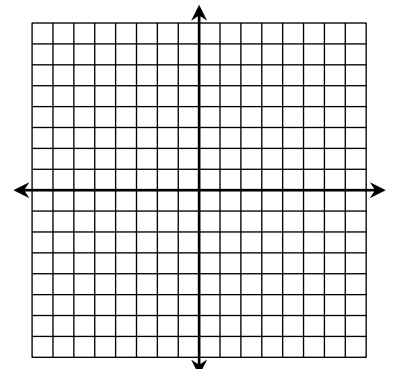
*Piecewise
Functions*

Graph each piecewise function, then identify the domain and range.

3. $f(x) = \begin{cases} x & \text{if } x \leq -3 \\ -2x + 1 & \text{if } x > -3 \end{cases}$

$D =$ _____

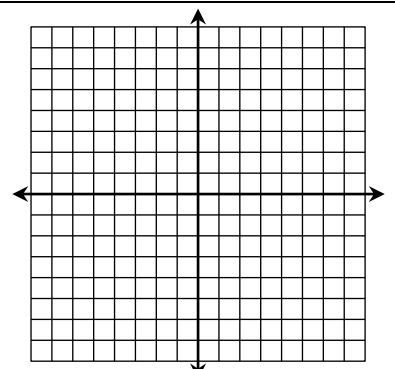
$R =$ _____



4. $g(x) = \begin{cases} -3x - 7 & \text{if } x < -1 \\ -5 & \text{if } x \geq -1 \end{cases}$

$D =$ _____

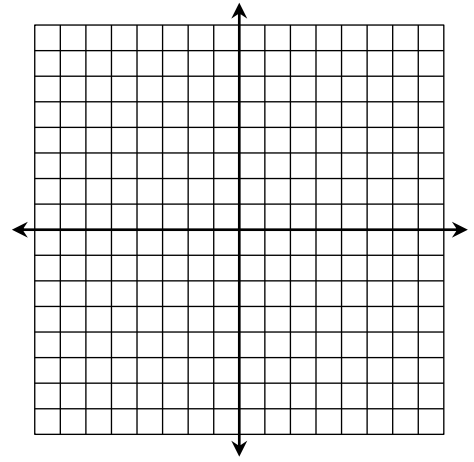
$R =$ _____



$$5. g(x) = \begin{cases} -\frac{5}{2}x - 1 & \text{if } x < 2 \\ x + 2 & \text{if } x > 2 \end{cases}$$

$D =$ _____

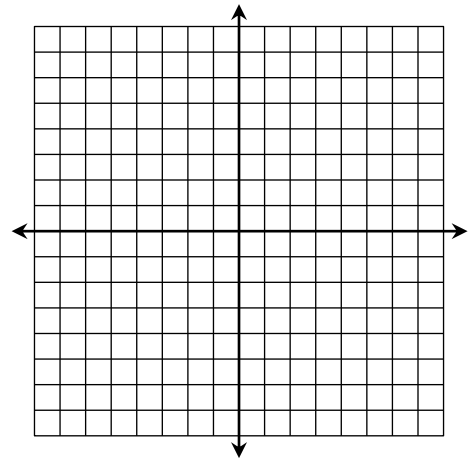
$R =$ _____



$$6. g(x) = \begin{cases} -1 & \text{if } x \leq -5 \\ -x - 3 & \text{if } -5 < x < 1 \\ 6 & \text{if } x \geq 1 \end{cases}$$

$D =$ _____

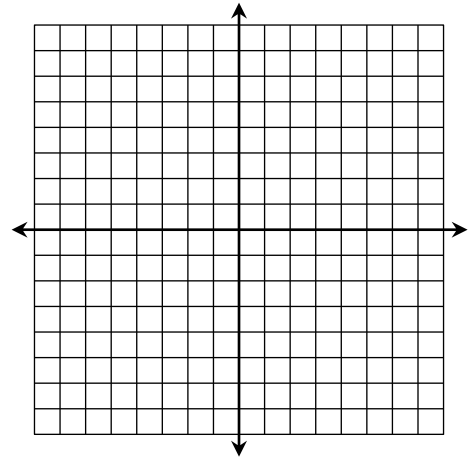
$R =$ _____



$$7. f(x) = \begin{cases} 2x & \text{if } x \leq -2 \\ -1 - x & \text{if } -2 < x < 4 \\ -3 & \text{if } x \geq 4 \end{cases}$$

$D =$ _____

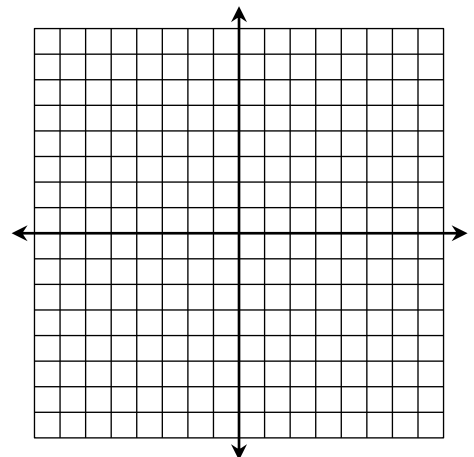
$R =$ _____



$$8. h(x) = \begin{cases} -3x & \text{if } x < 0 \\ -\frac{1}{2}x - 1 & \text{if } 0 \leq x < 6 \\ x - 7 & \text{if } x > 6 \end{cases}$$

$D =$ _____

$R =$ _____



Name: _____

Unit 3: Parent Functions & Transformations

Date: _____ Per: _____

Homework 1: Piecewise Functions

**** This is a 2-page document! ****

Given $a(x) = \begin{cases} |x - 8| & \text{if } x \leq -6 \\ 2x - x^2 & \text{if } -6 < x \leq 1 \\ -4x + 7 & \text{if } x > 1 \end{cases}$, find each function value.

1. $a(8)$

2. $a(1)$

3. $a(-7)$

4. $a(-3)$

5. $a\left(-\frac{1}{2}\right)$

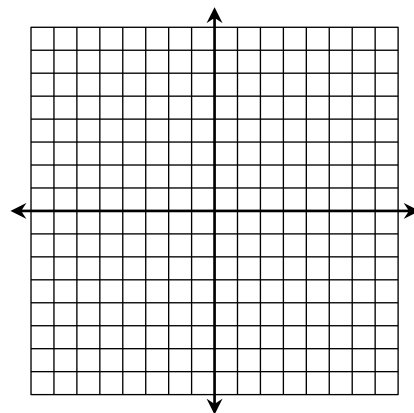
6. $a\left(\frac{9}{4}\right)$

Graph each function. Identify the domain and range.

7. $f(x) = \begin{cases} 5x - 2 & \text{if } x < 1 \\ x - 7 & \text{if } x \geq 1 \end{cases}$

$D =$ _____

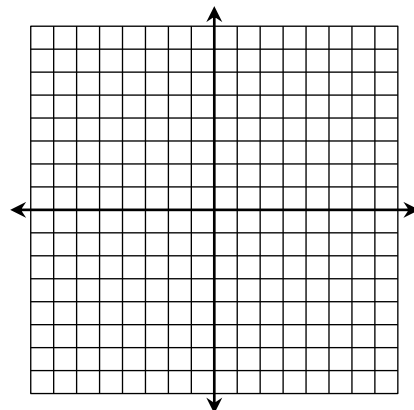
$R =$ _____



8. $g(x) = \begin{cases} -x & \text{if } x < -5 \\ 3 & \text{if } x > -5 \end{cases}$

$D =$ _____

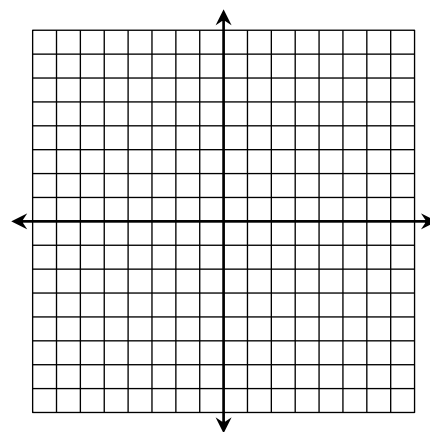
$R =$ _____



9. $h(x) = \begin{cases} \frac{4}{3}x - 2 & \text{if } x < 0 \\ -x + 1 & \text{if } x \geq 0 \end{cases}$

$D =$ _____

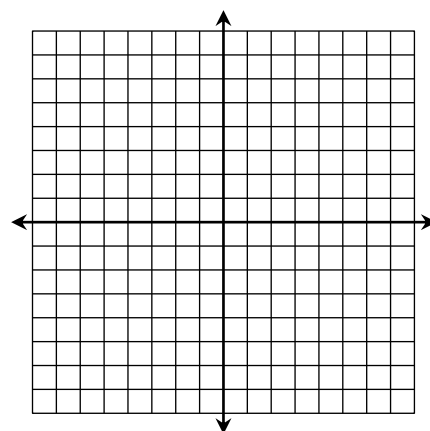
$R =$ _____



10. $p(x) = \begin{cases} -3x + 7 & \text{if } x \leq 3 \\ x & \text{if } 3 < x < 5 \\ -1 & \text{if } x \geq 5 \end{cases}$

$D =$ _____

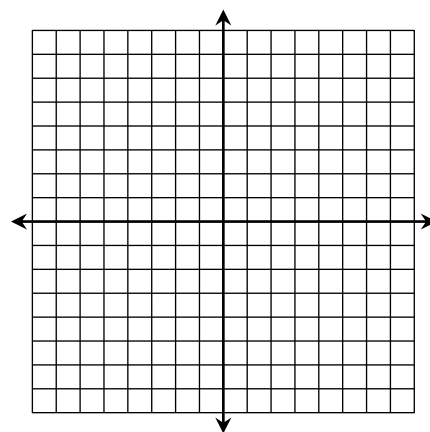
$R =$ _____



11. $k(x) = \begin{cases} x + 4 & \text{if } x < -1 \\ 5 & \text{if } -1 < x < 2 \\ -\frac{1}{2}x + 1 & \text{if } x \geq 2 \end{cases}$

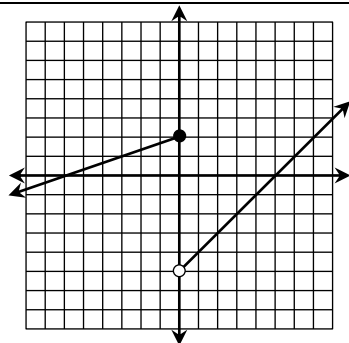
$D =$ _____

$R =$ _____

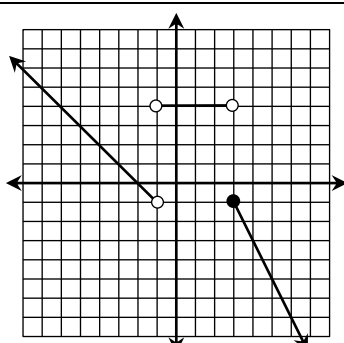


Write a piecewise function for each function graphed below.

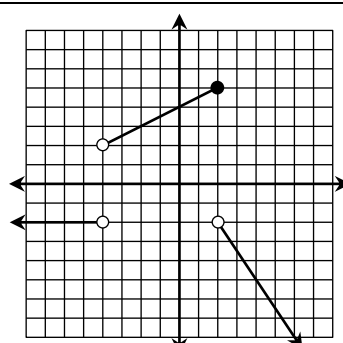
12.



13.



14.



Name:

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Main Ideas/Questions

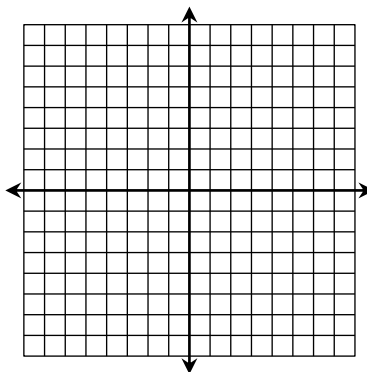
Notes/Examples

ABSOLUTE VALUE

Functions

- The **absolute value function** is written as _____
- Complete the table below to graph the function. Give the domain and range.

x	$ x $
-3	
-2	
-1	
0	
1	
2	
3	

 $D =$ _____ $R =$ _____

- The **graph** looks like a _____; The **turning point** is called the _____.

GRAPHING

by Table

To create a table of values, it's best to place the vertex in the middle, then include values on both sides.

①

Find the **x-value of the vertex**. Set the expression from the inside of the absolute value bars equal to 0 and solve.

②

PLACE THIS VALUE IN THE MIDDLE ROW OF YOUR TABLE.
Number up and down, then complete the table.

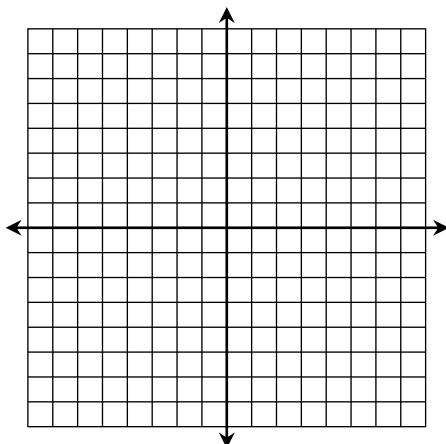
③

GRAPH!

Directions: Graph each function. Give the domain and range.

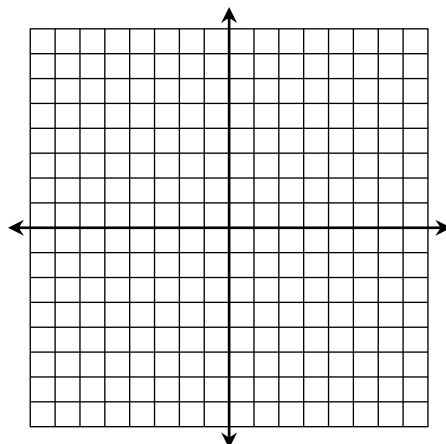
1. $f(x) = |x - 4|$

x	$f(x)$

 $D =$ _____ $R =$ _____

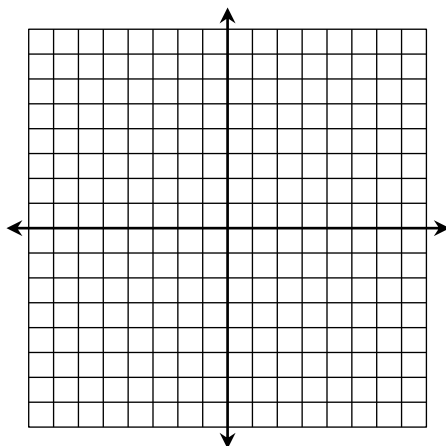
2. $f(x) = |2x + 6|$

x	$f(x)$

 $D =$ _____ $R =$ _____

3. $f(x) = |x| + 5$

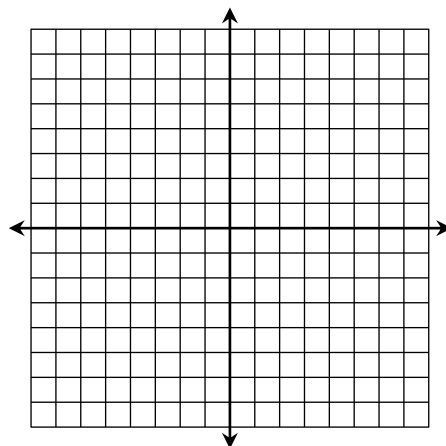
x	$f(x)$



$D =$ _____ $R =$ _____

4. $f(x) = |3x + 2|$

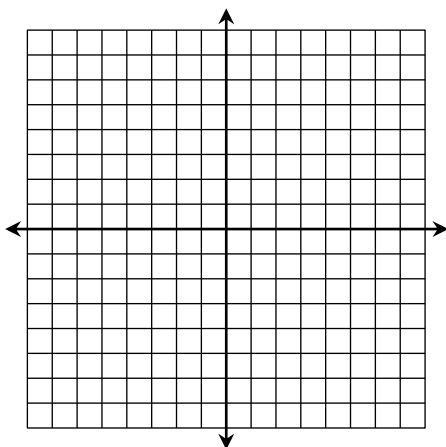
x	$f(x)$



$D =$ _____ $R =$ _____

5. $f(x) = -|x + 1|$

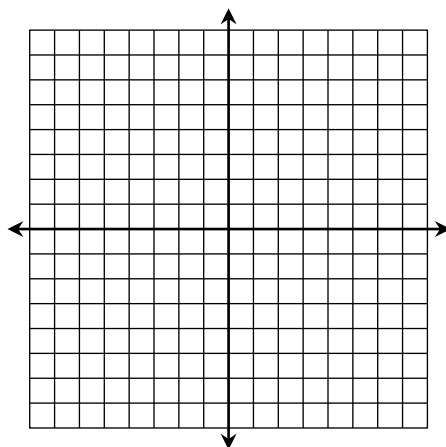
x	$f(x)$



$D =$ _____ $R =$ _____

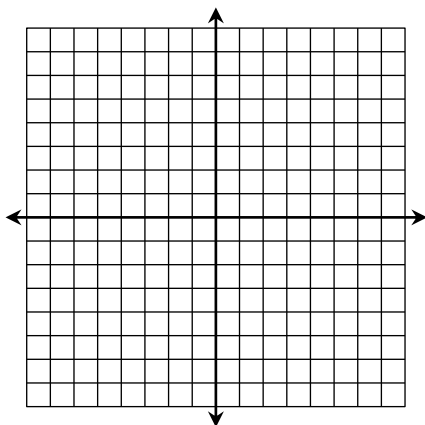
6. $f(x) = \left| \frac{1}{2}x + 1 \right| - 4$

x	$f(x)$



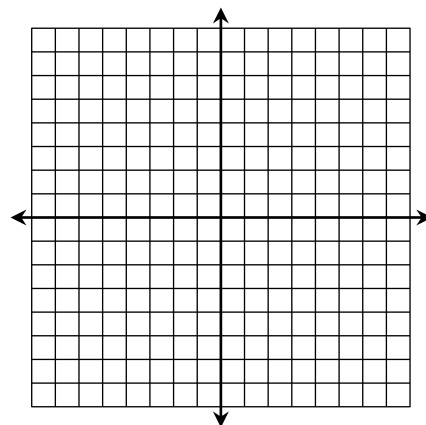
$D =$ _____ $R =$ _____

7. $f(x) = \begin{cases} -2x + 1 & \text{if } x < 4 \\ |x - 3| & \text{if } x \geq 4 \end{cases}$



$D =$ _____ $R =$ _____

8. $f(x) = \begin{cases} |2x| - 4 & \text{if } x \leq -3 \\ -5 & \text{if } -3 < x < 2 \\ x - 2 & \text{if } x \geq 2 \end{cases}$



$D =$ _____ $R =$ _____

Name:

Date:

Topic:

Class:

Main Ideas/Questions

Notes/Examples

GRAPHING ABSOLUTE VALUE

inequalities

①

Create a **TABLE OF VALUES**. Recall that the vertex should be in the middle row.

②

GRAPH! Use a dashed line for $<$ and $>$ symbols and a solid line for $\leq$ and $\geq$ symbols.

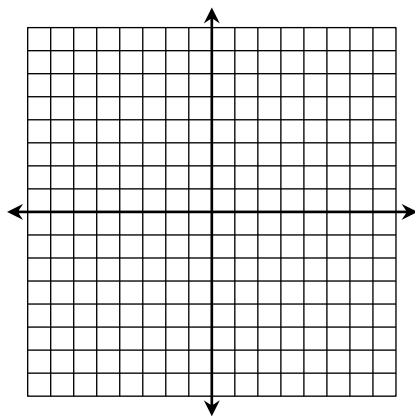
③

SHADE! Assuming y is to the left, shade above the line for $>$ or $\geq$ symbols and below the line for $<$ or $\leq$ symbols.

Directions: Graph each inequality.

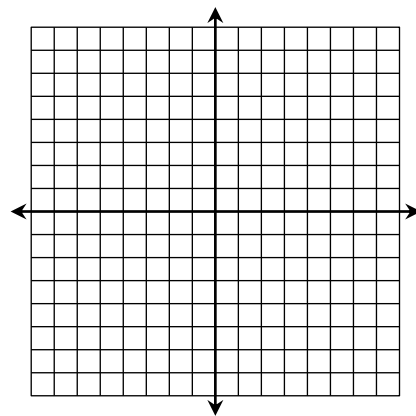
1. $y < |x + 1|$

x	y



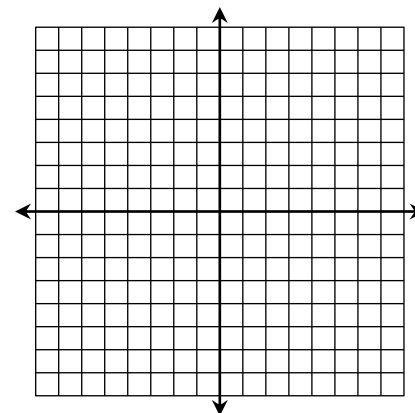
2. $y \geq |x - 5| - 2$

x	y



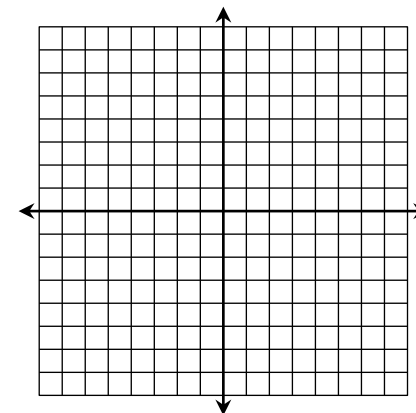
3. $y > |4x + 6|$

x	y



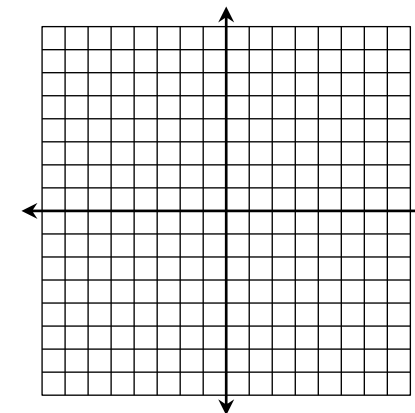
4. $y \leq -|2x| + 3$

x	y



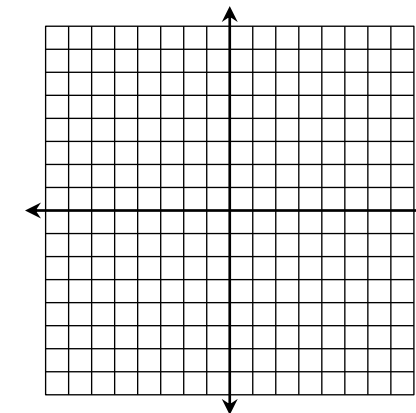
5. $y \geq |2x - 12| - 7$

x	y



6. $y < -|3x + 9| + 4$

x	y



Name: _____

Unit 3: Parent Functions & Transformations

Date: _____ Per: _____

Homework 2: Graphing Absolute Value Functions & Inequalities

**** This is a 2-page document! ****

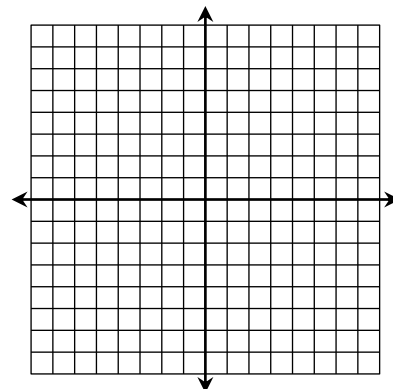
Graph each function. Identify the domain and range.

1. $f(x) = |x + 6|$

$D =$ _____

$R =$ _____

x	y

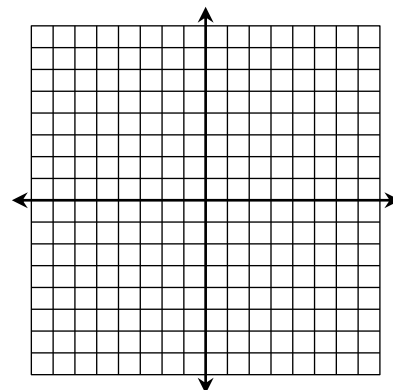


2. $f(x) = |x - 1| + 2$

$D =$ _____

$R =$ _____

x	y

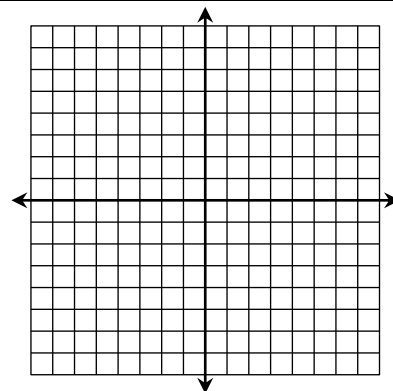


3. $f(x) = |3x - 4|$

$D =$ _____

$R =$ _____

x	y

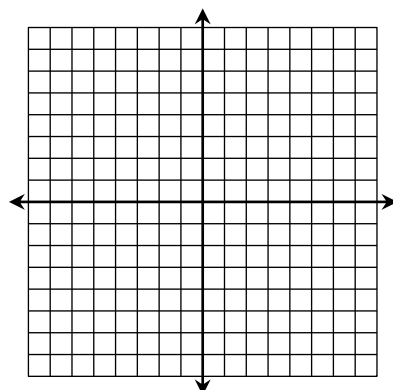


4. $f(x) = -|2x| + 7$

$D =$ _____

$R =$ _____

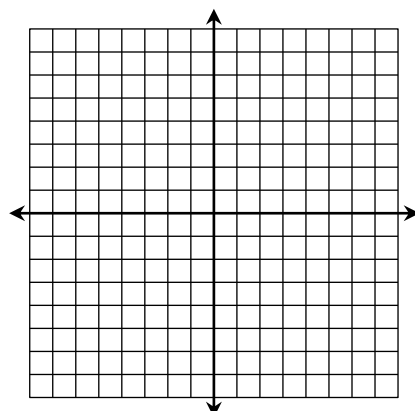
x	y



5. $f(x) = \begin{cases} -|x| + 3 & \text{if } x \leq -2 \\ 4x + 1 & \text{if } x > -2 \end{cases}$

$D =$ _____

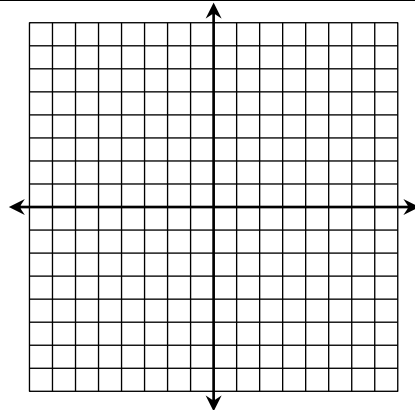
$R =$ _____



6. $f(x) = \begin{cases} -1 & \text{if } x \leq -6 \\ -x - 4 & \text{if } -6 < x \leq 0 \\ |2x - 5| & \text{if } x > 0 \end{cases}$

$D =$ _____

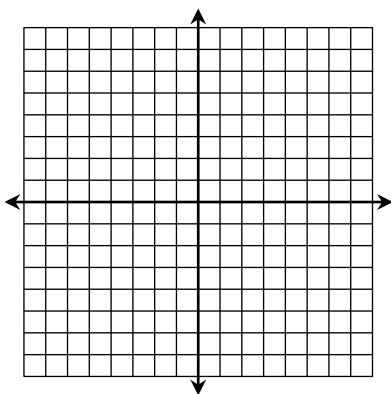
$R =$ _____



Graph each inequality.

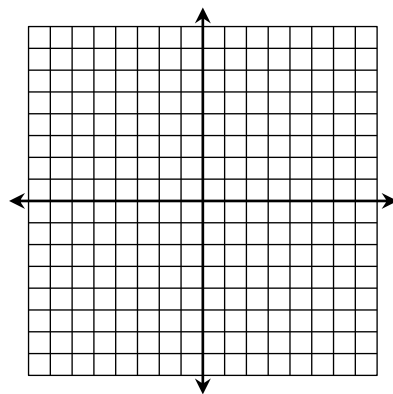
7. $y \geq |x| + 2$

x	y



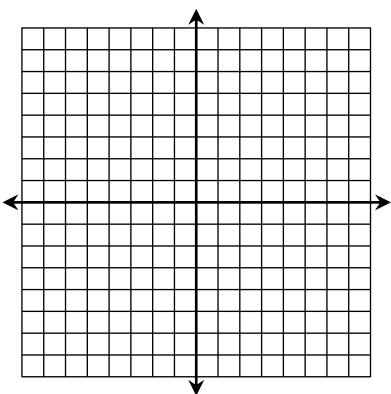
8. $y < -|x - 4|$

x	y



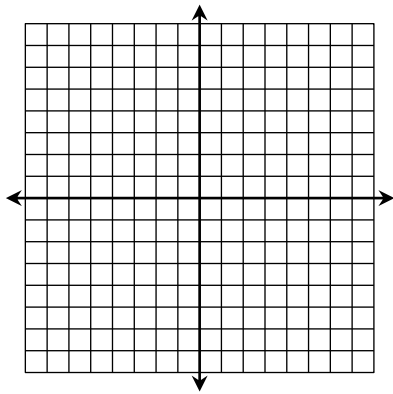
9. $y < |4x + 1| - 3$

x	y



10. $y \geq -|3x + 6| + 1$

x	y



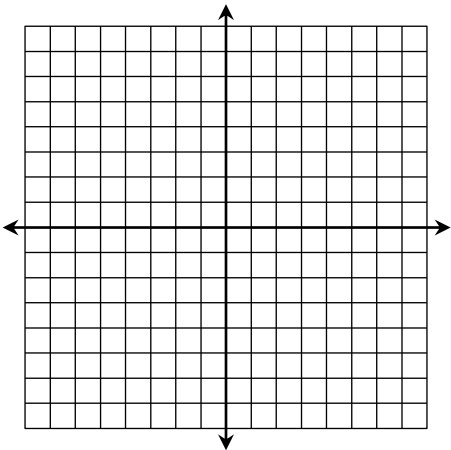
Quiz 3-1: Absolute Value Functions & Piecewise Functions

Use the table to graph each absolute value function. Identify the domain and range.

1. $f(x) = -|x - 1| + 4$

2. $f(x) = |2x + 5| - 3$

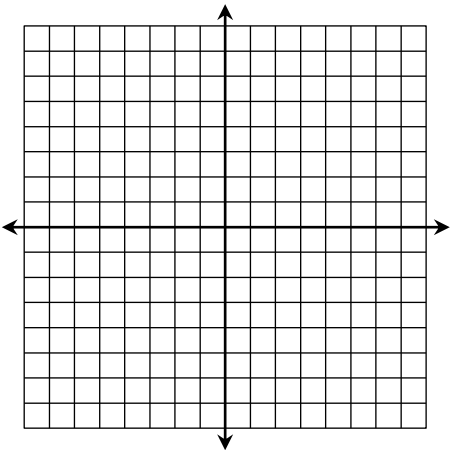
x	$f(x)$



D: _____

R: _____

x	$f(x)$



D: _____

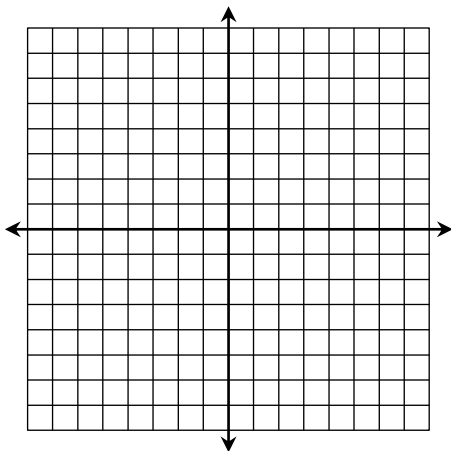
R: _____

Use the table to graph each absolute value inequality.

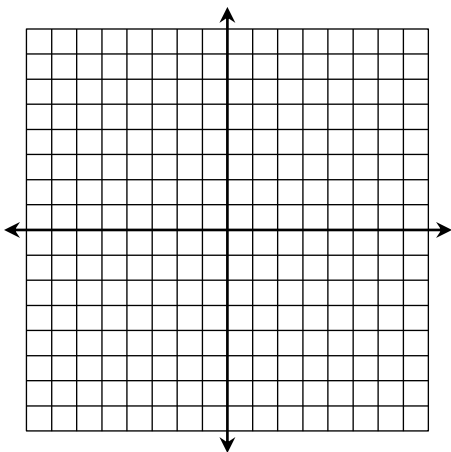
3. $y < -|3x| + 5$

4. $y \geq |x + 4| - 6$

x	y



x	y



Find each value given the piecewise function below.

$$f(x) = \begin{cases} |2x - 1| & \text{if } x < -3 \\ 4 - x & \text{if } -3 \leq x < 5 \\ -x^2 + 7 & \text{if } x \geq 5 \end{cases}$$

5. $f(-8)$

5. _____

6. $f(11)$

6. _____

7. $f(-3)$

7. _____

8. $f\left(-\frac{9}{2}\right)$

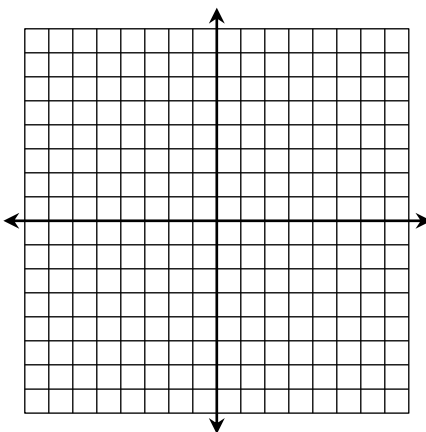
8. _____

9. $f\left(\frac{5}{4}\right)$

9. _____

Graph each piecewise function. State the domain and range.

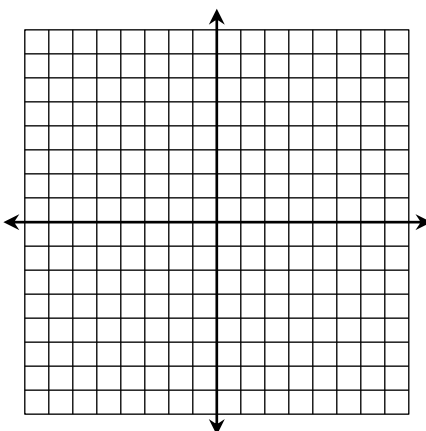
10. $f(x) = \begin{cases} 3x - 1 & \text{if } x \leq -1 \\ -\frac{1}{2}x - 5 & \text{if } x > -1 \end{cases}$



10. D : _____

R : _____

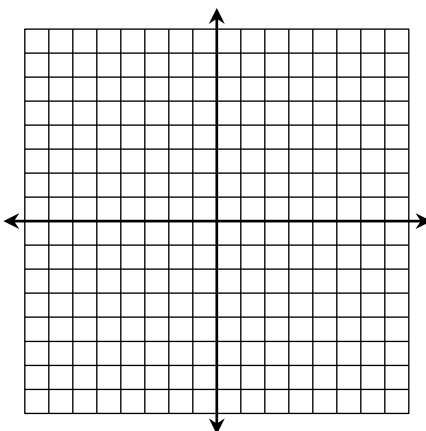
11. $f(x) = \begin{cases} -2x - 7 & \text{if } x < -4 \\ -|x - 1| & \text{if } x \geq -4 \end{cases}$



11. D : _____

R : _____

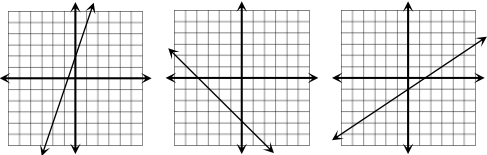
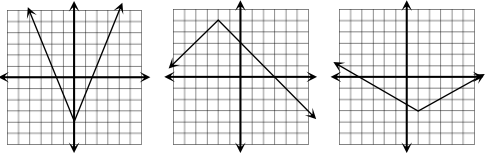
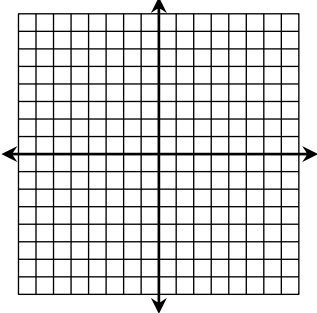
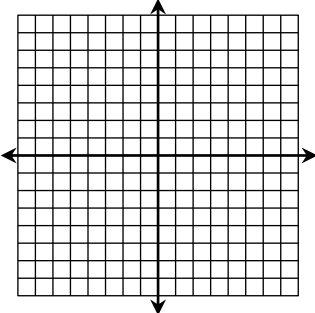
12. $f(x) = \begin{cases} 6 & \text{if } x < -5 \\ |x + 3| - 7 & \text{if } -5 \leq x < 2 \\ x - 2 & \text{if } x \geq 2 \end{cases}$



12. D : _____

R : _____

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples
FUNCTION FAMILY	A family of functions (or function family) is a set of functions whose equations and graphs have a similar form and shape.
	Name the function family for each set of graphs.
	1.  2. 
PARENT FUNCTION	A parent function is the simplest function of the family.
	Give the parent function for each function family. Then graph the function.
	LINEAR: _____  ABSOLUTE VALUE: _____ 
TRANSFORMATIONS	A transformation changes the position or size of a figure. Transformations can be applied to parent functions.
	Let's use the absolute value function as an example. Using your calculator, graph the following functions, then compare to the parent function. Give a brief description.
	3. $f(x) = x + 2 $
	4. $f(x) = x - 2 $
	5. $f(x) = x + 2$
	6. $f(x) = x - 2$
	7. $f(x) = 2 x $
	8. $f(x) = \frac{1}{2} x $
	9. $f(x) = - x $
Types of TRANSFORMATIONS	- A _____ slides (or shifts) a function. - A _____ stretches or compresses a function. - A _____ flips a function.

TRANSFORMATION <i>Rules</i>	We can tell from the function which transformation(s) from the parent function have taken place using the following rules:	
	TRANSLATIONS	$f(x + h)$
		$f(x - h)$
		$f(x) + k$
		$f(x) - k$
	DILATIONS	$a \cdot f(x)$ (when $ a > 1$)
		$a \cdot f(x)$ (when $ a < 1$)
	REFLECTIONS	$-f(x)$
IDENTIFYING <i>Transformations</i>	Describe how the following functions compare to the parent function.	
	10. $f(x) = x - 3$	11. $f(x) = x + 7 $
	12. $f(x) = - x - 4 $	13. $f(x) = 4 x + 1$
	14. $f(x) = \frac{1}{3} x $	15. $f(x) = x + 2 - 5$
	16. $f(x) = -2 x - 1 $	17. $f(x) = \frac{4}{3} x + 2$
WRITING <i>Functions</i>	Transformations to the absolute value parent function are described. Write the new function.	
	18. Translated 6 units right.	
	19. Reflected in the x-axis, then translated 2 units up.	
	20. Vertically stretched by a factor of 5, then translated 1 unit left.	
	21. Translated 3 units right and 4 units down.	
	22. Vertically compressed by a factor of $\frac{2}{3}$, then reflected in the x-axis.	
	23. Vertically stretched by a factor of 2, then translated 9 units down.	
	24. Vertically compressed by a factor of $\frac{1}{4}$, then shifted 1 unit left.	

Name: _____

Unit 3: Parent Functions & Transformations

Date: _____ Per: _____

Homework 3: Identifying Transformations & Writing Functions

Give the parent function of each function family.

1. Linear Functions

2. Absolute Value Functions

Describe the transformation of each function compared to its parent function.

3. $f(x) = |x| - 7$

4. $f(x) = \frac{1}{5}|x|$

5. $f(x) = -|x|$

6. $f(x) = |x + 6|$

7. $f(x) = 3|x|$

8. $f(x) = |x - 4| + 6$

9. $f(x) = -|x + 9|$

10. $f(x) = 4|x - 1|$

11. $f(x) = 2|x + 5| - 4$

12. $f(x) = -\frac{5}{3}|x| + 2$

Transformations to the absolute value parent function are described. Write the new function.

13. Translated 9 units right.

14. Reflected in the x-axis.

15. Vertically compressed by a factor of $\frac{3}{4}$.

16. Shifted 4 units up.

17. Reflected in the x-axis, then translated 2 units left.

18. Vertically stretched by a factor of 5, then translated 3 units right.

19. Translated 7 units down and 4 units left.

20. Vertically compressed by a factor of $\frac{1}{2}$, then reflected in the x-axis.

21. Vertically stretched by a factor of 4, then shifted 2 units right and 1 unit down.

22. Reflected in the x-axis, then translated 8 units left and 5 units up.

Name:

Date:

Topic:

Class:

Main Ideas/Questions

Notes/Examples

WARM-UP

- What is the parent function of the absolute value function family?
- Describe how the following functions compare to the parent function:
 - $f(x) = |x - 5|$
 - $f(x) = |x| + 9$
 - $f(x) = -|x|$
 - $f(x) = 3|x| - 2$
 - $f(x) = \frac{1}{4}|x + 6|$
 - $f(x) = -|x + 7| - 2$

The **horizontal** and **vertical shifts** describe the **location of the vertex**!

VERTEX FORM

of an Absolute Value Function

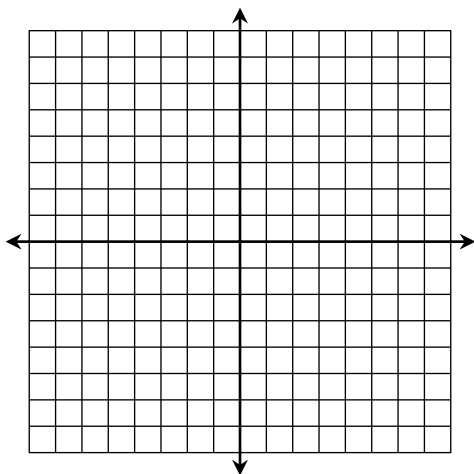
The **vertex form** of an **absolute value function** is written as

where _____ is the **vertex**,

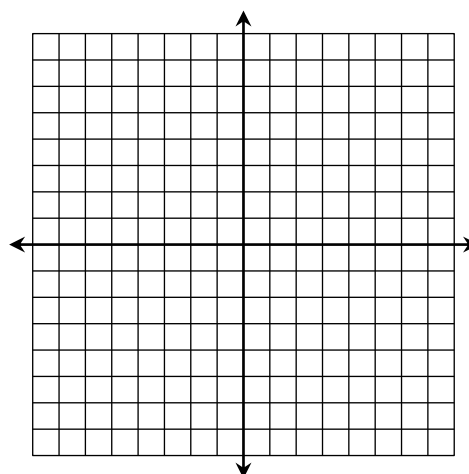
_____ is the **slope on the right side** and _____ is the **slope on the left side**

Directions: Give the vertex of each absolute value function. Then graph.

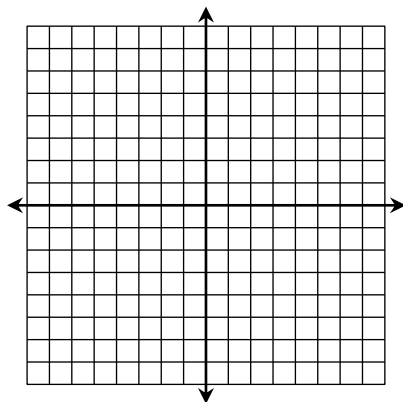
1. $f(x) = |x - 2| - 4$



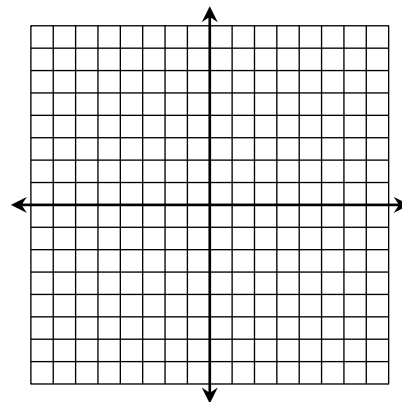
2. $f(x) = 3|x + 5|$



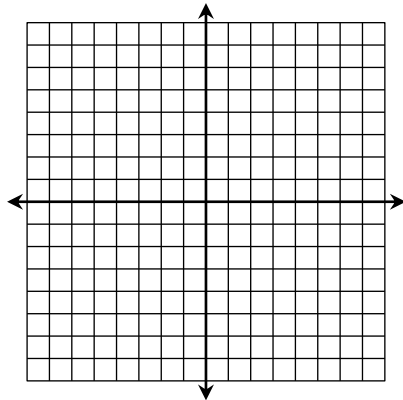
3. $f(x) = 2|x+1| - 2$



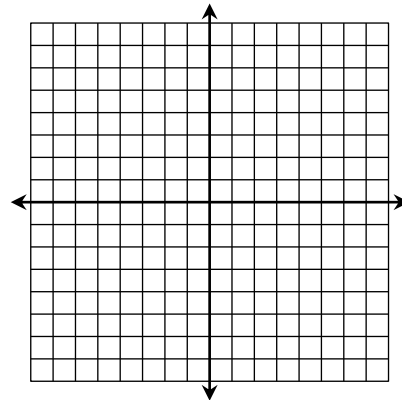
4. $f(x) = -|x-6|$



5. $f(x) = \frac{2}{3}|x+1| - 1$

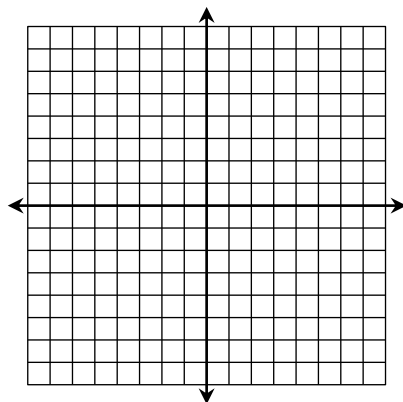


6. $f(x) = -\frac{3}{4}|x| + 7$

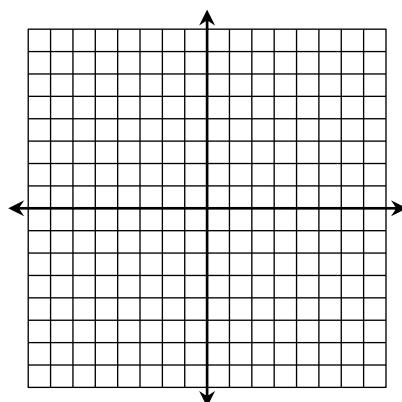


Directions: Graph each piecewise function.

7. $f(x) = \begin{cases} 2x+7 & \text{if } x < -4 \\ 2|x+3| & \text{if } x \geq -4 \end{cases}$

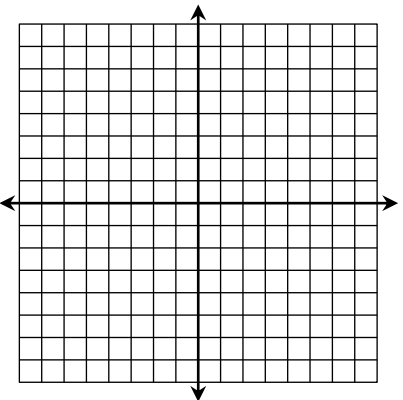


8. $f(x) = \begin{cases} -|x+1|+5 & \text{if } x \leq -4 \\ 3 & \text{if } -4 < x < 2 \\ -x+7 & \text{if } x \geq 2 \end{cases}$

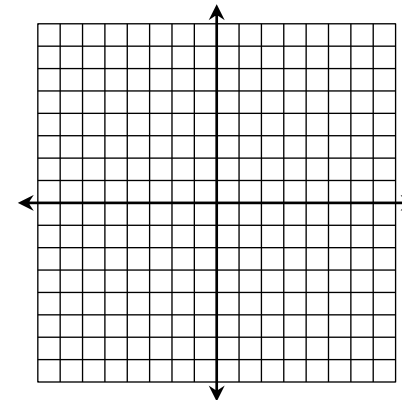


Directions: Graph each absolute value inequality by identifying the vertex.

9. $y \leq |x+4| - 6$



10. $y > -2|x-5| + 4$



Name: _____

Unit 3: Parent Functions & Transformations

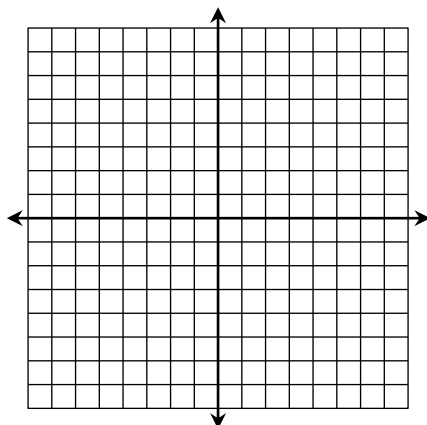
Date: _____ Per: _____

Homework 4: Vertex Form of Absolute Value Functions

**** This is a 2-page document! ****

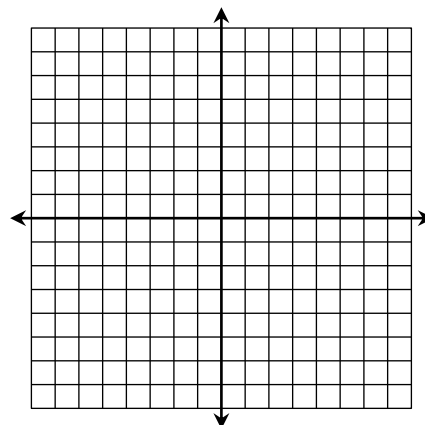
Give the vertex of each function, then graph. State the domain and range.

1. $f(x) = |x| + 3$



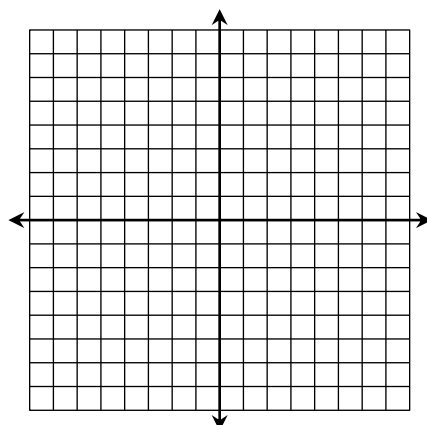
$D =$ _____ $R =$ _____

2. $f(x) = |x - 1| - 1$



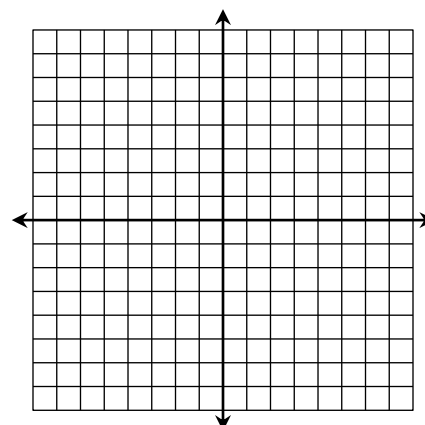
$D =$ _____ $R =$ _____

3. $f(x) = -|x + 5|$



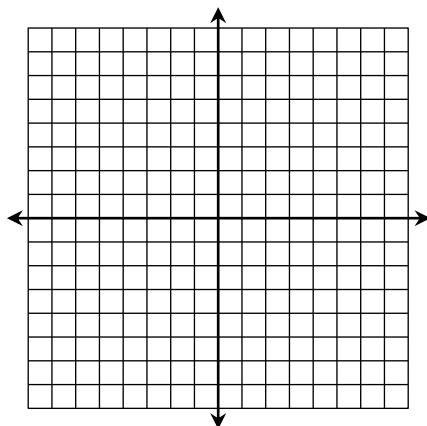
$D =$ _____ $R =$ _____

4. $f(x) = \frac{1}{3}|x| - 7$



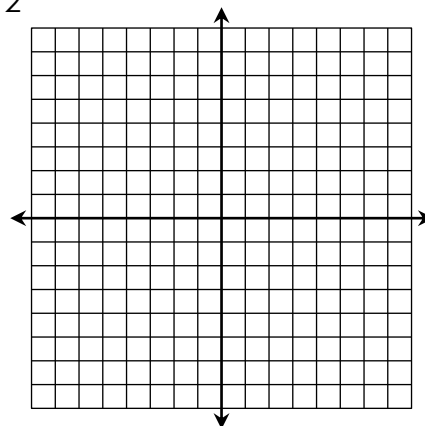
$D =$ _____ $R =$ _____

5. $f(x) = -|x + 3| + 6$



$D =$ _____ $R =$ _____

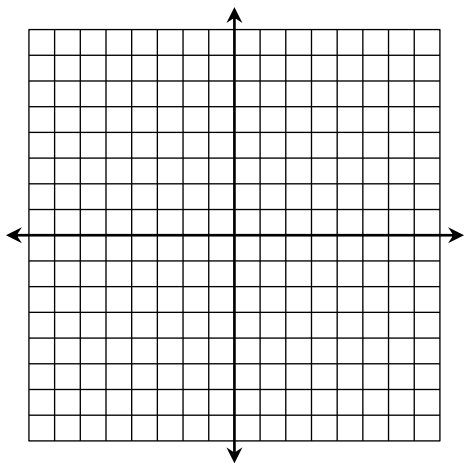
6. $f(x) = \frac{5}{2}|x - 4| + 2$



$D =$ _____ $R =$ _____

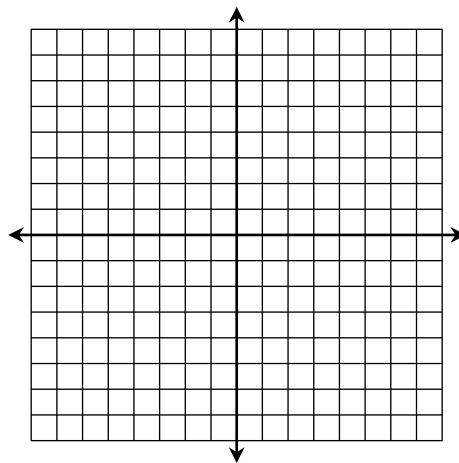
Graph each piecewise function and state the domain and range.

7. $f(x) = \begin{cases} -\frac{3}{2}|x| + 5 & \text{if } x \leq 4 \\ -2x + 3 & \text{if } x > 4 \end{cases}$



$D =$ _____ $R =$ _____

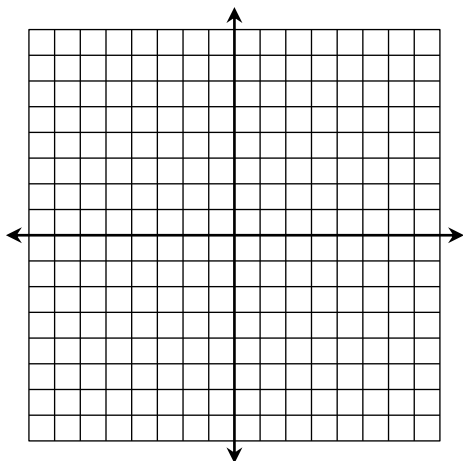
8. $f(x) = \begin{cases} 1 & \text{if } x < -5 \\ -x & \text{if } -5 \leq x \leq 3 \\ |x - 4| - 7 & \text{if } x > 3 \end{cases}$



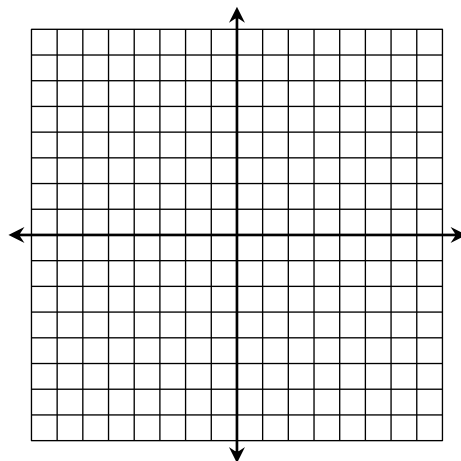
$D =$ _____ $R =$ _____

Graph each absolute value inequality.

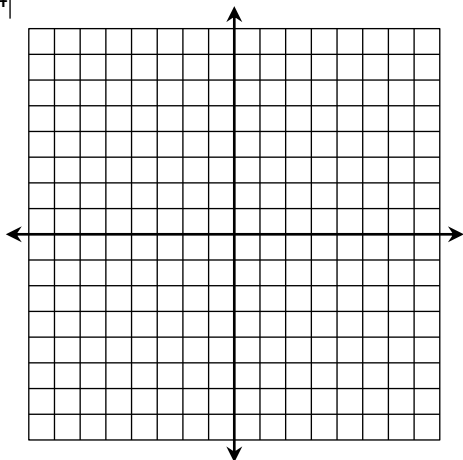
9. $y \geq |x + 2| - 3$



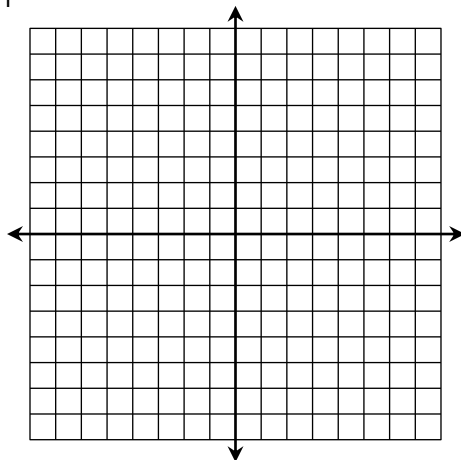
10. $y < -2|x| + 6$



11. $y \leq -\frac{5}{2}|x - 4|$



12. $y > \frac{1}{4}|x + 1| + 1$



Name: _____

Algebra 2

Date: _____ Per: _____

Unit 3: Parent Functions & Transformations

Quiz 3-2: Transformations & Vertex Form of Absolute Value Functions

Identify the parent function of each function family.

1. Linear

1. _____

2. Absolute Value

2. _____

Describe the transformations that were applied to the parent function.

3. $f(x) = -|x| - 5$

3. _____

4. $f(x) = 3|x + 8|$

4. _____

5. $f(x) = \frac{1}{2}|x - 1| - 1$

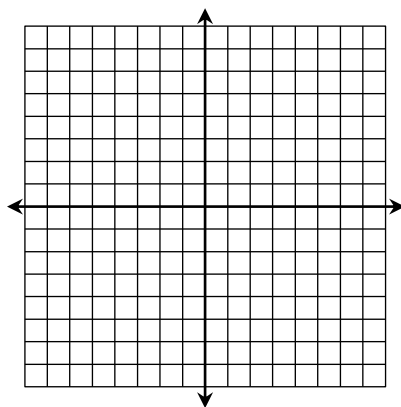
5. _____

6. Dale graphed the absolute value parent function. Then, he reflected the graph over the x -axis, shifted it four units to the right and three units up. Give the new equation.

6. _____

Identify the vertex, then graph. State the domain and range.

7. $f(x) = |x - 2| + 1$

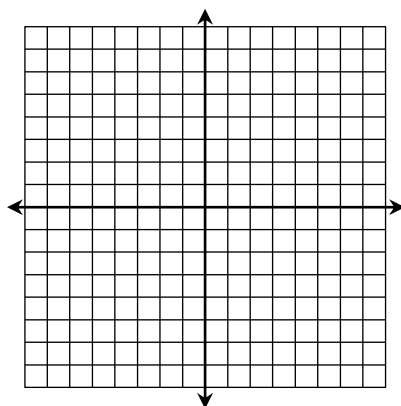


7. Vertex: _____

D : _____

R : _____

8. $f(x) = -\frac{5}{3}|x + 4|$



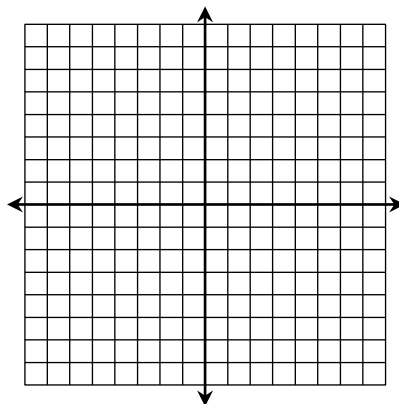
8. Vertex: _____

D : _____

R : _____

Graph each piecewise function. State the domain and range.

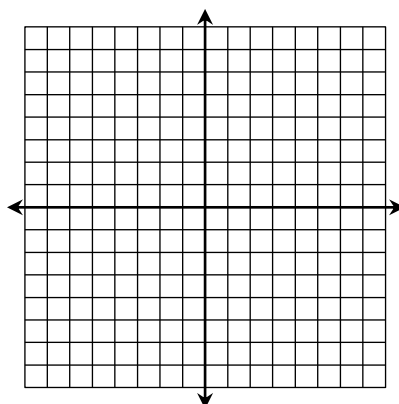
9. $f(x) = \begin{cases} \frac{2}{3}x + 4 & \text{if } x < -3 \\ -|x - 2| & \text{if } x > -3 \end{cases}$



9. D : _____

R : _____

10. $f(x) = \begin{cases} 2|x| - 7 & \text{if } x < -5 \\ 1 & \text{if } -5 \leq x \leq 2 \\ -x - 4 & \text{if } x > 2 \end{cases}$



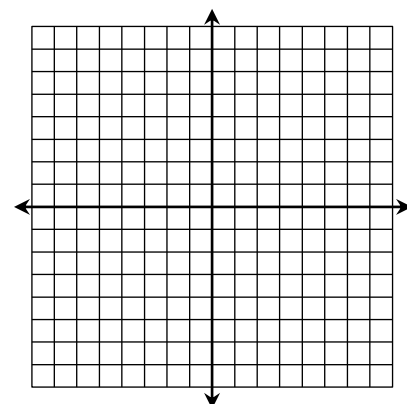
10. D : _____

R : _____

Graph each inequality.

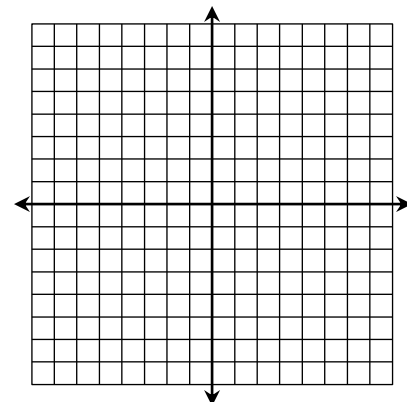
11. $y \geq -|x - 5| - 3$

11.

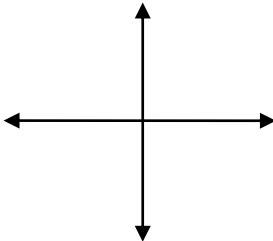
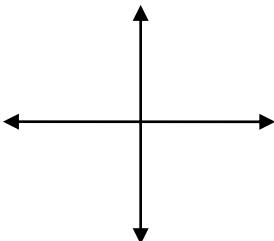


12. $y < \frac{3}{4}|x + 2| + 4$

12.



Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples
QUADRATIC FUNCTION	- Standard Form of a Quadratic Equation: _____ - A quadratic equation creates a U-shaped curve called a parabola.
Examples	Using your graphing calculator, sketch the following functions: $f(x) = x^2 + 2x - 5$  $f(x) = -x^2 + 3x + 7$  If "a" is positive, then the parabola will open up. If "a" is negative, then the parabola will open down.
Parts of a PARABOLA	- The vertical line that divides the parabola into two equal parts is called the axis of symmetry. Axis of Symmetry Formula: _____ - The turning point of the parabola is called the vertex. When the vertex is the lowest point, it's called a minimum, when it's the highest point, it's called a maximum. Label these parts on the graphs above.
Directions: Determine whether each function has a minimum or maximum. Then find the axis of symmetry and vertex.	
1. $f(x) = 4x^2$ Minimum / Maximum Axis of Symmetry: _____ Vertex: _____	2. $f(x) = -x^2 + 6x$ Minimum / Maximum Axis of Symmetry: _____ Vertex: _____
3. $f(x) = -2x^2 + 7$ Minimum / Maximum Axis of Symmetry: _____ Vertex: _____	4. $f(x) = x^2 - 8x + 9$ Minimum / Maximum Axis of Symmetry: _____ Vertex: _____

GRAPHING

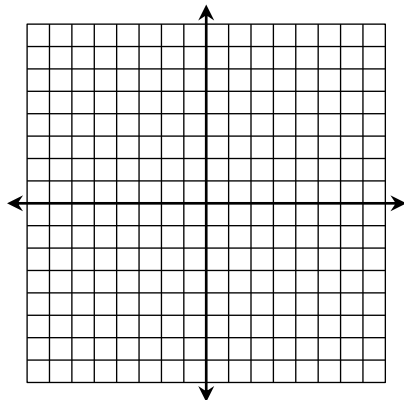
by Table

- ① Find the **AXIS OF SYMMETRY** and **VERTEX**.
- ② **PLACE THE VERTEX AS THE MIDDLE ROW.**
Number up and down, then complete the table.
- ③ **GRAPH!**

Directions: Graph each function. State the domain and range.

5. $f(x) = -3x^2$

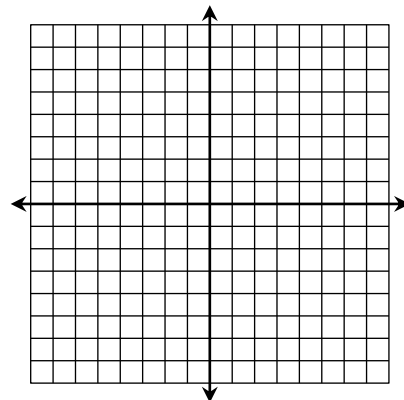
x	f(x)



D = _____ R = _____

6. $f(x) = x^2 - 6x + 11$

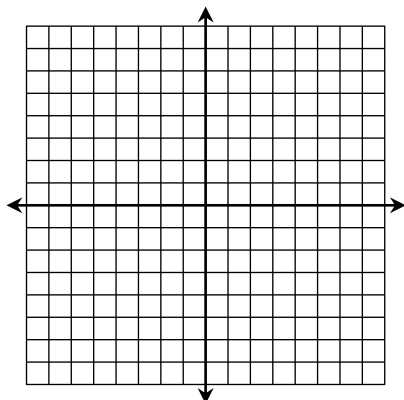
x	f(x)



D = _____ R = _____

7. $f(x) = 2x^2 - 8x + 6$

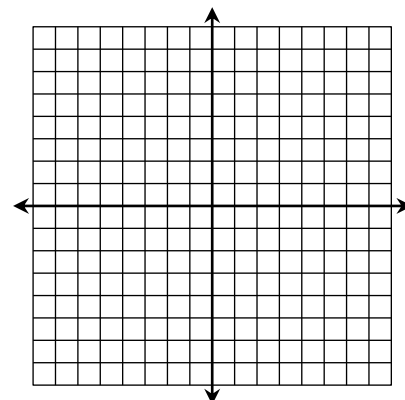
x	f(x)



D = _____ R = _____

8. $f(x) = -x^2 + 5$

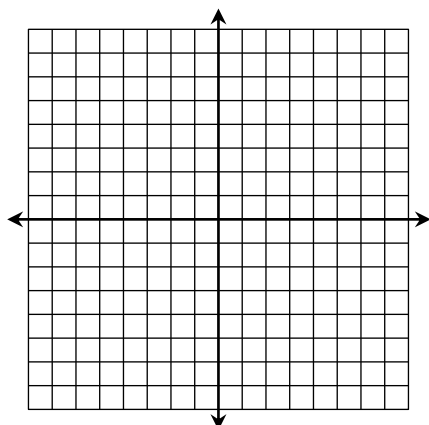
x	f(x)



D = _____ R = _____

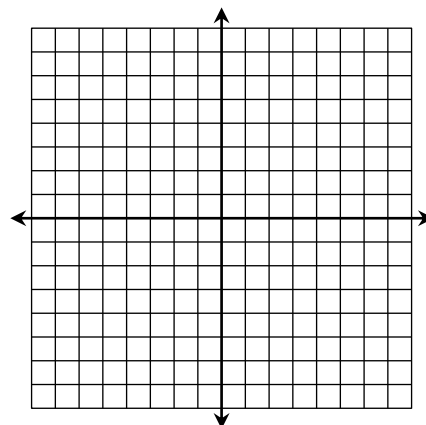
9. $f(x) = -\frac{1}{2}x^2 - x$

x	f(x)



D = _____ R = _____

10. $f(x) = \begin{cases} 2x + 5 & \text{if } x < -4 \\ x^2 + 3x & \text{if } x \geq -4 \end{cases}$



D = _____ R = _____

Name:

Date:

Topic:

Class:

Main Ideas/Questions

Notes/Examples

GRAPHING QUADRATIC

inequalities

①

Create a **TABLE OF VALUES**. Recall that the vertex should be in the middle row.

②

GRAPH! Use a dashed line for $<$ and $>$ symbols and a solid line for $\leq$ and $\geq$ symbols.

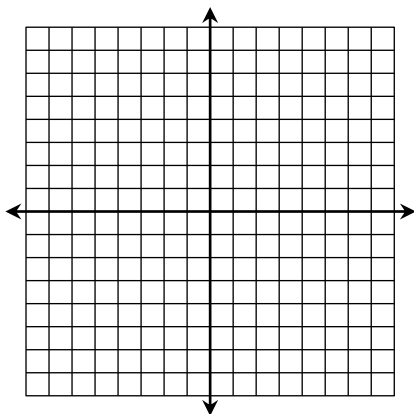
③

SHADE! Assuming y is to the left, shade above the line for $>$ or $\geq$ symbols and below the line for $<$ or $\leq$ symbols.

Directions: Graph each inequality.

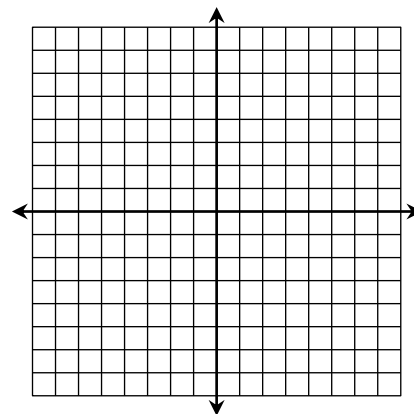
1. $y > x^2 + 2x - 2$

x	y



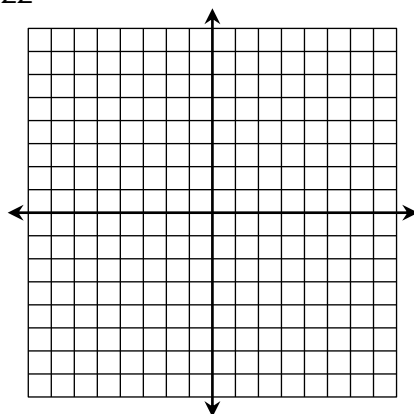
2. $y \leq -x^2 + 4x - 1$

x	y



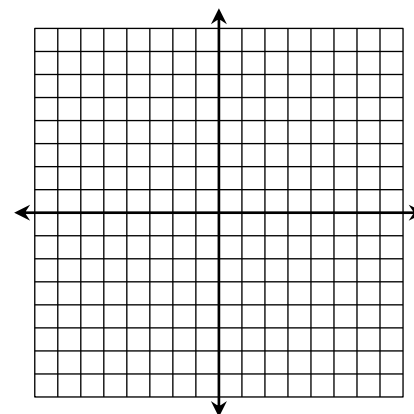
3. $y \geq -3x^2 + 18x - 22$

x	y



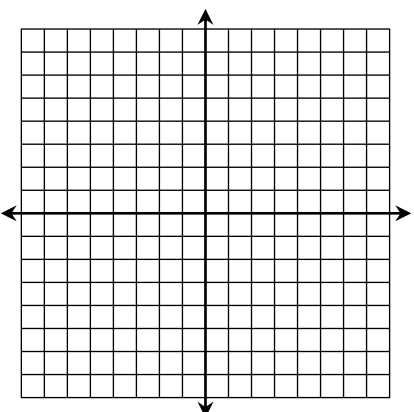
4. $y < x^2 + 6$

x	y



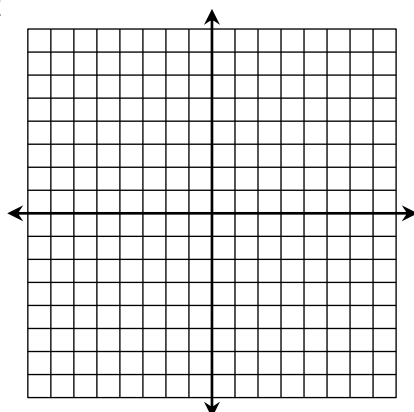
5. $y \leq -2x^2 - 2x$

x	y



6. $y > \frac{1}{2}x^2 - 4x + 12$

x	y



Name: _____

Unit 3: Parent Functions & Transformations

Date: _____ Per: _____

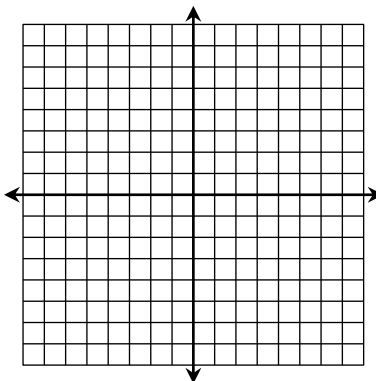
Homework 5: Graphing Quadratic Equations
& Inequalities (Standard Form)

** This is a 2-page document! **

Determine whether the function has a minimum or maximum. Find the axis of symmetry, vertex, then graph using a table of values. Give the domain and range.

1. $f(x) = x^2 - 8x + 17$

x	$f(x)$



Minimum / Maximum

Axis of Symmetry: _____

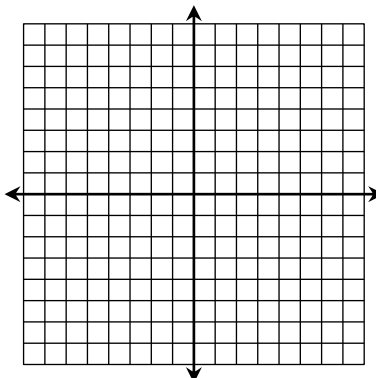
Vertex: _____

Domain: _____

Range: _____

2. $f(x) = -x^2 - 2x - 5$

x	$f(x)$



Minimum / Maximum

Axis of Symmetry: _____

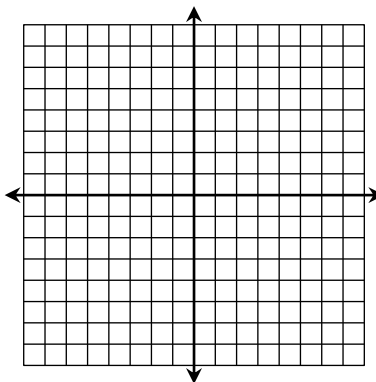
Vertex: _____

Domain: _____

Range: _____

3. $f(x) = 3x^2 + 1$

x	$f(x)$



Minimum / Maximum

Axis of Symmetry: _____

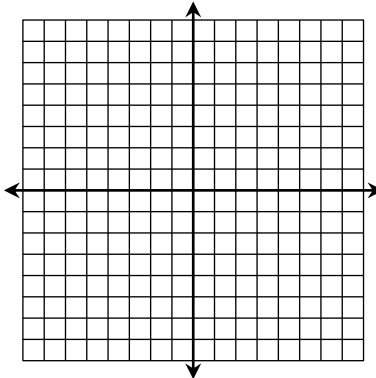
Vertex: _____

Domain: _____

Range: _____

4. $f(x) = -2x^2 + 8x - 11$

x	$f(x)$



Minimum / Maximum

Axis of Symmetry: _____

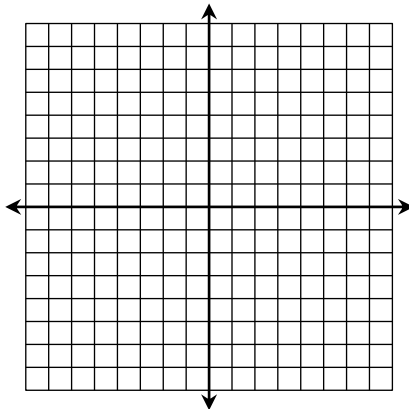
Vertex: _____

Domain: _____

Range: _____

Graph each function. Give the domain and range.

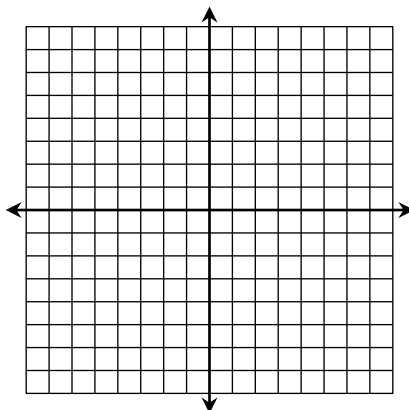
5. $f(x) = \begin{cases} x^2 + 4x & \text{if } x < -3 \\ 2 & \text{if } x \geq -3 \end{cases}$



Domain: _____

Range: _____

6. $f(x) = \begin{cases} -\frac{2}{3}x + 5 & \text{if } x \leq 0 \\ -x & \text{if } 0 < x \leq 4 \\ -x^2 + 4x + 4 & \text{if } x > 4 \end{cases}$



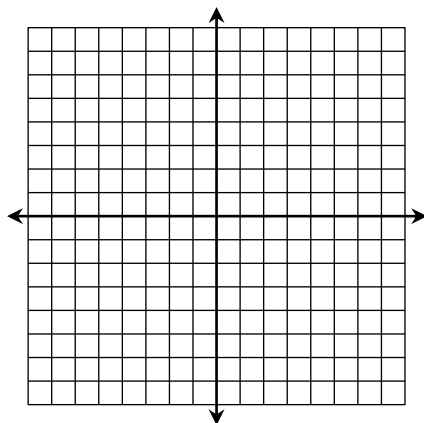
Domain: _____

Range: _____

Graph each inequality.

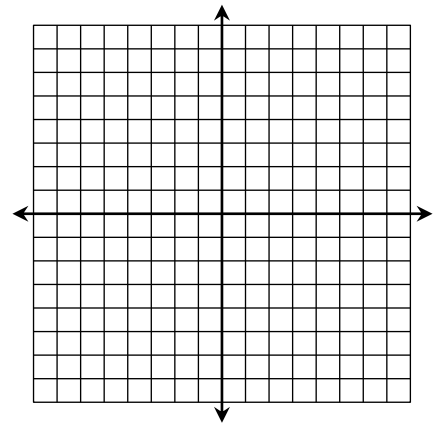
7. $y \leq x^2 + 6x + 5$

x	$f(x)$



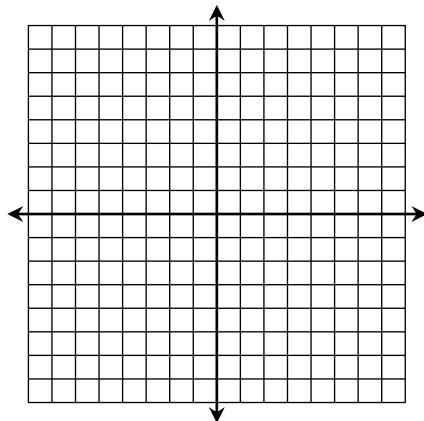
8. $y > -2x^2 + 8x - 12$

x	$f(x)$



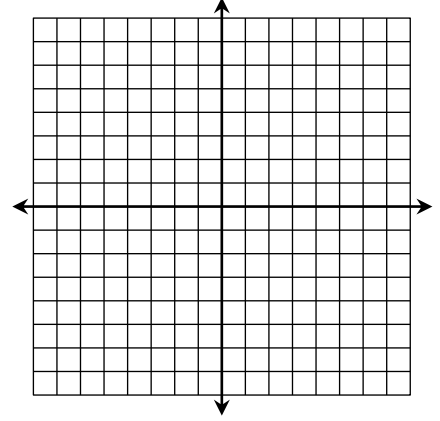
9. $y < -x^2 + 12x - 29$

x	$f(x)$



10. $y \geq 4x^2 - 16x + 15$

x	$f(x)$



Name:

Date:

Topic:

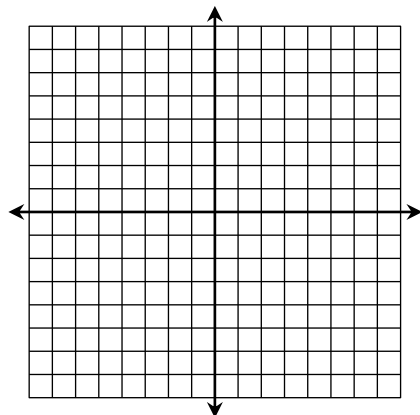
Class:

Main Ideas/Questions	Notes/Examples
VERTEX FORM <i>of a Quadratic Function</i>	• Vertex Form of a Quadratic Equation: _____ • _____ is the vertex; _____ is the axis of symmetry • _____ determines the width and direction of the parabola
WHY USE <i>Vertex Form?</i>	Vertex form shows the _____ on a quadratic equation in respect to its parent function, _____.
TRANSFORMATIONS <i>from the Parent Function</i>	Describe how each function compares to the parent function. 1) $f(x) = x^2 + 9$ 2) $f(x) = x^2 - 2$ 3) $f(x) = -(x + 5)^2$ 4) $f(x) = (x - 4)^2 - 1$ 5) $f(x) = \frac{1}{3}x^2 + 2$ 6) $f(x) = -2(x + 4)^2$ 7) Mason graphed the parent function of a quadratic equation. Then, he reflected the parabola over the x-axis, and translated it seven units up and three units to the right. What is the equation of the new parabola?

Directions: Give the vertex and axis of symmetry of each quadratic function. Then graph.

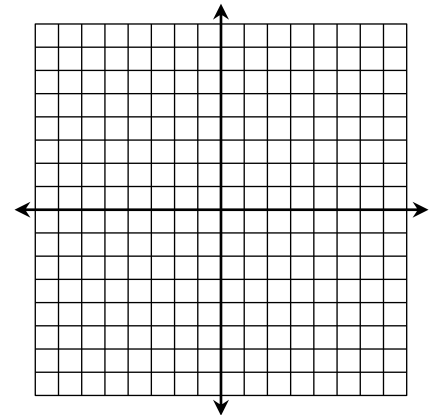
8. $f(x) = (x - 2)^2 + 3$

x	f(x)



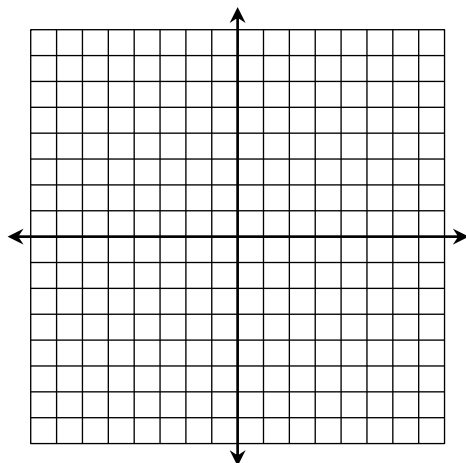
9. $f(x) = x^2 - 4$

x	f(x)



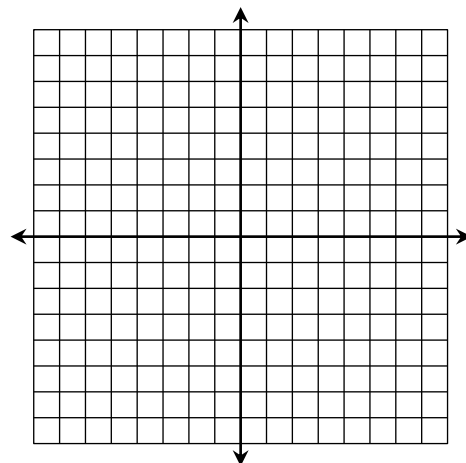
10. $f(x) = -(x+5)^2 - 1$

x	$f(x)$



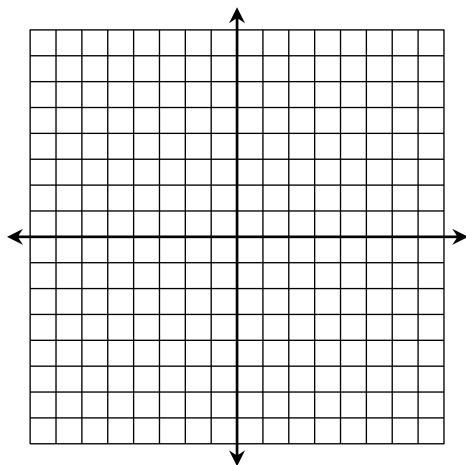
11. $f(x) = 3(x-4)^2$

x	$f(x)$

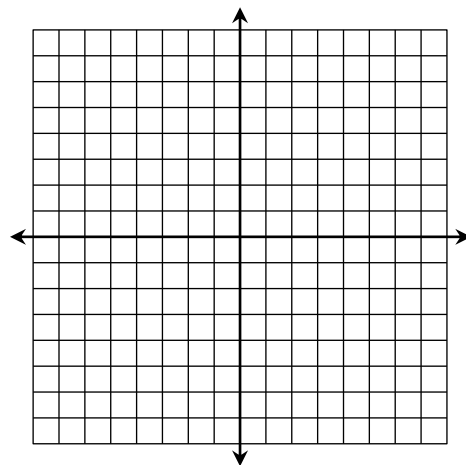


Directions: Graph each piecewise function.

12. $f(x) = \begin{cases} -\frac{1}{2}x - 6 & \text{if } x < -2 \\ (x-1)^2 - 5 & \text{if } x > -2 \end{cases}$



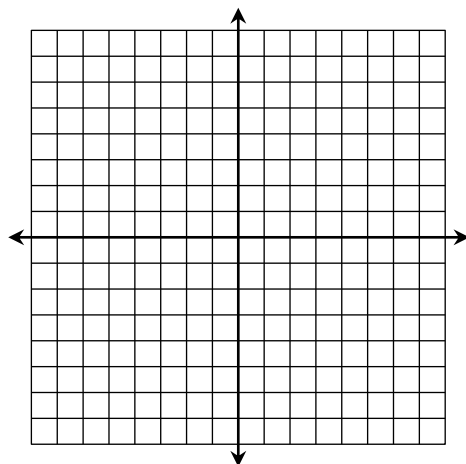
13. $f(x) = \begin{cases} -2x^2 + 6 & \text{if } x \leq 1 \\ -3x + 4 & \text{if } x > 1 \end{cases}$



Directions: Give the vertex and axis of symmetry of each quadratic inequality. Then graph.

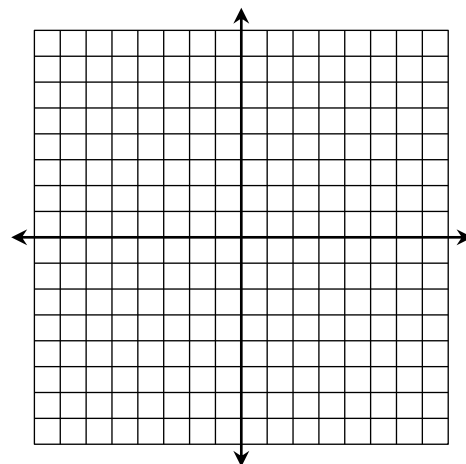
14. $y < -(x-3)^2$

x	$f(x)$



15. $y \geq 4(x+1)^2 - 7$

x	$f(x)$



Name: _____

Unit 3: Parent Functions & Transformations

Date: _____ Per: _____

Homework 6: Vertex Form of a Quadratic Function

**** This is a 2-page document! ****

Describe the transformation of each function compared to its parent function.

1. $f(x) = (x - 4)^2 + 9$

2. $f(x) = -2x^2 - 3$

3. $f(x) = \frac{1}{4}(x + 6)^2$

4. $f(x) = -(x - 7)^2 - 1$

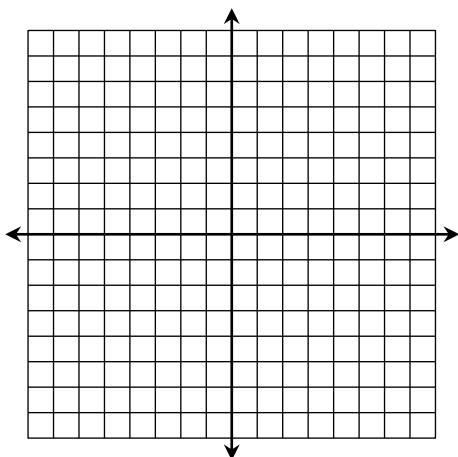
5. $f(x) = \frac{3}{2}x^2$

6. $f(x) = -\frac{2}{3}(x - 5)^2 + 2$

Give the vertex and axis of symmetry of each quadratic function, then graph.

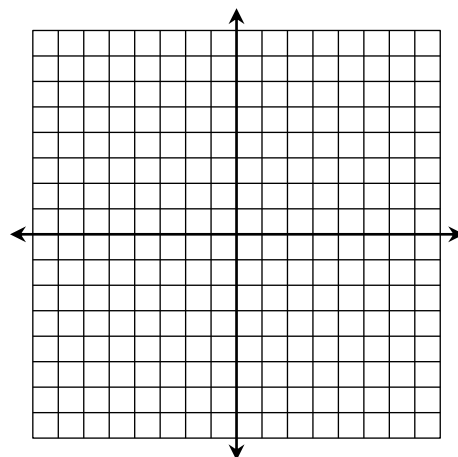
7. $f(x) = (x + 1)^2 - 8$

x	f(x)



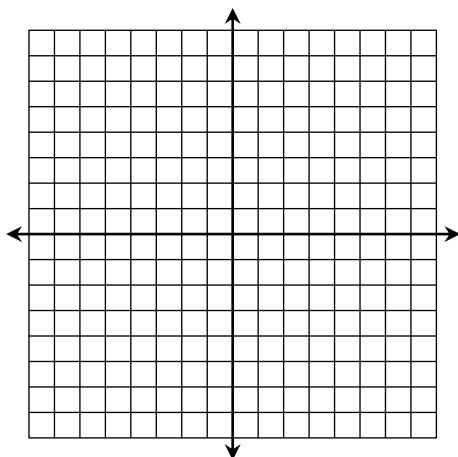
8. $f(x) = -(x + 5)^2 + 5$

x	f(x)



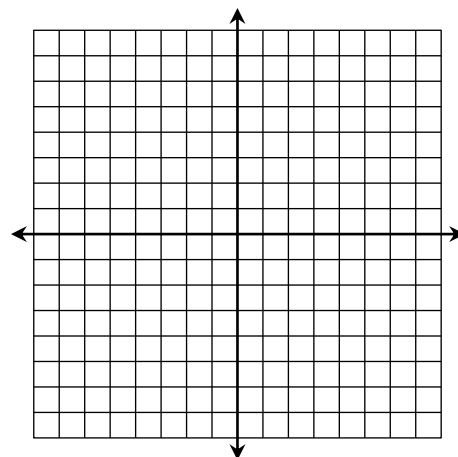
9. $f(x) = \frac{1}{2}x^2 - 3$

x	f(x)



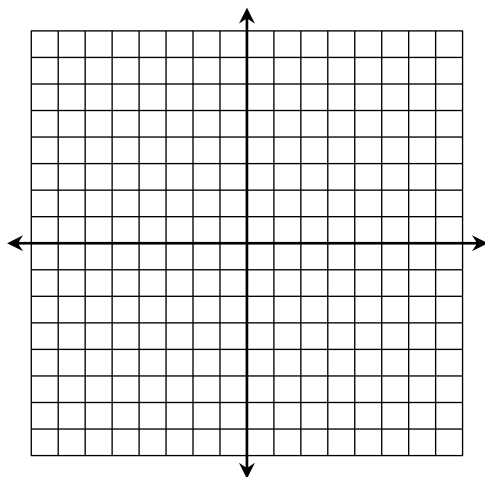
10. $f(x) = -3(x - 4)^2 + 1$

x	f(x)

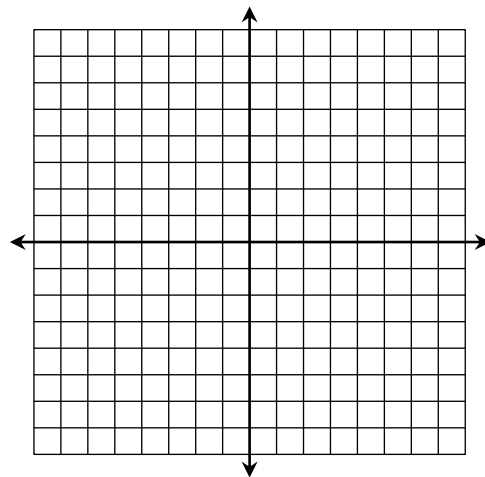


Graph each piecewise function.

11. $f(x) = \begin{cases} -x^2 + 8 & \text{if } x < 1 \\ 2x & \text{if } x > 1 \end{cases}$



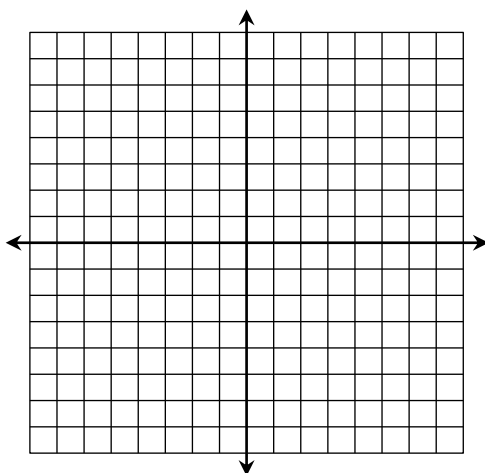
12. $f(x) = \begin{cases} -2 & \text{if } x < 4 \\ (x-4)^2 - 5 & \text{if } x \geq 4 \end{cases}$



Give the vertex and axis of symmetry of each quadratic inequality, then graph.

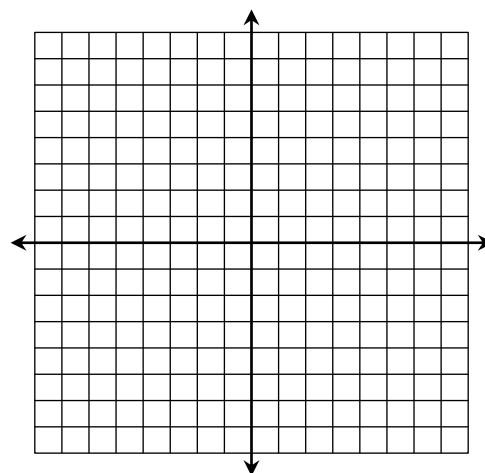
13. $y > -x^2$

x	$f(x)$



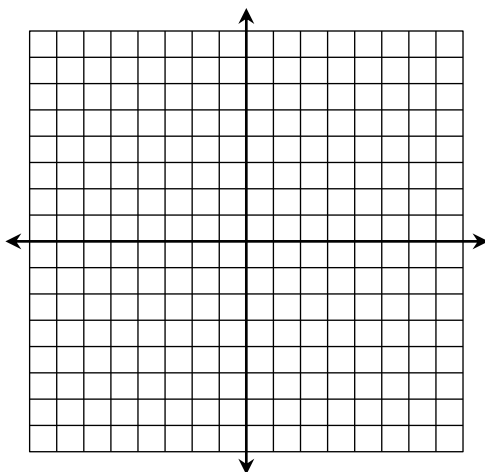
14. $y \geq 4x^2 - 7$

x	$f(x)$



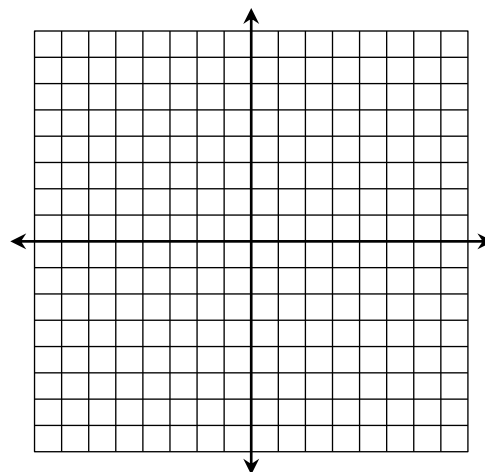
15. $y < (x-1)^2 - 1$

x	$f(x)$



16. $y \leq -\frac{1}{2}(x+3)^2 + 4$

x	$f(x)$



Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples	
ALGEBRA I REVIEW: <i>Perfect Square Trinomials</i>	A trinomial of the form $ax^2 + bx + c$ where _____. Factor the following trinomials: • $x^2 + 8x + 16 =$ _____ • $m^2 - 18m + 81 =$ _____ • $a^2 + 2a + 1 =$ _____ • $p^2 - 10p + 25 =$ _____	
	If the standard form equation is a perfect square (like ex. 1), simply factor to convert to vertex form. If not, you can use a process called completing the square to write the equation in vertex form.	
STANDARD FORM $f(x) = ax^2 + bx + c$  VERTEX FORM $f(x) = a(x - h)^2 + k$	①	GROUP $(ax^2 + bx)$
	②	If $a \neq 1$, FACTOR it outside so it becomes $a(x^2 + bx)$
	③	COMPLETE THE SQUARE by taking half of b and square it. This is the new " c ". Add this to " $x^2 + bx$ ". This is the perfect square trinomial.
	④	SUBTRACT $a \cdot c$ from the end of the equation.
	⑤	FACTOR the trinomial and SIMPLIFY the end of the equation.
Directions: Write each function in vertex form. Give the vertex.		
1. $f(x) = x^2 + 4x + 4$	2. $f(x) = x^2 - 8x + 18$	
3. $f(x) = -x^2 + 6x - 5$	4. $f(x) = 2x^2 - 12x + 13$	

5. $f(x) = x^2 - 3x - 2$

6. $f(x) = -x^2 - 9x - 22$

7. $f(x) = x^2 + 5x$

8. $f(x) = 4x^2 - 8x$

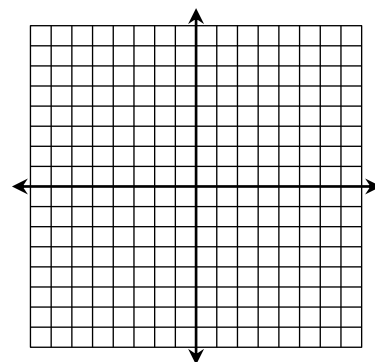
9. $f(x) = -2x^2 + 20x - 46$

10. $f(x) = -\frac{1}{2}x^2 - 4x - 5$

Directions: Convert to vertex form. Identify the vertex, axis of symmetry, then graph.

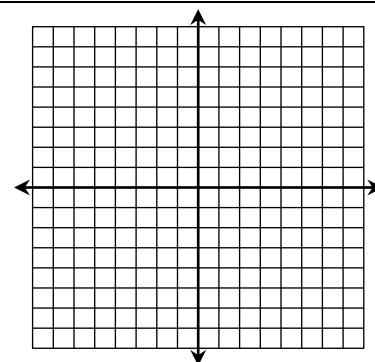
11. $f(x) = x^2 - 12x + 39$

x	$f(x)$



12. $f(x) = -3x^2 - 12x - 8$

x	$f(x)$



Name: _____

Unit 3: Parent Functions & Transformations

Date: _____ Per: _____

Homework 7: Converting to Vertex Form

**** This is a 2-page document! ****

Write each function in vertex form. Give the vertex.

1. $f(x) = x^2 - 12x + 36$

2. $f(x) = x^2 + 4x - 1$

3. $f(x) = x^2 - 2x + 5$

4. $f(x) = 3x^2 + 6x - 4$

5. $f(x) = x^2 + 7x - 10$

6. $f(x) = -x^2 + 6x - 10$

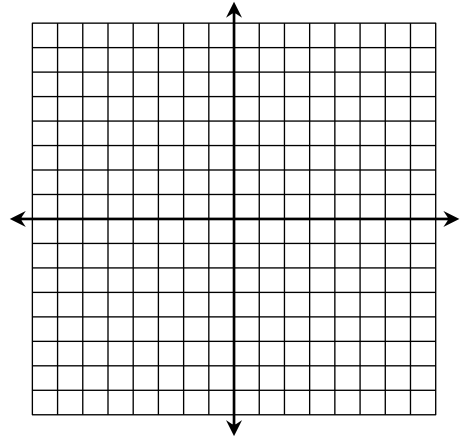
7. $f(x) = -6x^2 + 42x$

8. $f(x) = \frac{1}{2}x^2 - 4x + 7$

Write each function in vertex form. Identify the vertex and axis of symmetry, then graph.

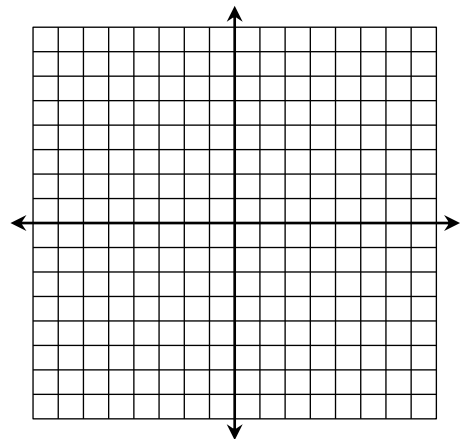
9. $f(x) = x^2 + 10x + 18$

x	$f(x)$



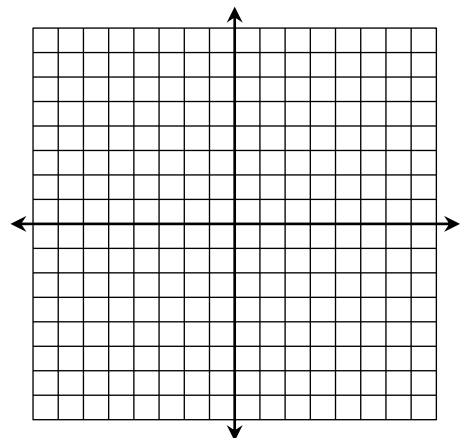
10. $f(x) = -x^2 + 4x - 6$

x	$f(x)$



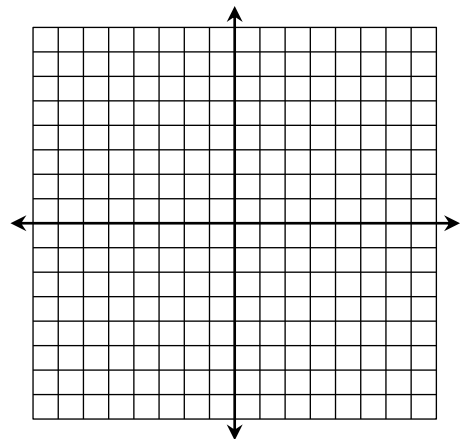
11. $f(x) = -2x^2 + 8x - 1$

x	$f(x)$



12. $f(x) = 3x^2 + 6x$

x	$f(x)$

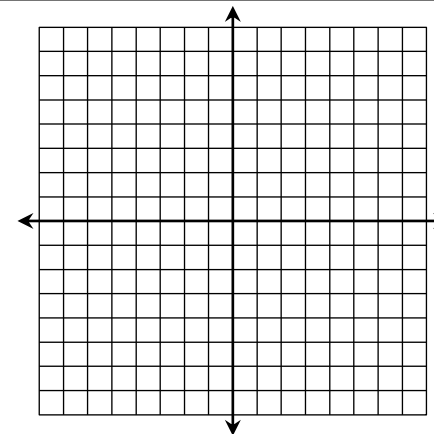


QUADRATIC FUNCTIONS

PARENT FUNCTION:

- Creates a U-shape curve called a _____.
- The _____ is a vertical line that divides the parabola into two equal parts.
- The turning point is called the _____.

x	f(x)

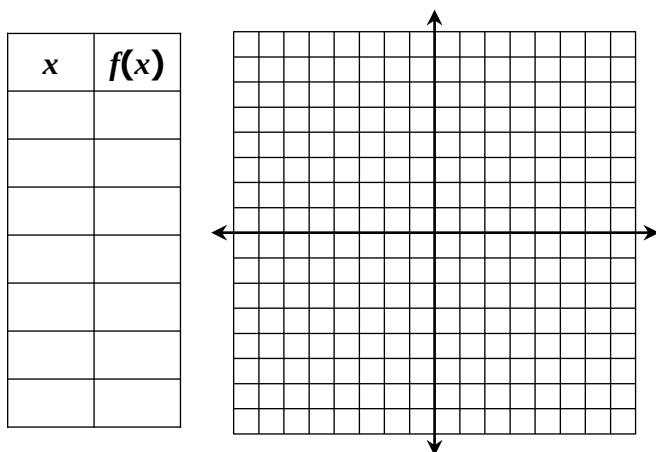


STANDARD FORM:

Axis of Symmetry: _____

Plug back in to find the vertex!

EXAMPLE: $y = x^2 - 4x + 1$

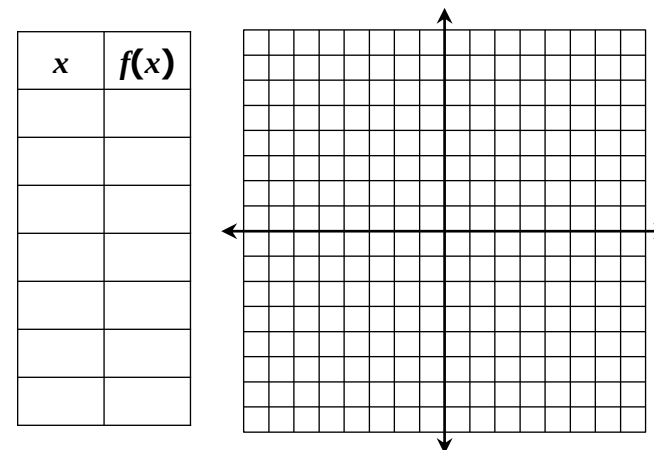


VERTEX FORM:

Axis of Symmetry: _____

Vertex: _____

EXAMPLE: $y = -2(x + 1)^2 + 6$



Quiz 3-3: Graphing Quadratic Functions

Identify the following forms in respect to quadratic functions.

1. Parent Function
1. _____
2. Standard Form
2. _____
3. Vertex Form
3. _____

Give the axis of symmetry, vertex, then graph the function. Give the domain and range.

4. $f(x) = -x^2 + 10x - 17$

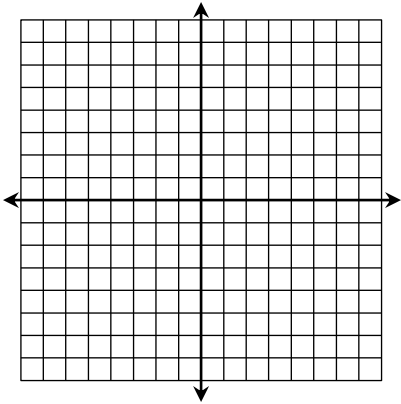
Axis of Symmetry: _____

Vertex: _____

Domain: _____

Range: _____

x	f(x)



5. $f(x) = 2x^2 + 8x + 2$

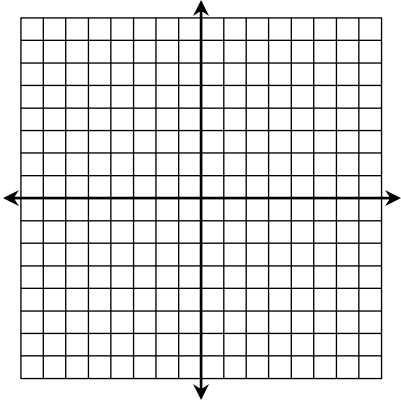
Axis of Symmetry: _____

Vertex: _____

Domain: _____

Range: _____

x	f(x)



6. $f(x) = x^2 - 6$

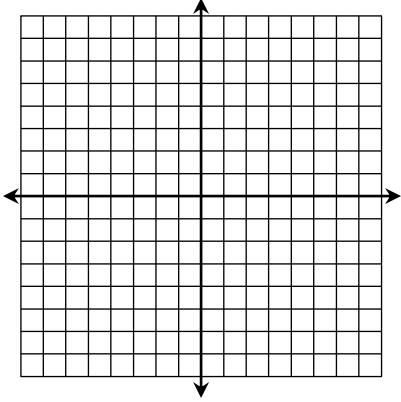
Axis of Symmetry: _____

Vertex: _____

Domain: _____

Range: _____

x	f(x)



7. $f(x) = (x + 4)^2 - 5$

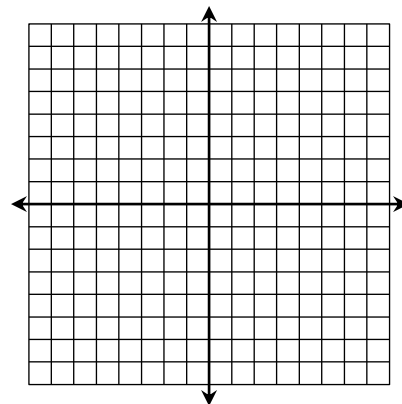
Axis of Symmetry: _____

Vertex: _____

Domain: _____

Range: _____

x	$f(x)$



8. $f(x) = -\frac{1}{2}(x + 3)^2$

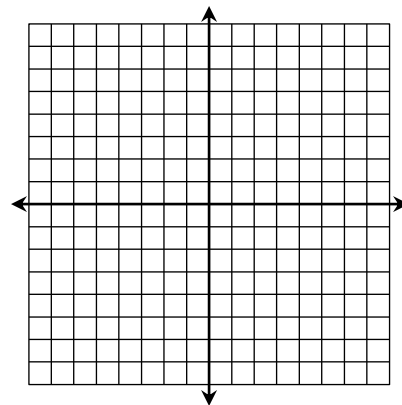
Axis of Symmetry: _____

Vertex: _____

Domain: _____

Range: _____

x	$f(x)$



9. The parent function of a quadratic equation is vertically stretched by a factor of 2, then translated 5 units right and 1 unit down. Write the equation of the new function in vertex form.

9. _____

Write each function in vertex form.

10. $f(x) = x^2 - 16x + 64$

10. _____

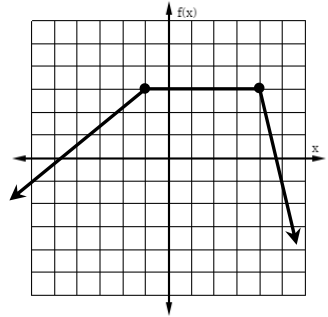
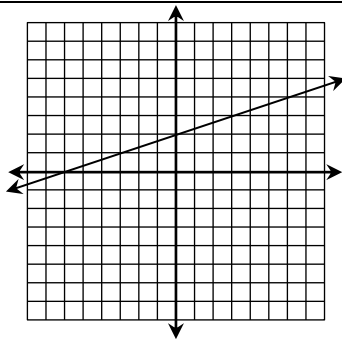
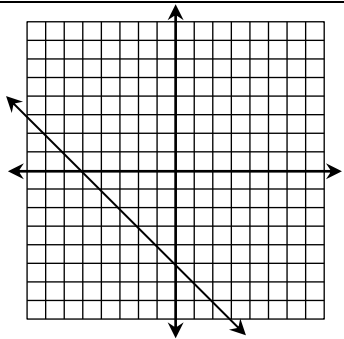
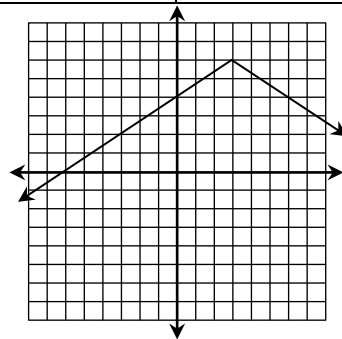
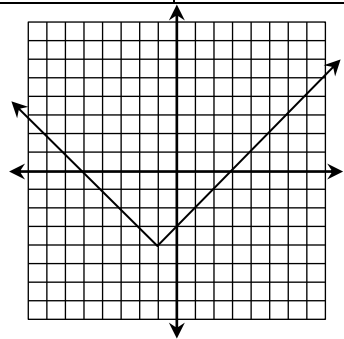
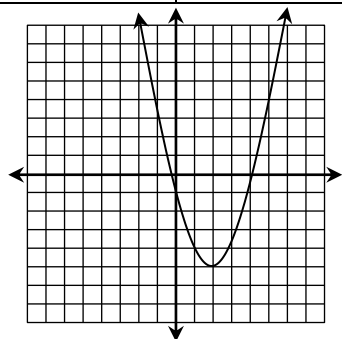
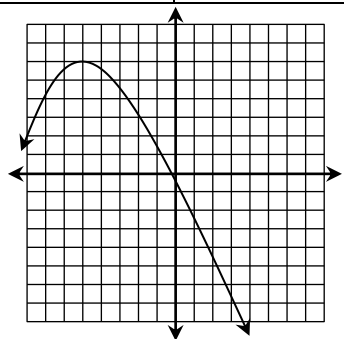
11. $f(x) = x^2 + 4x - 5$

11. _____

12. $f(x) = -3x^2 + 6x + 4$

12. _____

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples			
INCREASING & DECREASING Intervals	Moving from left to right, a function may increase, decrease, or remain constant on certain intervals. For example, the function to the right is increasing on the interval _____. constant on the interval _____. decreasing on the interval _____.			
	EXAMPLES	Identify the intervals on which each function increases or decreases.		
1. 		2. 		
Increasing Interval(s): Decreasing Interval(s):		Increasing Interval(s): Decreasing Interval(s):		
3. 		4. 		
Increasing Interval(s): Decreasing Interval(s):		Increasing Interval(s): Decreasing Interval(s):		
5. 		6. 		
Increasing Interval(s): Decreasing Interval(s):		Increasing Interval(s): Decreasing Interval(s):		

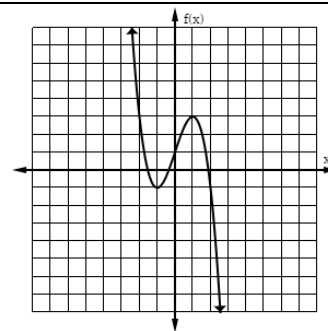
END BEHAVIOR

As x moves towards positive infinity and negative infinity, end behavior describes what is happening to the value of $f(x)$.

Describe the end behavior of the graph to the right:

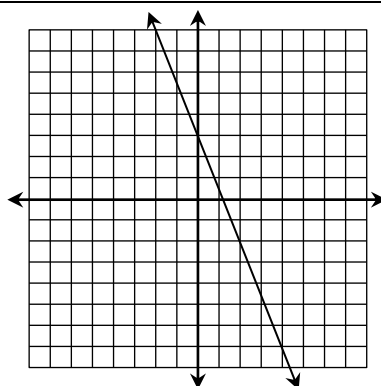
As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____



Describe the end behavior of each function.

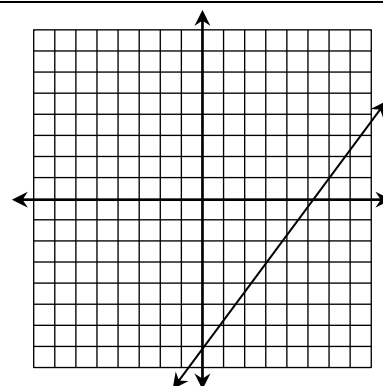
7.



As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

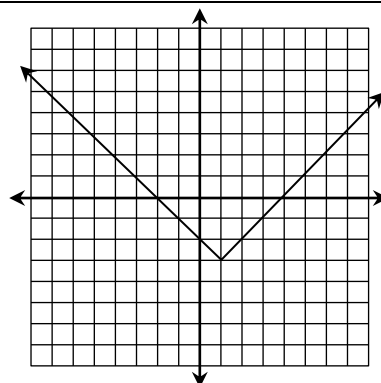
8.



As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

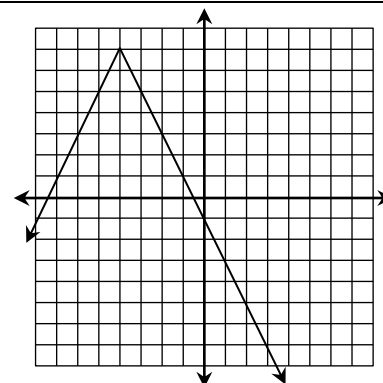
9.



As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

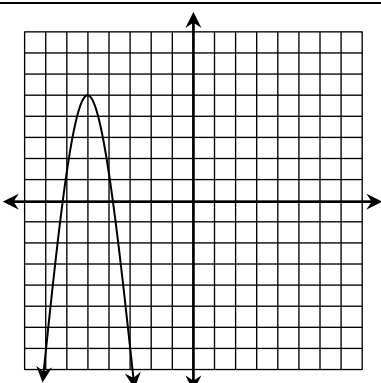
10.



As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

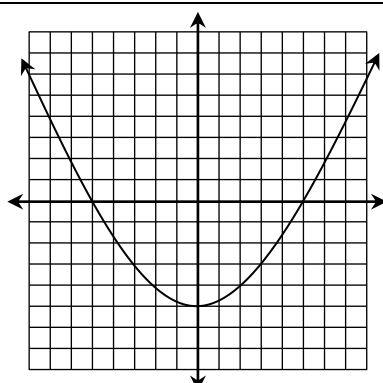
11.



As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

12.



As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

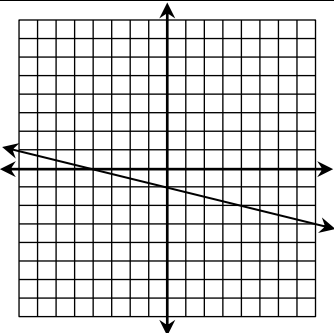
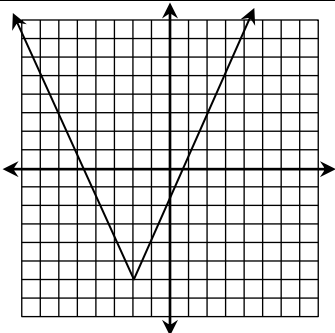
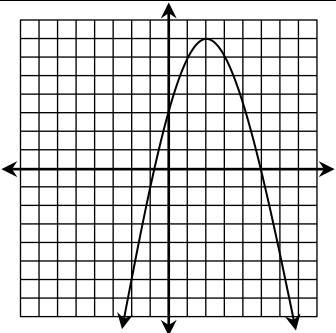
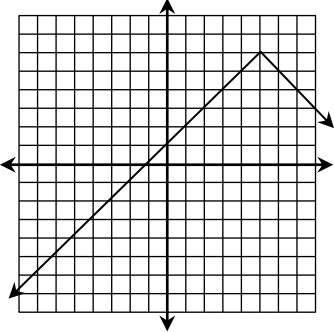
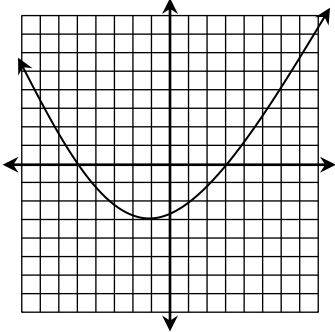
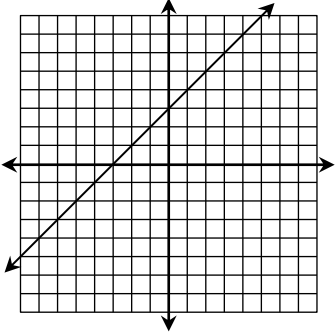
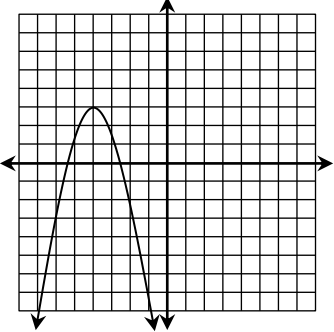
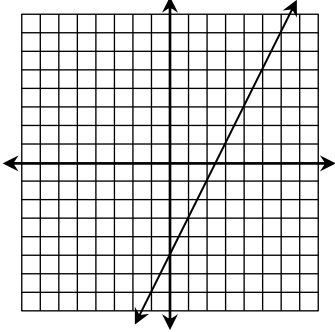
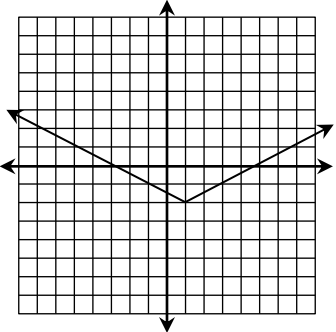
As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

Name: _____

Unit 3: Parent Functions & Transformations

Date: _____ Per: _____

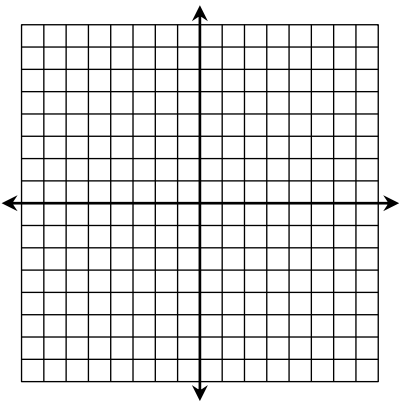
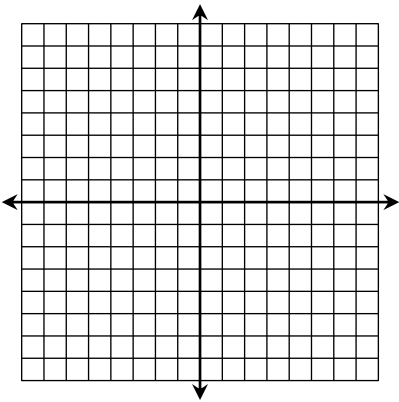
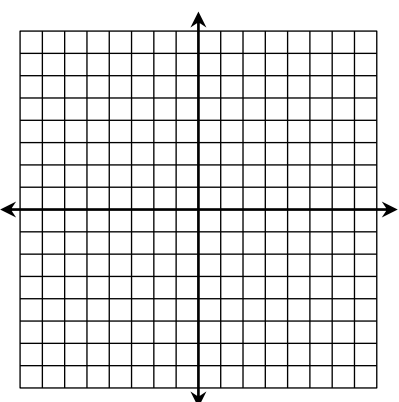
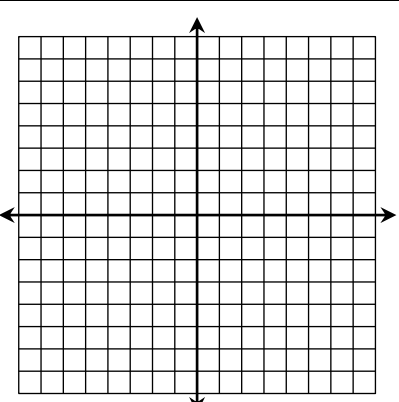
**Homework 8: Increasing/Decreasing Intervals
& End Behavior**

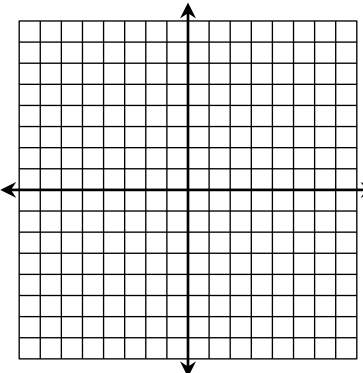
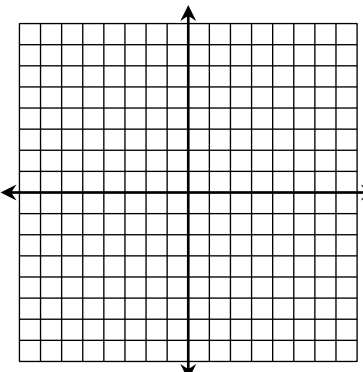
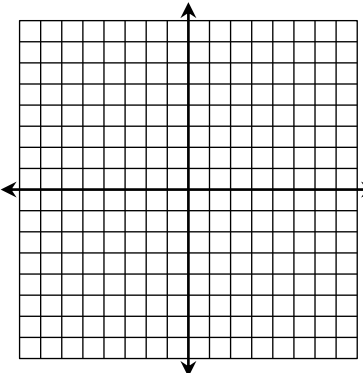
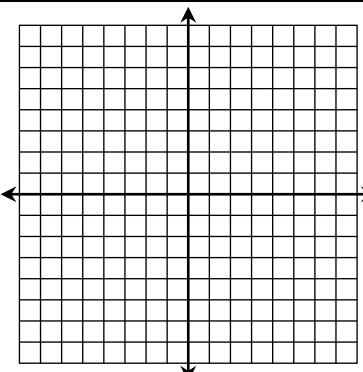
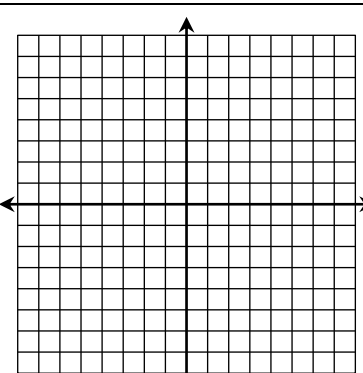
Identify the increasing and decreasing intervals and the end behavior of each graph.					
1. 		2. 		3. 	
Increasing Int(s):		Decreasing Int(s):		Increasing Int(s):	
				Decreasing Int(s):	
End Behavior:		End Behavior:		End Behavior:	
As $x \rightarrow \infty$, $f(x) \rightarrow$ _____		As $x \rightarrow \infty$, $f(x) \rightarrow$ _____		As $x \rightarrow \infty$, $f(x) \rightarrow$ _____	
As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____		As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____		As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____	
4. 		5. 		6. 	
Increasing Int(s):		Decreasing Int(s):		Increasing Int(s):	
				Decreasing Int(s):	
End Behavior:		End Behavior:		End Behavior:	
As $x \rightarrow \infty$, $f(x) \rightarrow$ _____		As $x \rightarrow \infty$, $f(x) \rightarrow$ _____		As $x \rightarrow \infty$, $f(x) \rightarrow$ _____	
As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____		As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____		As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____	
7. 		8. 		9. 	
Increasing Int(s):		Decreasing Int(s):		Increasing Int(s):	
				Decreasing Int(s):	
End Behavior:		End Behavior:		End Behavior:	
As $x \rightarrow \infty$, $f(x) \rightarrow$ _____		As $x \rightarrow \infty$, $f(x) \rightarrow$ _____		As $x \rightarrow \infty$, $f(x) \rightarrow$ _____	
As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____		As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____		As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____	

PARENT FUNCTIONS

A **parent function** is the simplest function of the family.

A **family of graphs** displays similar characteristics.

PARENT FUNCTION	GRAPH	CHARACTERISTICS
LINEAR 		• D: _____; R: _____ • x-intercept: _____; y-intercept: _____ • End Behavior: As $x \rightarrow$ _____, $f(x) \rightarrow$ _____ As $x \rightarrow$ _____, $f(x) \rightarrow$ _____ • Increasing Interval(s): _____ • Decreasing Interval(s): _____
ABSOLUTE VALUE 		• D: _____; R: _____ • x-intercept: _____; y-intercept: _____ • End Behavior: As $x \rightarrow$ _____, $f(x) \rightarrow$ _____ As $x \rightarrow$ _____, $f(x) \rightarrow$ _____ • Increasing Interval(s): _____ • Decreasing Interval(s): _____
QUADRATIC 		• D: _____; R: _____ • x-intercept: _____; y-intercept: _____ • End Behavior: As $x \rightarrow$ _____, $f(x) \rightarrow$ _____ As $x \rightarrow$ _____, $f(x) \rightarrow$ _____ • Increasing Interval(s): _____ • Decreasing Interval(s): _____
CUBIC 		• D: _____; R: _____ • x-intercept: _____; y-intercept: _____ • End Behavior: As $x \rightarrow$ _____, $f(x) \rightarrow$ _____ As $x \rightarrow$ _____, $f(x) \rightarrow$ _____ • Increasing Interval(s): _____ • Decreasing Interval(s): _____

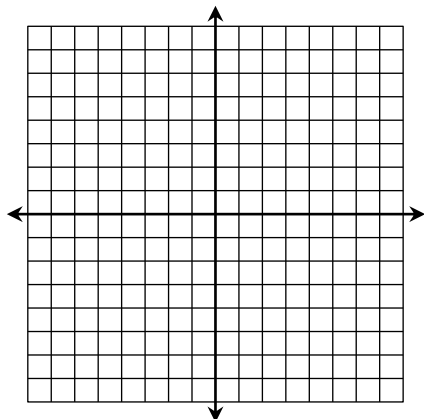
SQUARE ROOT		• D: _____; R: _____ • x-intercept: _____; y-intercept: _____ • End Behavior: As $x \rightarrow$ _____, $f(x) \rightarrow$ _____ As $x \rightarrow$ _____, $f(x) \rightarrow$ _____ • Increasing Interval(s): _____ • Decreasing Intervals(s): _____
CUBE ROOT		• D: _____; R: _____ • x-intercept: _____; y-intercept: _____ • End Behavior: As $x \rightarrow$ _____, $f(x) \rightarrow$ _____ As $x \rightarrow$ _____, $f(x) \rightarrow$ _____ • Increasing Interval(s): _____ • Decreasing Intervals(s): _____
RECIPROCAL		• D: _____; R: _____ • x-intercept: _____; y-intercept: _____ • End Behavior: As $x \rightarrow$ _____, $f(x) \rightarrow$ _____ As $x \rightarrow$ _____, $f(x) \rightarrow$ _____ • Increasing Interval(s): _____ • Decreasing Intervals(s): _____ • Asymptote(s): _____
EXPONENTIAL		• D: _____; R: _____ • x-intercept: _____; y-intercept: _____ • End Behavior: As $x \rightarrow$ _____, $f(x) \rightarrow$ _____ As $x \rightarrow$ _____, $f(x) \rightarrow$ _____ • Increasing Interval(s): _____ • Decreasing Intervals(s): _____ • Asymptote(s): _____
LOGARITHMIC		• D: _____; R: _____ • x-intercept: _____; y-intercept: _____ • End Behavior: As $x \rightarrow$ _____, $f(x) \rightarrow$ _____ As $x \rightarrow$ _____, $f(x) \rightarrow$ _____ • Increasing Interval(s): _____ • Decreasing Intervals(s): _____ • Asymptote(s): _____

FUNCTION FAMILIES REVIEW

{Linear, Absolute Value, and Quadratic}

Graph each function, then identify the characteristics.

1. $f(x) = -3x + 5$



Function Family: _____

Parent Function: _____

D: _____; R: _____

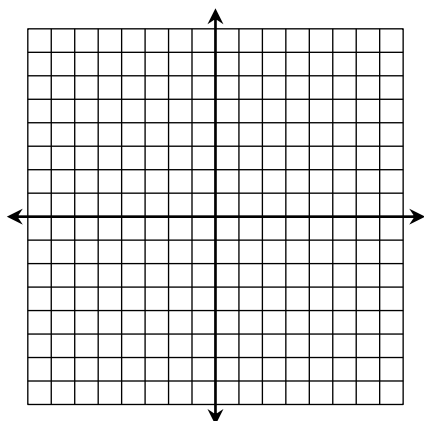
End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

Increasing Interval(s): _____

Decreasing Interval(s): _____

2. $f(x) = |x - 1| - 4$



Function Family: _____

Parent Function: _____

D: _____; R: _____

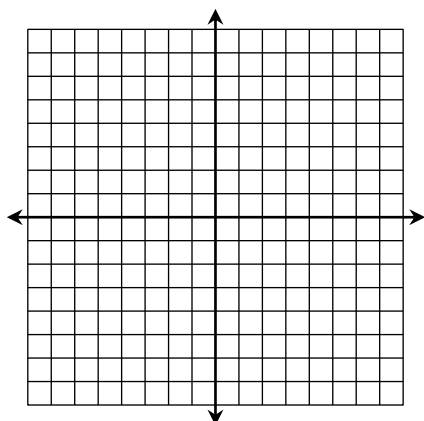
End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

Increasing Interval(s): _____

Decreasing Interval(s): _____

3. $f(x) = -x^2 + 8x - 11$



Function Family: _____

Parent Function: _____

D: _____; R: _____

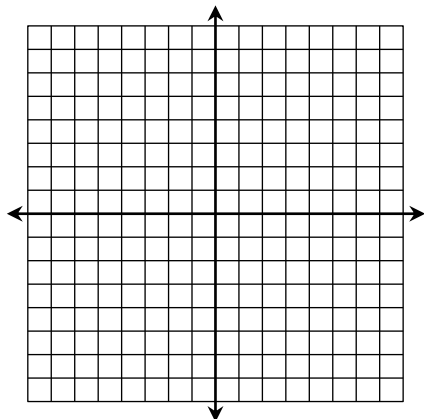
End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

Increasing Interval(s): _____

Decreasing Interval(s): _____

4. $f(x) = -|x| + 6$



Function Family: _____

Parent Function: _____

D: _____; R: _____

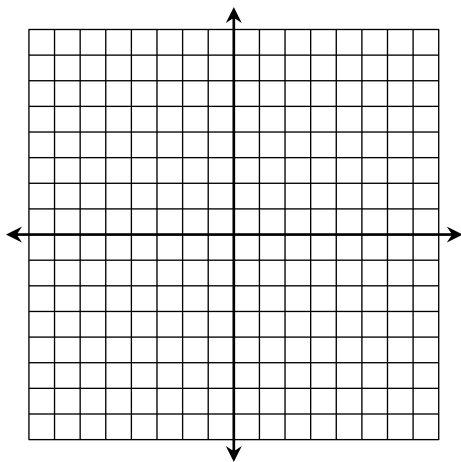
End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

Increasing Interval(s): _____

Decreasing Interval(s): _____

5. $f(x) = 2(x+2)^2 - 7$



Function Family: _____

Parent Function: _____

D: _____; R: _____

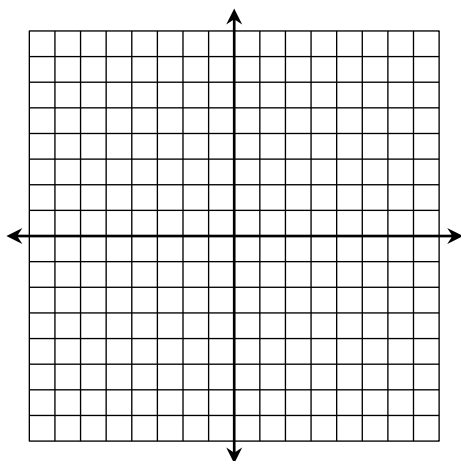
End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

Increasing Interval(s): _____

Decreasing Interval(s): _____

6. $f(x) = \frac{5}{2}x - 1$



Function Family: _____

Parent Function: _____

D: _____; R: _____

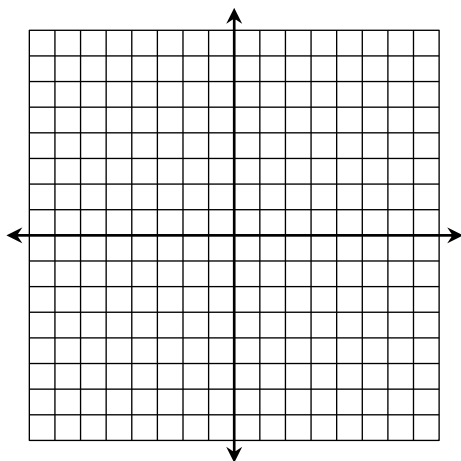
End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

Increasing Interval(s): _____

Decreasing Interval(s): _____

7. $f(x) = -(x-5)^2 + 6$



Function Family: _____

Parent Function: _____

D: _____; R: _____

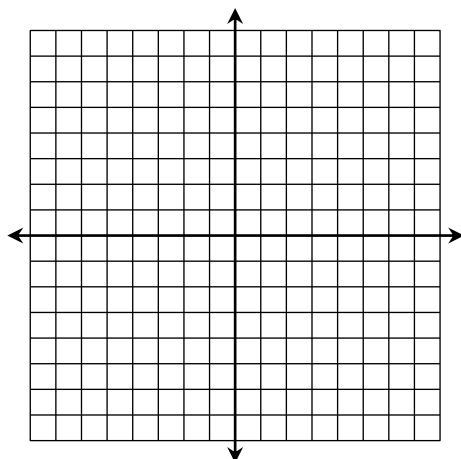
End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

Increasing Interval(s): _____

Decreasing Interval(s): _____

8. $f(x) = \frac{1}{2}|x+3| + 3$



Function Family: _____

Parent Function: _____

D: _____; R: _____

End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

Increasing Interval(s): _____

Decreasing Interval(s): _____

Name: _____

Unit 3: Parent Functions & Transformations

Date: _____ Per: _____

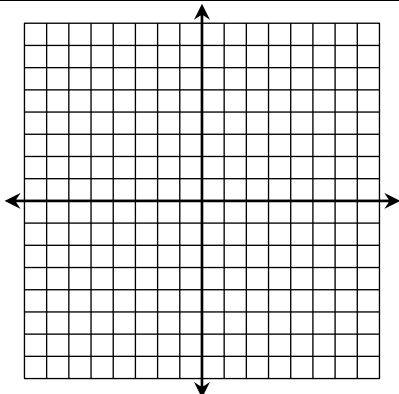
Homework 9: Function Families Review:

Linear, Absolute Value, & Quadratic

** This is a 2-page document! **

Graph each function, then identify the characteristics.

1. $f(x) = (x + 5)^2 - 2$



Function Family: _____

Parent Function: _____

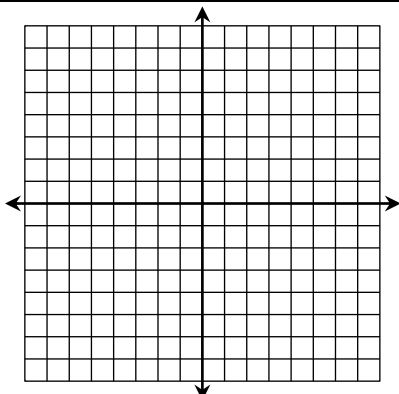
D: _____; R: _____

End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow$ _____As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

Increasing Interval(s): _____

Decreasing Interval(s): _____

2. $f(x) = -\frac{2}{3}x$



Function Family: _____

Parent Function: _____

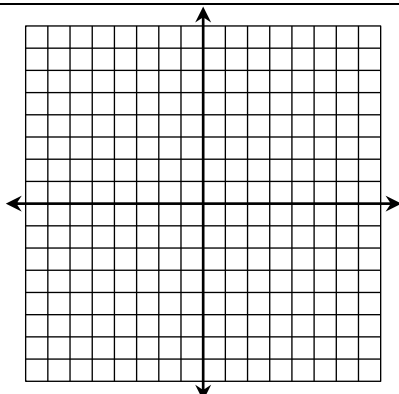
D: _____; R: _____

End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow$ _____As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

Increasing Interval(s): _____

Decreasing Interval(s): _____

3. $f(x) = -3x^2 + 12x - 8$



Function Family: _____

Parent Function: _____

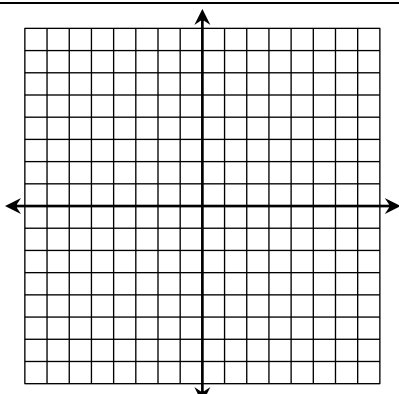
D: _____; R: _____

End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow$ _____As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

Increasing Interval(s): _____

Decreasing Interval(s): _____

4. $f(x) = 2|x + 4|$



Function Family: _____

Parent Function: _____

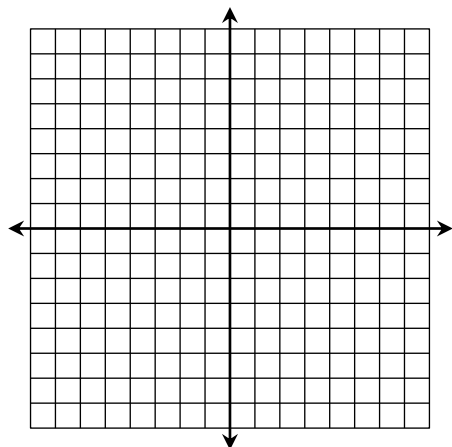
D: _____; R: _____

End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow$ _____As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

Increasing Interval(s): _____

Decreasing Interval(s): _____

5. $f(x) = -x^2 + 7$



Function Family: _____

Parent Function: _____

D: _____; R: _____

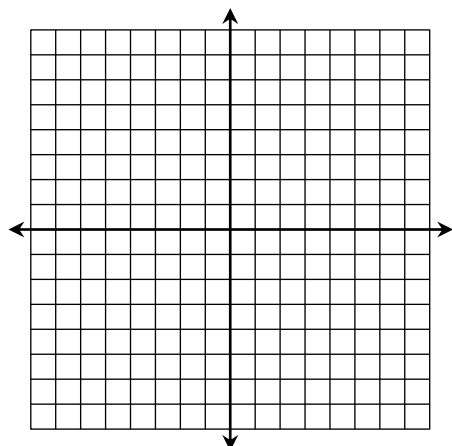
End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

Increasing Interval(s): _____

Decreasing Interval(s): _____

6. $f(x) = -\frac{1}{3}|x-2| + 4$



Function Family: _____

Parent Function: _____

D: _____; R: _____

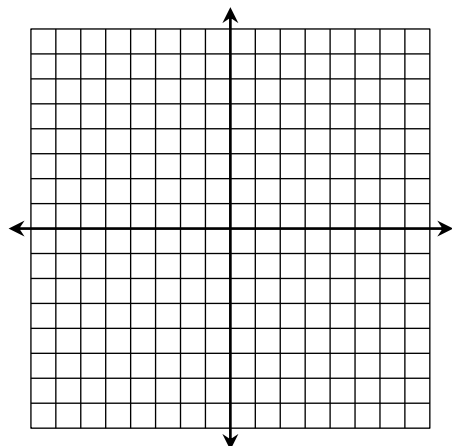
End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

Increasing Interval(s): _____

Decreasing Interval(s): _____

7. $f(x) = 5x + 4$



Function Family: _____

Parent Function: _____

D: _____; R: _____

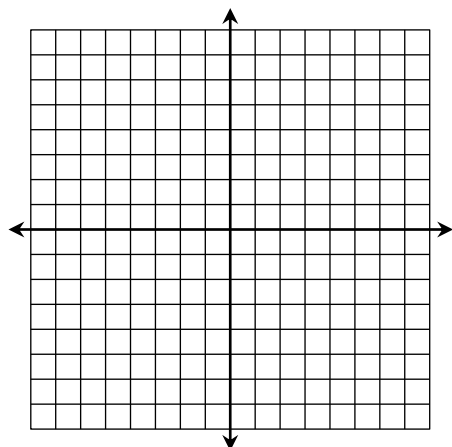
End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

Increasing Interval(s): _____

Decreasing Interval(s): _____

8. $f(x) = -|x+3| + 6$



Function Family: _____

Parent Function: _____

D: _____; R: _____

End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

Increasing Interval(s): _____

Decreasing Interval(s): _____

Name:

Date:

Topic:

Class:

Main Ideas/Questions

Notes/Examples

GREATEST INTEGER

Function

- The **greatest integer function** is written as _____
- The symbol _____ means _____
- In short, this means it will always **round the value down to the nearest integer**. It is also frequently called a **floor function**.

EVALUATING

1. Given $f(x) = \llbracket x \rrbracket$, find each value.

a) $f(7)$

b) $f(2.3)$

c) $f(-4.8)$

2. Given $f(x) = \llbracket x - 5 \rrbracket$, find each value.

a) $f(9)$

b) $f(0.5)$

c) $f(-8.1)$

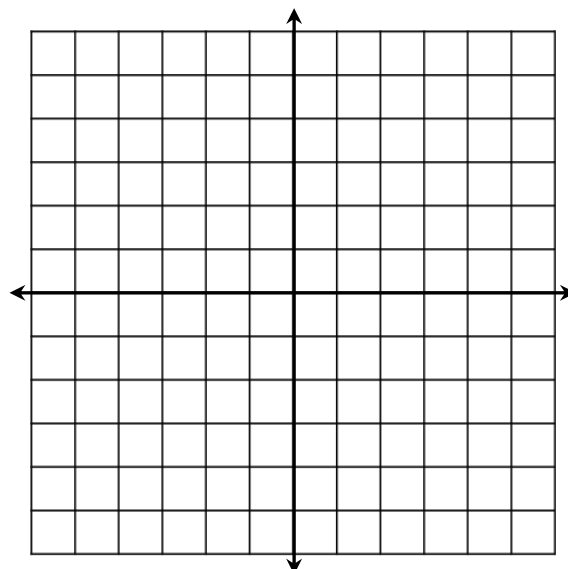
GREATEST INTEGER

Parent Function

Use a table to help generate a pattern, then graph the greatest integer parent function. Identify the domain and range.

3. $f(x) = \llbracket x \rrbracket$

x	y



$D =$ _____

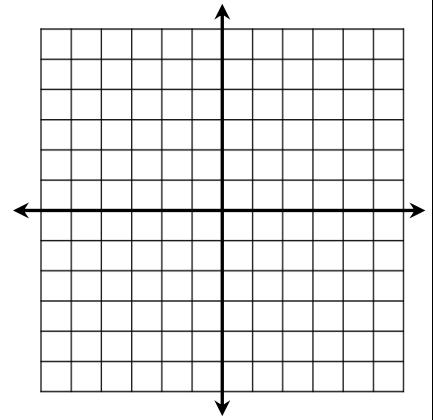
$R =$ _____

Graph each function. Then identify the domain and range.

4. $f(x) = \lfloor x \rfloor - 5$

$D =$ _____

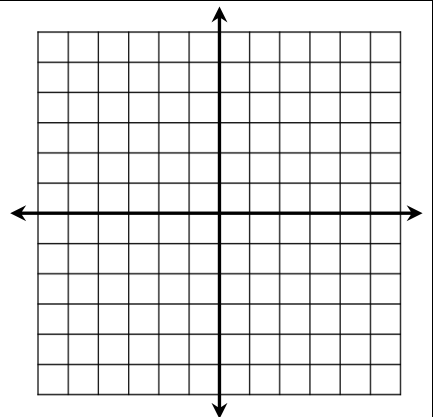
$R =$ _____



5. $f(x) = \left\lfloor \frac{1}{2}x + 4 \right\rfloor$

$D =$ _____

$R =$ _____



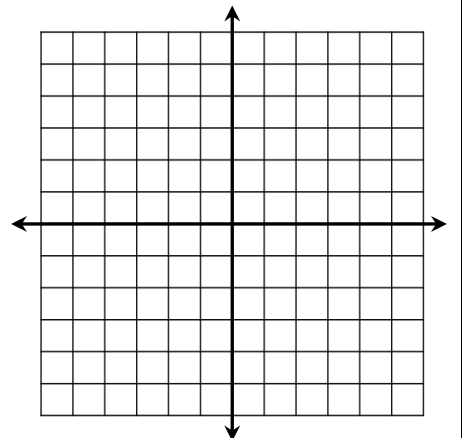
Transformation(s) from the parent function are described. Write the new function, then graph and identify the domain and range.

6. Shifted up 2 units

Function: _____

$D =$ _____

$R =$ _____

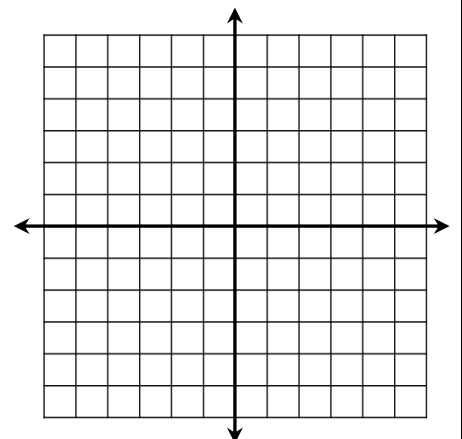


7. Vertical stretch by a factor of 2, then shifted right 1 unit

Function: _____

$D =$ _____

$R =$ _____



TRANSFORMATIONS

x	y

x	y

Name: _____

Unit 3: Parent Functions & Transformations

Date: _____ Per: _____

Homework 10: Greatest Integer Function

** This is a 2-page document! **

Evaluate the following.

1. $\llbracket 6.8 \rrbracket$

2. $\llbracket 17.21 \rrbracket$

3. $\llbracket 2.99 \rrbracket$

4. $\llbracket -12.7 \rrbracket$

5. $\llbracket -38.15 \rrbracket$

6. $\llbracket 495 \rrbracket$

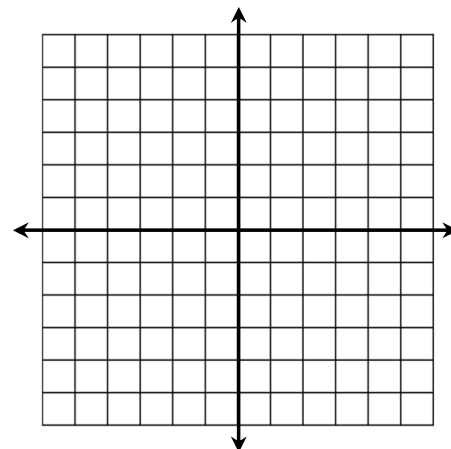
Graph each function, then identify the domain and range.

7. $f(x) = \llbracket x + 4 \rrbracket$

$D =$ _____

$R =$ _____

x	y

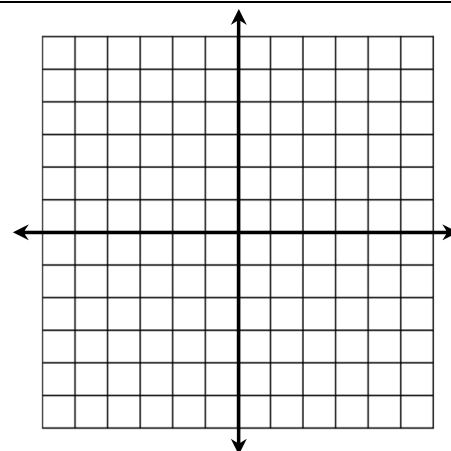


8. $f(x) = \frac{3}{2} \llbracket x \rrbracket + 1$

$D =$ _____

$R =$ _____

x	y

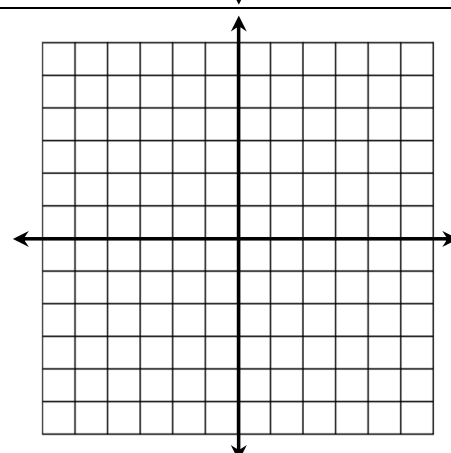


9. $f(x) = \llbracket 2x - 3 \rrbracket$

$D =$ _____

$R =$ _____

x	y



Transformation(s) from the greatest integer parent function are described. Write the new function, then graph and identify the domain and range.

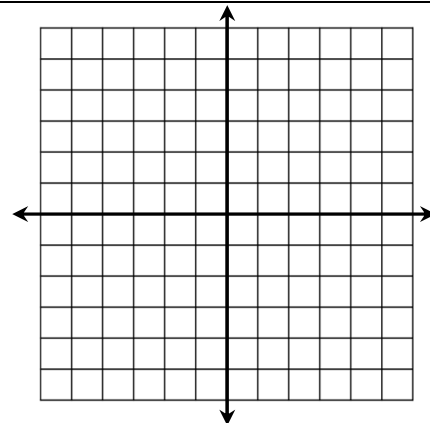
- 10.** Reflected in the x -axis, then shifted down 5 units.

Function: _____

D : _____

R : _____

x	y



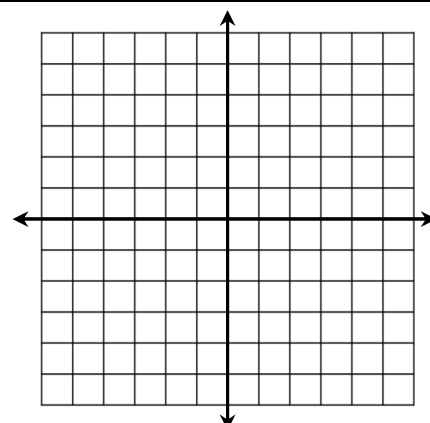
- 11.** Vertically compressed by a factor of $\frac{1}{2}$, then shifted right 3 units.

Function: _____

D : _____

R : _____

x	y



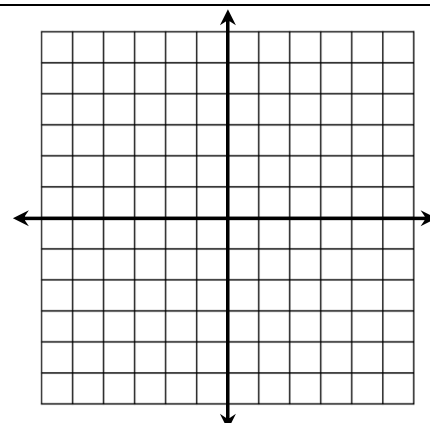
- 12.** Shifted left 1 unit and up 4 units.

Function: _____

D : _____

R : _____

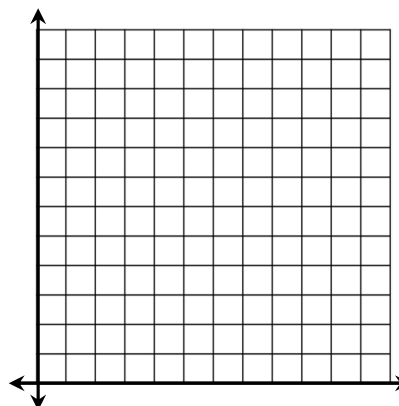
x	y



- 13.** The cost to mail a package is \$4 for up to one pound, then \$1.50 for each additional pound or fraction of a pound. Write a function to represent the cost, C , to mail a package weighing w pounds, then complete the table and graph the function.

Function: _____

w	C
$0 < w \leq 1$	
$1 < w \leq 2$	
$2 < w \leq 3$	
$3 < w \leq 4$	
$4 < w \leq 5$	
$5 < w \leq 6$	



Unit 3 Test Study Guide

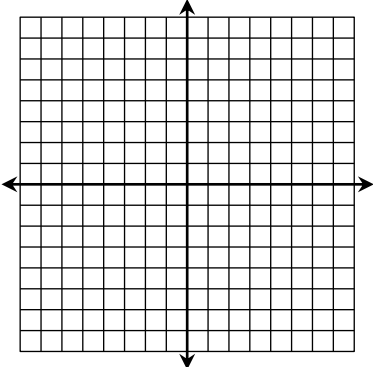
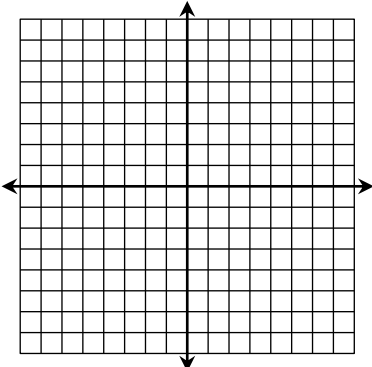
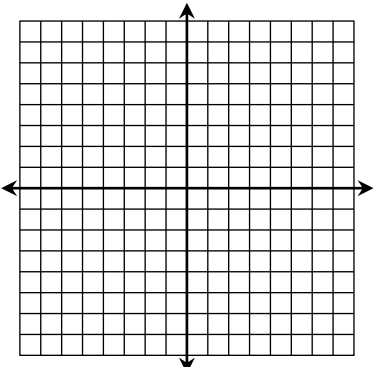
(Parent Functions & Transformations)

Name: _____

Date: _____ Per: _____

Topic 1: Introduction to Parent Functions - Linear, Absolute Value, and Quadratic

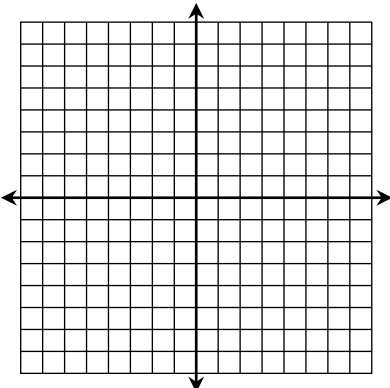
Identify the parent function of each function family, then graph:

Linear	Absolute Value	Quadratic
		

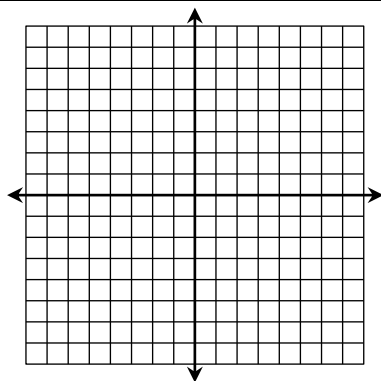
Topic 2: Transformation on Functions

Describe each transformation rule:			
$f(x + h)$		$f(x - h)$	
$f(x) + k$		$f(x) - k$	
$a \cdot f(x)$ when $ a > 1$		$a \cdot f(x)$ when $ a < 1$	
$-f(x)$			

Topic 3: Absolute Value Functions

Vertex Form of an Absolute Value Function:	
Give the vertex, then graph each function. State the domain and range.	
1. $f(x) = x - 2 - 7$ 	Vertex: _____ Domain: _____ Range: _____

2. $f(x) = -2|x| + 4$



Vertex: _____

Domain: _____

Range: _____

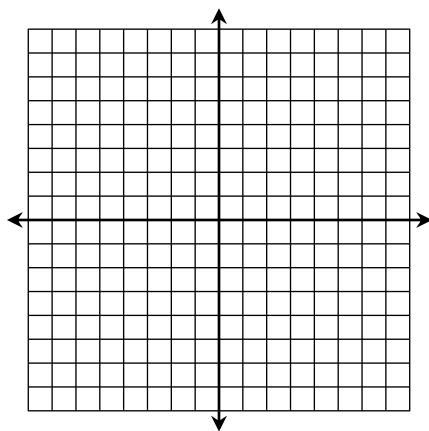
The parent function of an absolute value equation is transformed as described. Write the new equation in vertex form and identify the vertex.

3. Reflected about the x-axis, then shifted five units left.

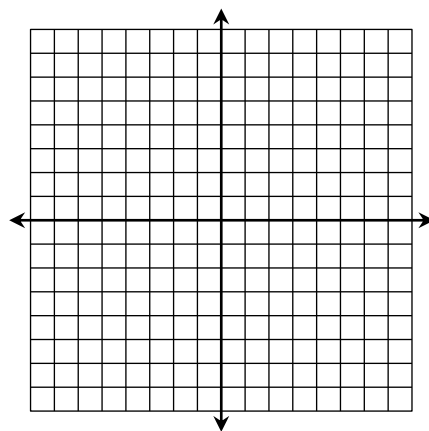
4. Vertically compressed by a factor of one-third, then shifted one unit up and two units right.

Graph each inequality.

5. $y < |x + 6| - 4$



6. $y \geq -2|x + 3|$



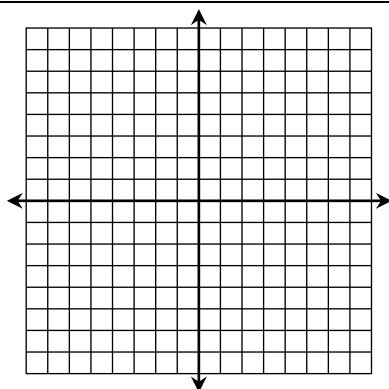
Topic 4: Quadratic Functions

Standard Form of a Quadratic Function:

Vertex Form of a Quadratic Function:

Give the vertex and axis of symmetry, then graph each function. State the domain and range.

7. $f(x) = x^2 - 2x - 7$



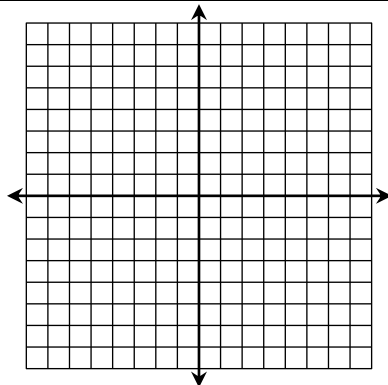
Axis of Symmetry: _____

Vertex: _____

Domain: _____

Range: _____

8. $f(x) = -3x^2 + 5$



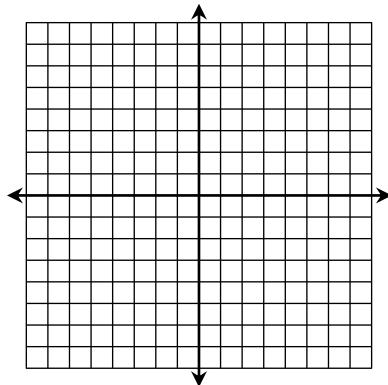
Axis of Symmetry: _____

Vertex: _____

Domain: _____

Range: _____

9. $f(x) = (x + 3)^2 + 2$



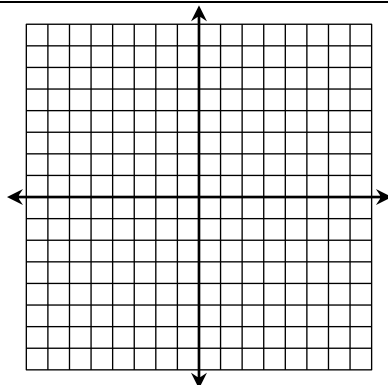
Axis of Symmetry: _____

Vertex: _____

Domain: _____

Range: _____

10. $f(x) = -(x - 4)^2$



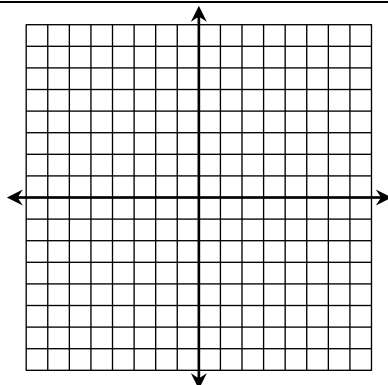
Axis of Symmetry: _____

Vertex: _____

Domain: _____

Range: _____

11. $f(x) = \frac{1}{2}(x - 5)^2 - 6$



Axis of Symmetry: _____

Vertex: _____

Domain: _____

Range: _____

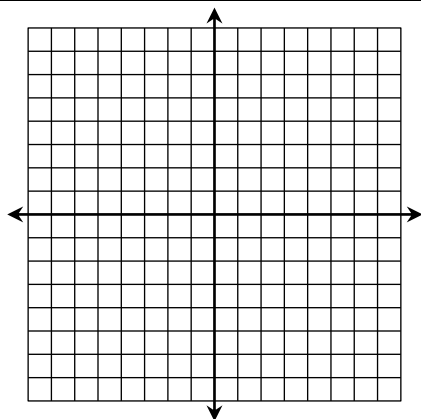
The parent function of a quadratic equation is transformed as described. Write the new equation in vertex form and identify the vertex.

12. Vertically stretched by a factor of four and shifted seven units right.

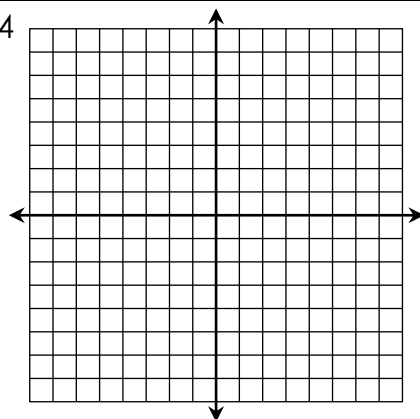
13. Reflected across the x-axis, vertically compressed by a factor of one-half, then shifted four units left and six units down.

Graph each inequality.

14. $y \geq -x^2 - 6x - 1$



15. $y < 2(x + 4)^2 - 4$



Convert each function to vertex form. Then state the vertex.

16. $f(x) = x^2 - 20x + 100$

17. $f(x) = x^2 + 2x - 8$

18. $f(x) = -x^2 + 10x - 23$

19. $f(x) = -2x^2 + 12x - 19$

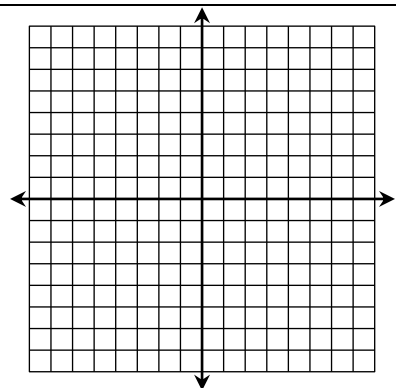
Topic 5: Piecewise Functions

Graph each function. State the domain and range.

20. $f(x) = \begin{cases} 2x + 7 & \text{if } x < -4 \\ -\frac{1}{4}x & \text{if } x \geq -4 \end{cases}$

Domain: _____

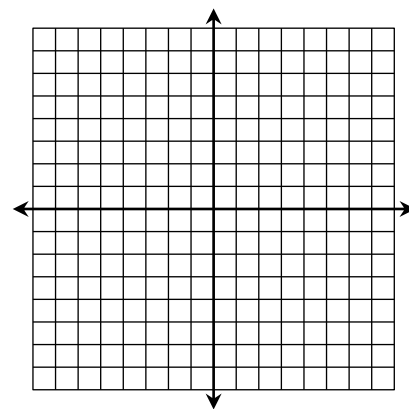
Range: _____



$$21. g(x) = \begin{cases} -|x+2| & \text{if } x < 1 \\ 3x-5 & \text{if } x > 1 \end{cases}$$

Domain: _____

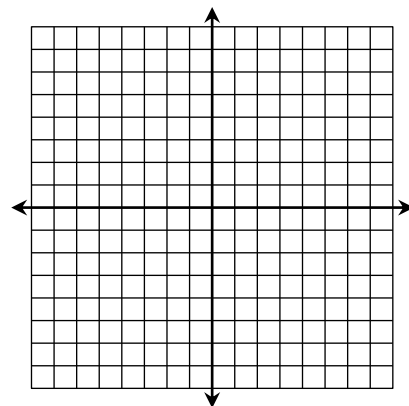
Range: _____



$$22. p(x) = \begin{cases} 5 & \text{if } x \leq 0 \\ -x^2 + 3 & \text{if } x > 0 \end{cases}$$

Domain: _____

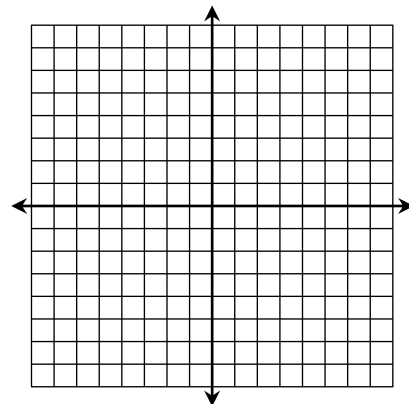
Range: _____



$$23. h(x) = \begin{cases} -x-9 & \text{if } x < -2 \\ (x+1)^2 - 5 & \text{if } -2 \leq x < 2 \\ 2|x-3| & \text{if } x \geq 2 \end{cases}$$

Domain: _____

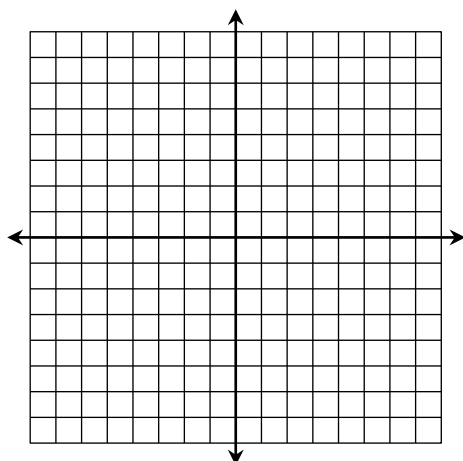
Range: _____



Topic 6: Mixed Review - Linear, Absolute Value, and Quadratic

Graph each function and identify its characteristics.

$$24. f(x) = |x-3| + 4$$



Function Family: _____

Parent Function: _____

D: _____; R: _____

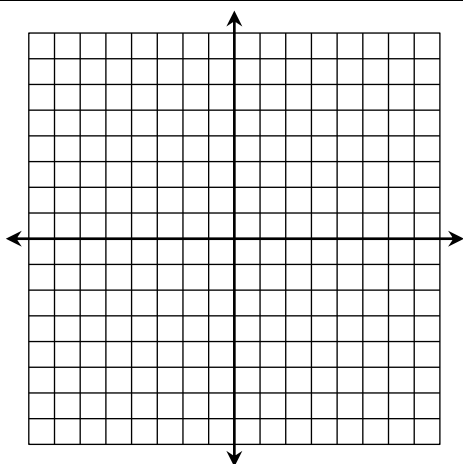
End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

Increasing Interval: _____

Decreasing Interval: _____

25. $f(x) = -\frac{6}{5}x + 2$



Function Family: _____

Parent Function: _____

D: _____; R: _____

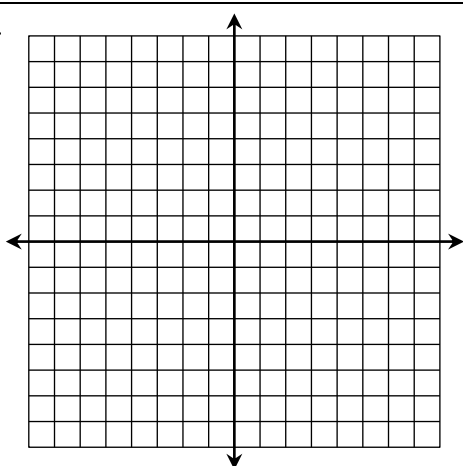
End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

Increasing Interval: _____

Decreasing Interval: _____

26. $f(x) = -x^2 + 8x - 14$



Function Family: _____

Parent Function: _____

D: _____; R: _____

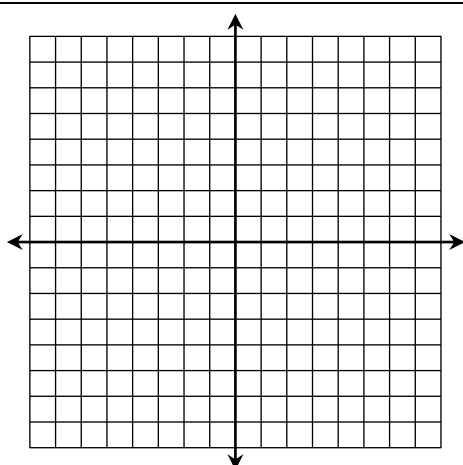
End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

Increasing Interval: _____

Decreasing Interval: _____

27. $f(x) = x - 4$



Function Family: _____

Parent Function: _____

D: _____; R: _____

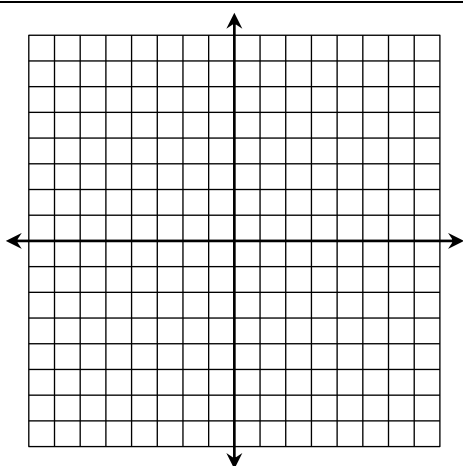
End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

Increasing Interval: _____

Decreasing Interval: _____

28. $f(x) = -\frac{1}{2}(x+1)^2$



Function Family: _____

Parent Function: _____

D: _____; R: _____

End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

Increasing Interval: _____

Decreasing Interval: _____

Name: _____

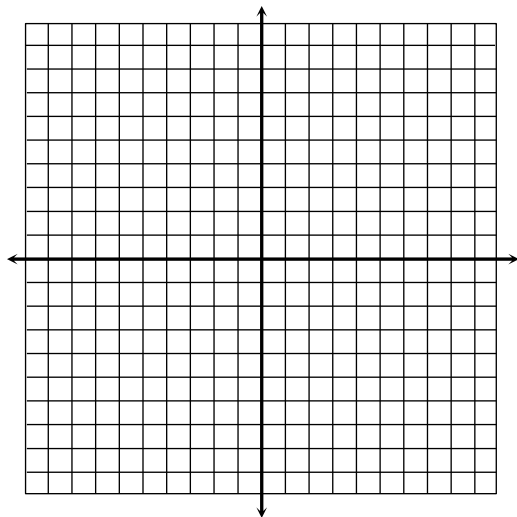
Unit 3 Test

Date: _____ Per: _____

Parent Functions & Transformations
(Linear, Absolute Value, and Quadratic)

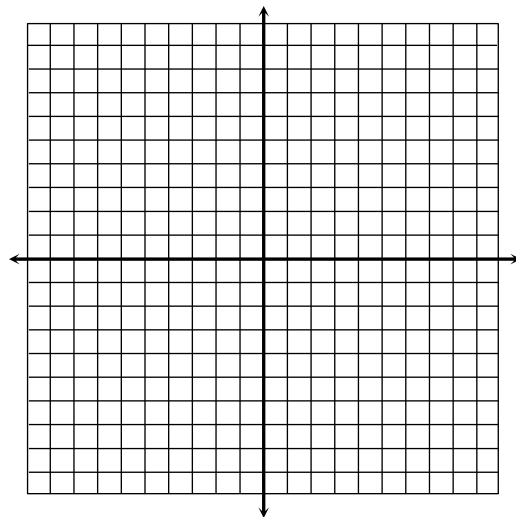
Identify the vertex and graph each function. State the domain and range.

1. $f(x) = 2|x - 6|$ Vertex: _____



D: _____ R: _____

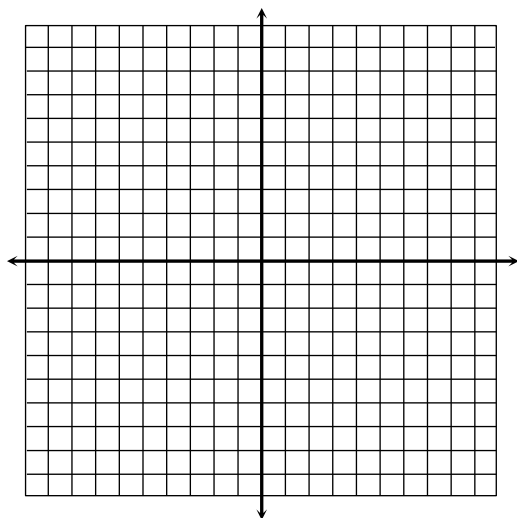
2. $f(x) = -|x + 1| + 9$ Vertex: _____



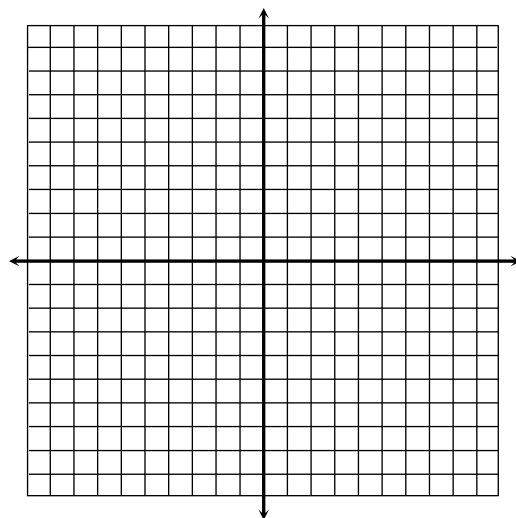
D: _____ R: _____

Identify the vertex, then graph each inequality.

3. $y > |x| - 3$ Vertex: _____



4. $y \leq -\frac{1}{2}|x - 4| + 8$ Vertex: _____



5. The graph of the absolute value parent function is vertically stretched by a factor of three, then shifted two units left and five units down. Write the equation of the new function in vertex form.

6. Which function has a vertex at $(-1, -3)$?

A. $f(x) = |x + 1| + 3$

B. $f(x) = |x - 1| + 3$

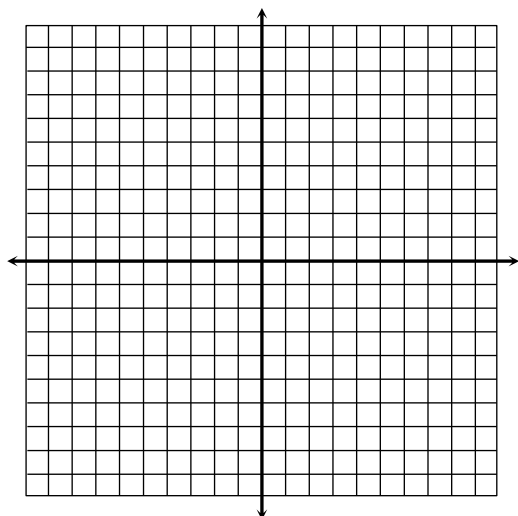
C. $f(x) + 3 = |x + 1|$

D. $f(x) + 3 = |x - 1|$

Identify the axis of symmetry and vertex for each function, then graph. State the domain and range.

7. $f(x) = x^2 + 12x + 31$

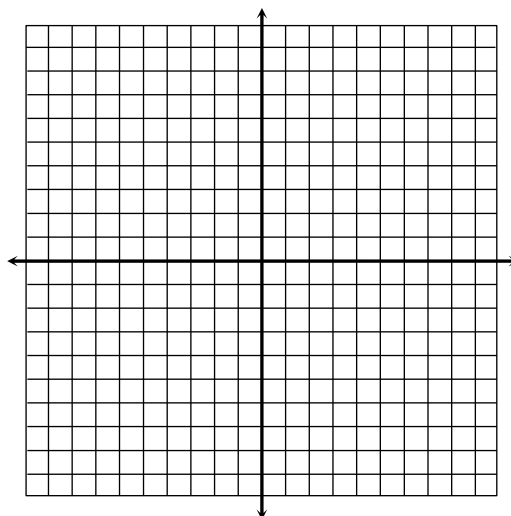
Axis of Symmetry: _____ Vertex: _____



D: _____ R: _____

8. $f(x) = -2(x - 3)^2 + 4$

Axis of Symmetry: _____ Vertex: _____

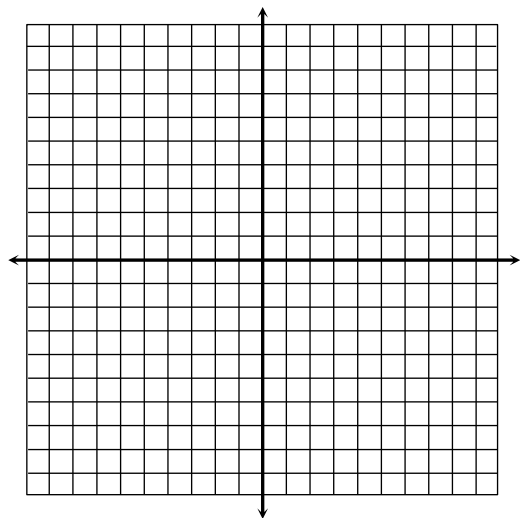


D: _____ R: _____

Identify the vertex, axis of symmetry, then graph each inequality.

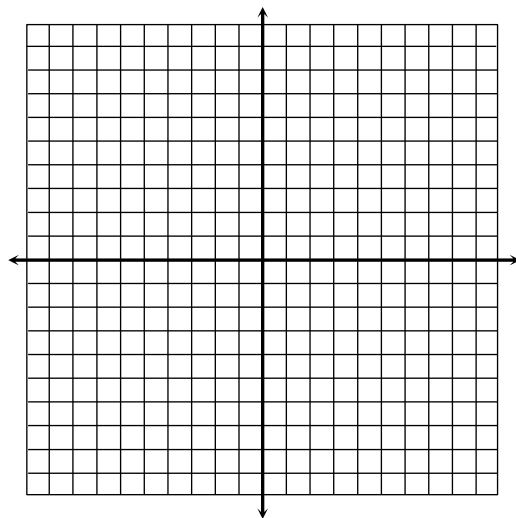
9. $y \geq -(x + 4)^2$

Axis of Symmetry: _____ Vertex: _____



10. $y < 3(x - 2)^2 - 5$

Axis of Symmetry: _____ Vertex: _____



11. The graph of the quadratic parent function is reflected across the x-axis, then shifted five units down and four units right. Write the equation of the new function in vertex form.

12. Which function has an axis of symmetry of $x = -2$?

- A. $f(x) = x^2 - 4x - 1$
- B. $f(x) = -3x^2 - 12x + 4$
- C. $f(x) = (x - 2)^2 - 8$
- D. $f(x) = -2(x + 4)^2 - 3$

13. What transformations take place by graphing the function below in respect to its parent function? Check all that apply.

$$f(x) = \frac{3}{2}(x + 7)^2 + 2$$

☐ Up 7 units	☐ Up 2 units	☐ Vertical Stretch
☐ Down 7 units	☐ Down 2 units	☐ Vertical Compression
☐ Left 7 units	☐ Left 2 units	☐ Reflection in x-axis
☐ Right 7 units	☐ Right 2 units	☐ Reflection in y-axis

For questions 14-16, convert each function to vertex form. State the vertex.

14. $f(x) = x^2 + 6x + 9$

Function:

Vertex:

15. $f(x) = x^2 - 10x + 18$

Function:

Vertex:

16. $f(x) = -3x^2 + 12x - 7$

Function:

Vertex:

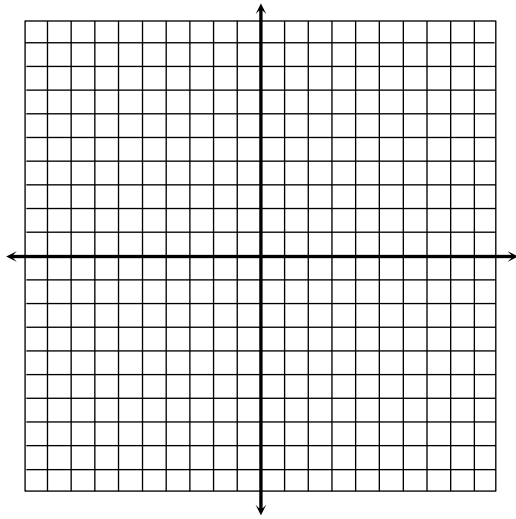
17. The function $f(x) = -x^2 + 8x - 7$ is translated five units down. Write the new equation in vertex form and state the vertex.

Function:

Vertex:

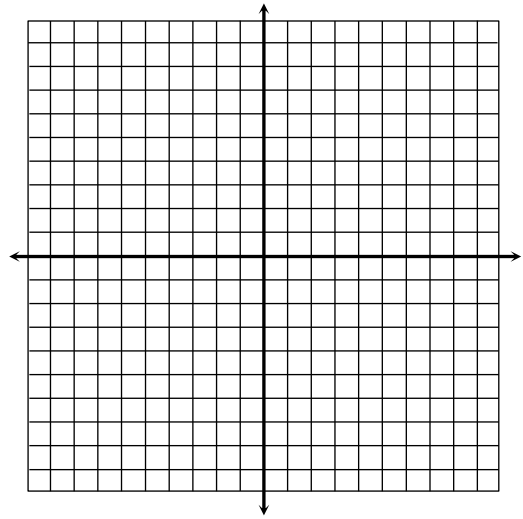
Graph each function. State the domain and range.

18. $f(x) = \begin{cases} x^2 + 3 & \text{if } x < 1 \\ 2x - 5 & \text{if } x > 1 \end{cases}$



D: _____ R: _____

19. $f(x) = \begin{cases} -x - 7 & \text{if } x < -5 \\ -3 & \text{if } -5 \leq x < -2 \\ -|x - 1| + 4 & \text{if } x \geq -2 \end{cases}$



D: _____ R: _____

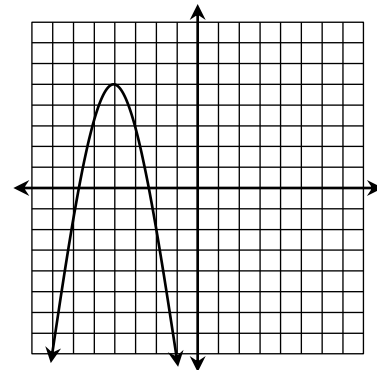
20. Identify the function family and give the equation of the parent function for the set of functions below.

$\left\{ f(x) = -x + 1, g(x) = \frac{1}{4}x + 3, h(x) = 2x - 9 \right\}$

Function Family:

Parent Function:

21. The graph of a function is shown below. Identify its function family and the equation of the parent function.



Function Family:

Parent Function:

22. Which parent functions have a domain of all real numbers? Check all that apply.

☐ Linear

☐ Absolute Value

☐ Quadratic

23. Which function has a range of all real numbers greater than or equal to -4?

A. $f(x) = x - 4$

B. $f(x) = -4x$

C. $f(x) = (x + 1)^2 - 4$

D. $f(x) = -|x - 3| - 4$

24. Given the function below, which of the following describes the end behavior as x approaches positive infinity?

$$f(x) = -\frac{4}{3}x + 1$$

- A. $f(x)$ approaches positive infinity
 B. $f(x)$ approaches negative infinity
 C. $f(x)$ approaches 1
 D. $f(x)$ approaches -1

25. Given the function below, which of the following describes the end behavior as x approaches negative infinity?

$$f(x) = (x - 6)^2 + 2$$

- A. $f(x)$ approaches positive infinity
 B. $f(x)$ approaches negative infinity
 C. $f(x)$ approaches 6
 D. $f(x)$ approaches -6

26. Which of the following describes the behavior of the function below as x approaches positive or negative infinity?

$$f(x) = -|x - h| + k$$

- A. $f(x)$ approaches positive infinity
 B. $f(x)$ approaches negative infinity
 C. $f(x)$ approaches h
 D. $f(x)$ approaches k

27. Over what interval is the function below decreasing?

$$f(x) = 2(x + 3)^2 - 5$$

- A. $-\infty < x < -3$
 B. $-\infty < x < 3$
 C. $-3 < x < \infty$
 D. $3 < x < \infty$

28. Which of the following describes the function $f(x) = -|x - 5| + 2$?

- A. Increasing over the interval $-\infty < x < -2$; Decreasing over the interval $-2 < x < \infty$
 B. Increasing over the interval $-\infty < x < 2$; Decreasing over the interval $2 < x < \infty$
 C. Increasing over the interval $-\infty < x < -5$; Decreasing over the interval $-5 < x < \infty$
 D. Increasing over the interval $-\infty < x < 5$; Decreasing over the interval $5 < x < \infty$

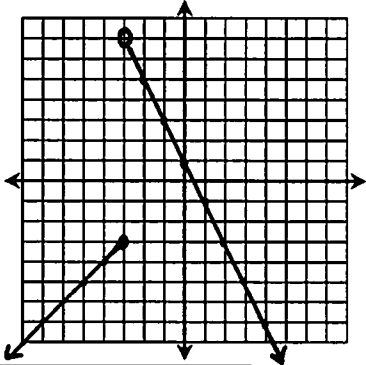
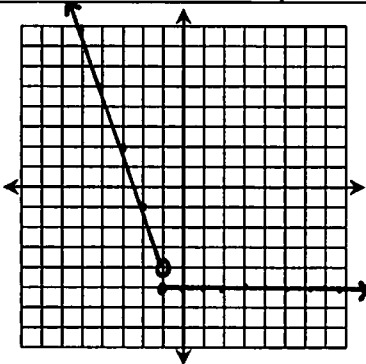
BONUS: Write the function below in vertex form and state the vertex.

$$f(x) = -\frac{1}{2}x^2 + \frac{3}{2}x - 2$$

Equation:

Vertex:

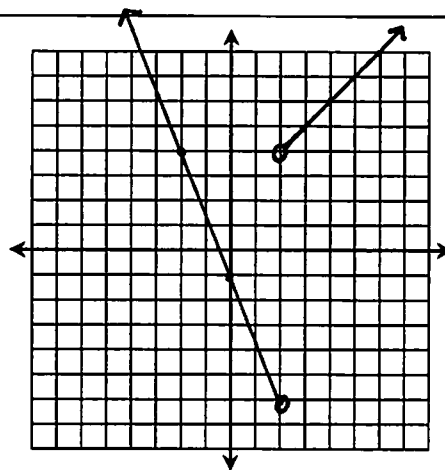
Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples
PIECEWISE FUNCTIONS	A piecewise function is made up of 2 or more functions, each with its own domain.
EVALUATING <i>Piecewise Functions</i>	1. Given $f(x) = \begin{cases} x^2 - 1 & \text{if } x < -2 \\ 5x + 3 & \text{if } x \geq -2 \end{cases}$, find each value. a) $f(-5)$ $(-5)^2 - 1$ $25 - 1$ $= \boxed{24}$ b) $f(-2)$ $5(-2) + 3$ $-10 + 3$ $= \boxed{-7}$ c) $f(7)$ $5(7) + 3$ $35 + 3$ $= \boxed{38}$ 2. Given $g(x) = \begin{cases} \frac{1}{2}x + 3 & \text{if } x \leq -4 \\ -x - 1 & \text{if } -4 < x < 1 \\ 2x^3 + 9 & \text{if } x \geq 1 \end{cases}$, find each value. a) $g(2)$ $2(2)^3 + 9$ $2(8) + 9$ $16 + 9 = \boxed{25}$ b) $g(-1)$ $-(-1) - 1$ $1 - 1$ $= \boxed{0}$ c) $g(-6)$ $\frac{1}{2}(-6) + 3$ $-3 + 3$ $= \boxed{0}$
GRAPHING <i>Piecewise Functions</i>	Graph each piecewise function, then identify the domain and range. 3. $f(x) = \begin{cases} x & \text{if } x \leq -3 \\ -2x + 1 & \text{if } x > -3 \end{cases}$ $D = \underline{\{ \mathbb{R} \}}$ $R = \underline{\{ y \mid y < 7 \}}$  4. $g(x) = \begin{cases} -3x - 7 & \text{if } x < -1 \\ -5 & \text{if } x \geq -1 \end{cases}$ $D = \underline{\{ \mathbb{R} \}}$ $R = \underline{\{ y \mid y = -5 \cup y > -4 \}}$ 

$$5. g(x) = \begin{cases} -\frac{5}{2}x - 1 & \text{if } x < 2 \\ x + 2 & \text{if } x > 2 \end{cases}$$

$$D = \{x \mid x \neq 2\}$$

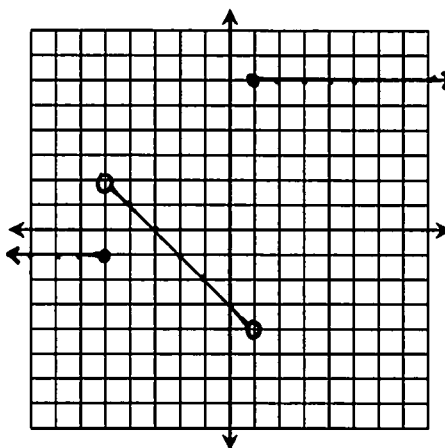
$$R = \{y \mid y > -6\}$$



$$6. g(x) = \begin{cases} -1 & \text{if } x \leq -5 \\ -x - 3 & \text{if } -5 < x < 1 \\ 6 & \text{if } x \geq 1 \end{cases}$$

$$D = \{R\}$$

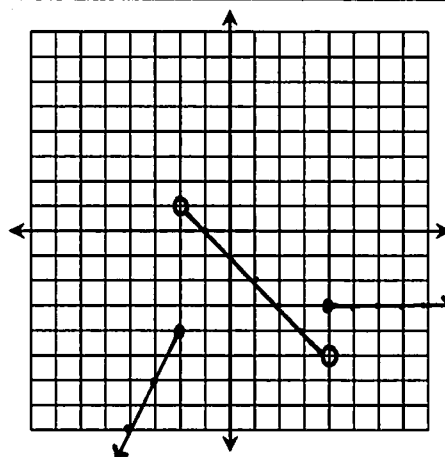
$$R = \{y \mid -4 < y < 2 \cup y = 6\}$$



$$7. f(x) = \begin{cases} 2x & \text{if } x \leq -2 \\ -1 - x & \text{if } -2 < x < 4 \\ -3 & \text{if } x \geq 4 \end{cases}$$

$$D = \{R\}$$

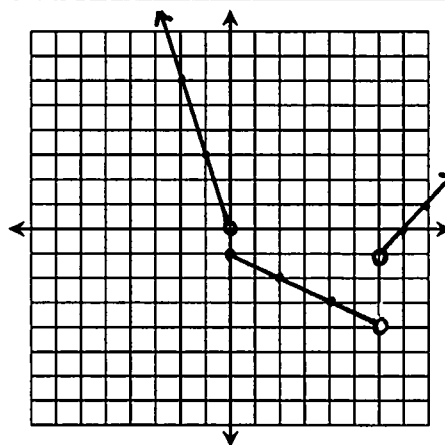
$$R = \{y \mid y < 1\}$$



$$8. h(x) = \begin{cases} -3x & \text{if } x < 0 \\ -\frac{1}{2}x - 1 & \text{if } 0 \leq x < 6 \\ x - 7 & \text{if } x > 6 \end{cases}$$

$$D = \{x \mid x \neq 6\}$$

$$R = \{y \mid y > -4\}$$



Name: _____

Unit 3: Parent Functions & Transformations

Date: _____ Per: _____

Homework 1: Piecewise Functions

** This is a 2-page document! **

Given $a(x) = \begin{cases} |x-8| & \text{if } x \leq -6 \\ 2x - x^2 & \text{if } -6 < x \leq 1 \\ -4x + 7 & \text{if } x > 1 \end{cases}$, find each function value.

1. $a(8)$

$$\begin{aligned} & -4(8) + 7 \\ & -32 + 7 \\ & = \boxed{-25} \end{aligned}$$

2. $a(1)$

$$\begin{aligned} & 2(1) - (1)^2 \\ & 2 - 1 \\ & = \boxed{1} \end{aligned}$$

3. $a(-7)$

$$\begin{aligned} & |-7 - 8| \\ & |-15| = \boxed{15} \end{aligned}$$

4. $a(-3)$

$$\begin{aligned} & 2(-3) - (-3)^2 \\ & -6 - 9 \\ & = \boxed{-15} \end{aligned}$$

5. $a\left(-\frac{1}{2}\right)$

$$\begin{aligned} & 2\left(-\frac{1}{2}\right) - \left(-\frac{1}{2}\right)^2 \\ & -1 - \frac{1}{4} = \boxed{-\frac{5}{4}} \end{aligned}$$

6. $a\left(\frac{9}{4}\right)$

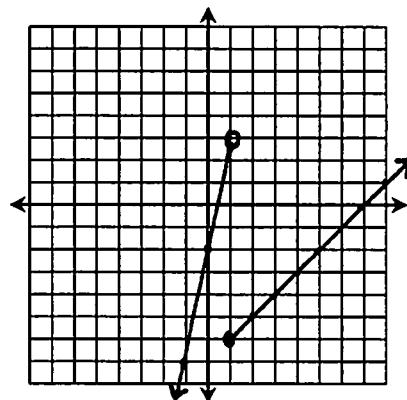
$$\begin{aligned} & -4\left(\frac{9}{4}\right) + 7 \\ & -9 + 7 \\ & = \boxed{-2} \end{aligned}$$

Graph each function. Identify the domain and range.

$$7. f(x) = \begin{cases} 5x - 2 & \text{if } x < 1 \\ x - 7 & \text{if } x \geq 1 \end{cases}$$

$$D = \underline{\{ \mathbb{R} \}}$$

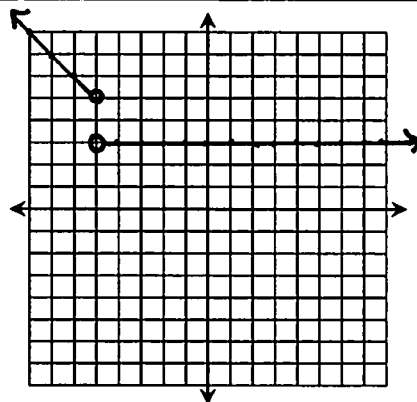
$$R = \underline{\{ \mathbb{R} \}}$$



$$8. g(x) = \begin{cases} -x & \text{if } x < -5 \\ 3 & \text{if } x \geq -5 \end{cases}$$

$$D = \underline{\{x \mid x \neq -5\}}$$

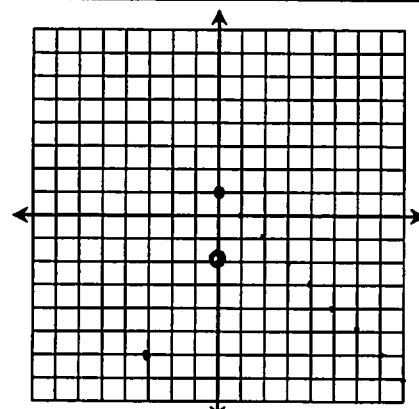
$$R = \underline{\{y \mid y = 3 \vee y > 5\}}$$



$$9. h(x) = \begin{cases} \frac{4}{3}x - 2 & \text{if } x < 0 \\ -x + 1 & \text{if } x \geq 0 \end{cases}$$

$$D = \mathbb{R}$$

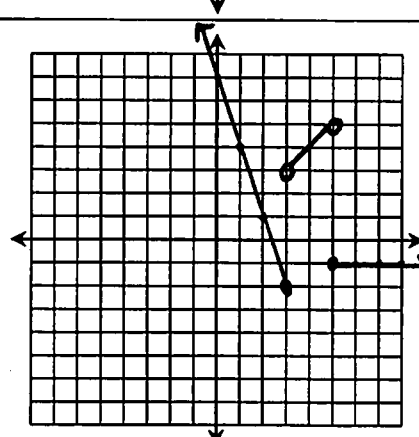
$$R = \{y \mid y \leq 1\}$$



$$10. p(x) = \begin{cases} -3x + 7 & \text{if } x \leq 3 \\ x & \text{if } 3 < x < 5 \\ -1 & \text{if } x \geq 5 \end{cases}$$

$$D = \mathbb{R}$$

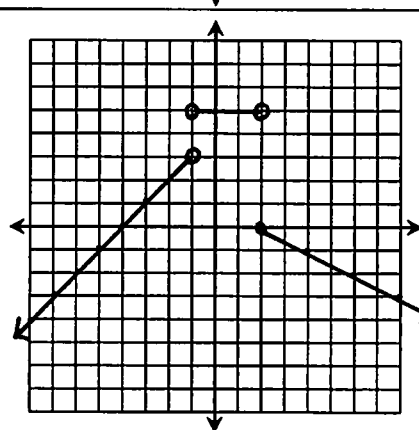
$$R = \{y \mid y \geq -2\}$$



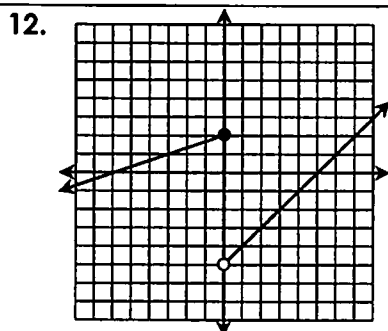
$$11. k(x) = \begin{cases} x + 4 & \text{if } x < -1 \\ 5 & \text{if } -1 < x < 2 \\ -\frac{1}{2}x + 1 & \text{if } x \geq 2 \end{cases}$$

$$D = \{x \mid x \neq -1\}$$

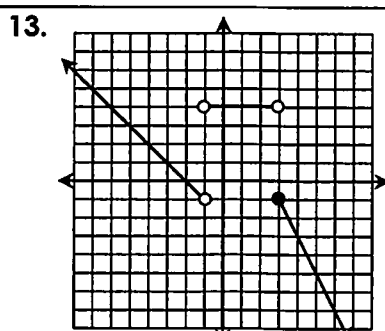
$$R = \{y \mid y < 3 \cup y = 5\}$$



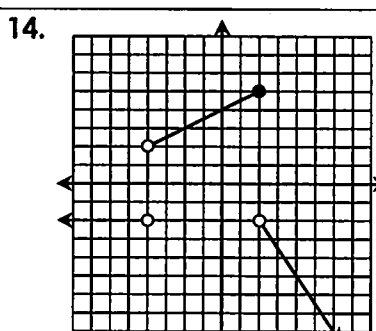
Write a piecewise function for each function graphed below.



$$f(x) = \begin{cases} \frac{1}{3}x + 2 & \text{if } x \leq 0 \\ x - 5 & \text{if } x > 0 \end{cases}$$



$$f(x) = \begin{cases} -x - 2 & \text{if } x < -1 \\ 4 & \text{if } -1 < x < 3 \\ -2x + 5 & \text{if } x \geq 3 \end{cases}$$



$$f(x) = \begin{cases} -2 & \text{if } x < -4 \\ \frac{1}{2}x + 4 & \text{if } -4 < x \leq 2 \\ -\frac{3}{2}x + 1 & \text{if } x > 2 \end{cases}$$

Name:

Date:

Topic:

Class:

Main Ideas/Questions

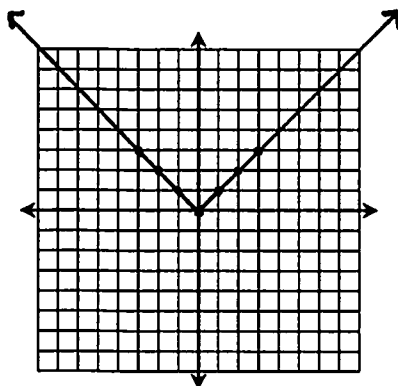
Notes/Examples

ABSOLUTE VALUE

Functions

- The **absolute value function** is written as $f(x) = |x|$
- Complete the table below to graph the function. Give the domain and range.

x	$ x $
-3	3
-2	2
-1	1
0	0
1	1
2	2
3	3



$$D = \{\mathbb{R}\}$$

$$R = \{y | y \geq 0\}$$

- The **graph** looks like a ∇ ; The **turning point** is called the **vertex**.

GRAPHING

by Table

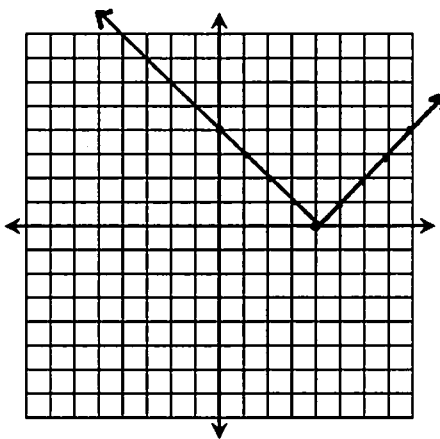
To create a table of values, it's best to place the vertex in the middle, then include values on both sides.

- Find the **x-value of the vertex**. Set the expression from the inside of the absolute value bars equal to 0 and solve.
- PLACE THIS VALUE IN THE MIDDLE ROW OF YOUR TABLE.** Number up and down, then complete the table.
- GRAPH!**

Directions: Graph each function. Give the domain and range.

1. $f(x) = |x - 4|$

x	$f(x)$
1	3
2	2
3	1
4	0
5	1
6	2
7	3

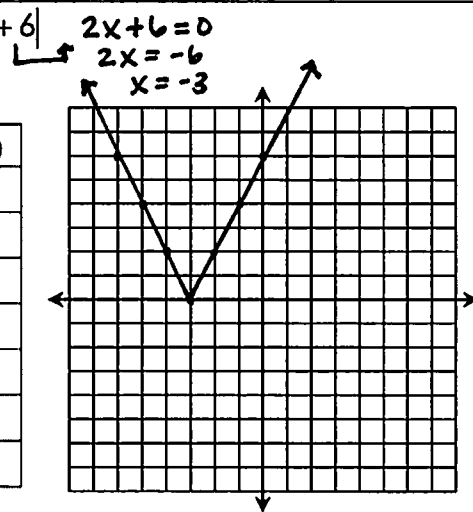


$$D = \{\mathbb{R}\}$$

$$R = \{y | y \geq 0\}$$

2. $f(x) = |2x + 6|$

x	$f(x)$
-6	6
-5	4
-4	2
-3	0
-2	2
-1	4
0	6

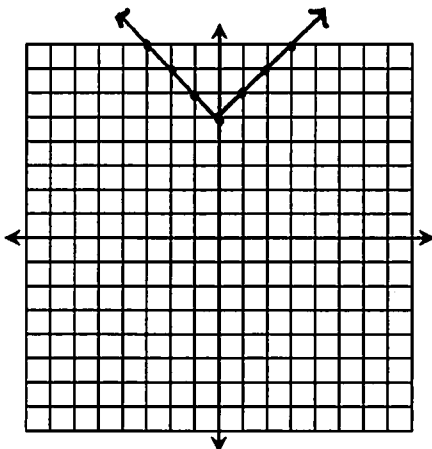


$$D = \{\mathbb{R}\}$$

$$R = \{y | y \geq 0\}$$

3. $f(x) = |x| + 5$

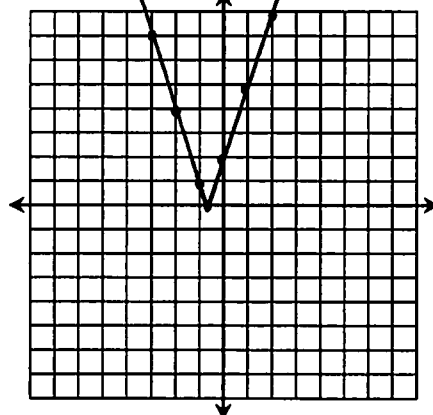
x	$f(x)$
-3	8
-2	7
-1	6
0	5
1	6
2	7
3	8



$D = \{\mathbb{R}\}$ $R = \{y | y \geq 5\}$

4. $f(x) = |3x + 2|$

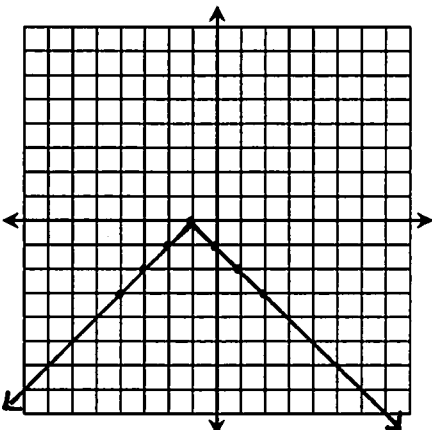
x	$f(x)$
-3	7
-2	4
-1	1
$-\frac{2}{3}$	0
0	2
1	5
2	8



$D = \{\mathbb{R}\}$ $R = \{y | y \geq 0\}$

5. $f(x) = -|x + 1|$
 $x + 1 = 0$
 $x = -1$

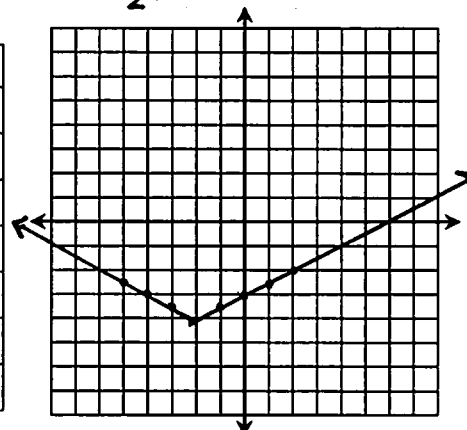
x	$f(x)$
-4	-3
-3	-2
-2	-1
-1	0
0	-1
1	-2
2	-3



$D = \{\mathbb{R}\}$ $R = \{y | y \leq 0\}$

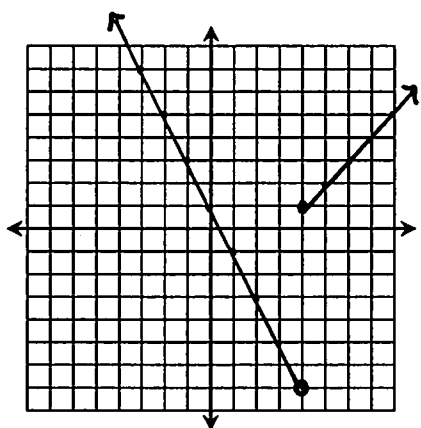
6. $f(x) = \left|\frac{1}{2}x + 1\right| - 4$
 $\frac{1}{2}x + 1 = 0$
 $\frac{1}{2}x = -1$
 $x = -2$

x	$f(x)$
-5	-2.5
-4	-3
-3	-3.5
-2	-4
-1	-3.5
0	-3
1	-2.5



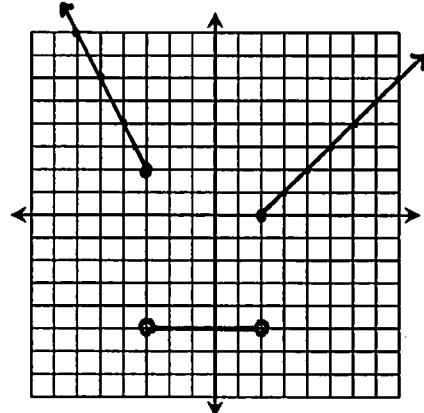
$D = \{\mathbb{R}\}$ $R = \{y | y \geq -4\}$

7. $f(x) = \begin{cases} -2x + 1 & \text{if } x < 4 \\ |x - 3| & \text{if } x \geq 4 \end{cases}$



$D = \{\mathbb{R}\}$ $R = \{y | y > -7\}$

8. $f(x) = \begin{cases} |2x| - 4 & \text{if } x \leq -3 \\ -5 & \text{if } -3 < x < 2 \\ x - 2 & \text{if } x \geq 2 \end{cases}$



$D = \{\mathbb{R}\}$ $R = \{y | y = -5 \cup y \geq 0\}$

Name:

Date:

Topic:

Class:

Main Ideas/Questions

Notes/Examples

GRAPHING ABSOLUTE VALUE inequalities

- ① Create a **TABLE OF VALUES**. Recall that the vertex should be in the middle row.
- ② **GRAPH!** Use a dashed line for $<$ and $>$ symbols and a solid line for $\leq$ and $\geq$ symbols.
- ③ **SHADE!** Assuming y is to the left, shade above the line for $>$ or $\geq$ symbols and below the line for $<$ or $\leq$ symbols.

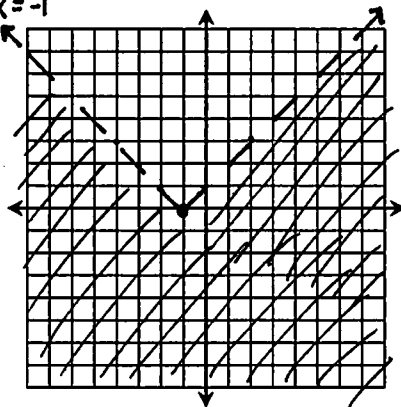
Directions: Graph each inequality.

1. $y < |x + 1|$

$$x + 1 = 0$$

$$x = -1$$

x	y
-4	3
-3	2
-2	1
-1	0
0	1
1	2
2	3

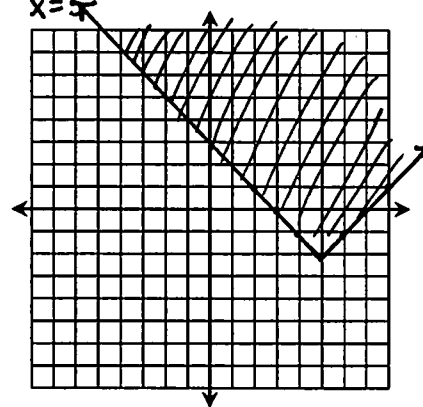


2. $y \geq |x - 5| - 2$

$$x - 5 = 0$$

$$x = 5$$

x	y
2	1
3	0
4	-1
5	-2
6	-1
7	0
8	1



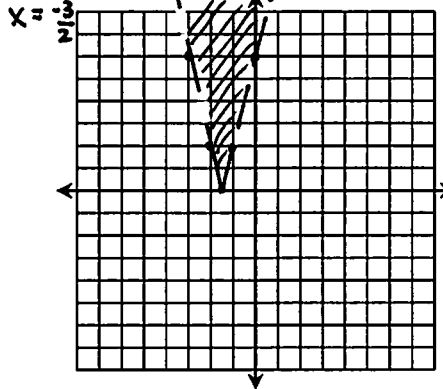
3. $y > |4x + 6|$

$$4x + 6 = 0$$

$$4x = -6$$

$$x = -\frac{3}{2}$$

x	y
-4	10
-3	6
-2	2
-3/2	0
-1	2
0	6
1	10

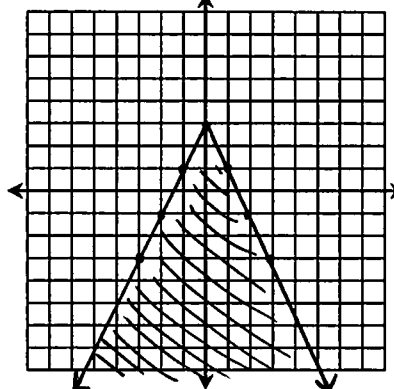


4. $y \leq -|2x| + 3$

$$2x = 0$$

$$x = 0$$

x	y
-3	-3
-2	-1
-1	1
0	3
1	1
2	-1
3	-3



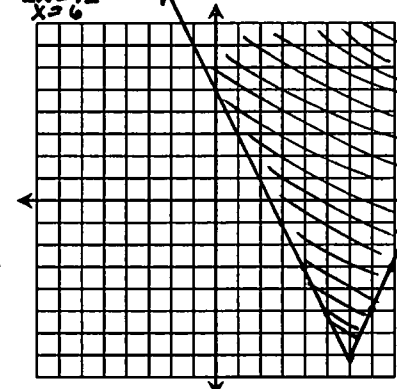
5. $y \geq |2x - 12| - 7$

$$2x - 12 = 0$$

$$2x = 12$$

$$x = 6$$

x	y
3	-1
4	-3
5	-5
6	-7
7	-5
8	-3
9	-1



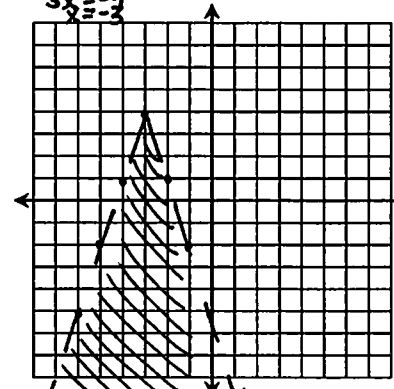
6. $y < -|3x + 9| + 4$

$$3x + 9 = 0$$

$$3x = -9$$

$$x = -3$$

x	y
-6	-5
-5	-2
-4	1
-3	4
-2	1
-1	-2
0	-5



Name: _____

Unit 3: Parent Functions & Transformations

Date: _____ Per: _____

Homework 2: Graphing Absolute Value Functions & Inequalities

**** This is a 2-page document! ****

Graph each function. Identify the domain and range.

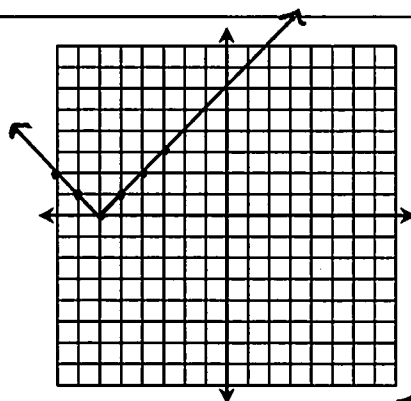
1. $f(x) = |x+6|$

$$\begin{aligned} \hookrightarrow x+6 &= 0 \\ x &= -6 \end{aligned}$$

$D = \{ \mathbb{R} \}$

$R = \{ y \mid y \geq 0 \}$

x	y
-9	3
-8	2
-7	1
-6	0
-5	1
-4	2
-3	3



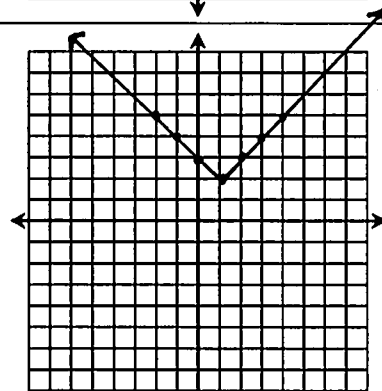
2. $f(x) = |x-1| + 2$

$$\begin{aligned} \hookrightarrow x-1 &= 0 \\ x &= 1 \end{aligned}$$

$D = \{ \mathbb{R} \}$

$R = \{ y \mid y \geq 2 \}$

x	y
-2	5
-1	4
0	3
1	2
2	3
3	4
4	5



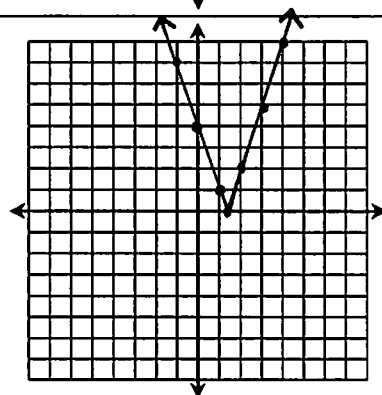
3. $f(x) = |3x-4|$

$$\begin{aligned} \hookrightarrow 3x-4 &= 0 \\ 3x &= 4 \\ x &= \frac{4}{3} \end{aligned}$$

$D = \{ \mathbb{R} \}$

$R = \{ y \mid y \geq 0 \}$

x	y
-1	7
0	4
1	1
$\frac{4}{3}$	0
2	2
3	5
4	8

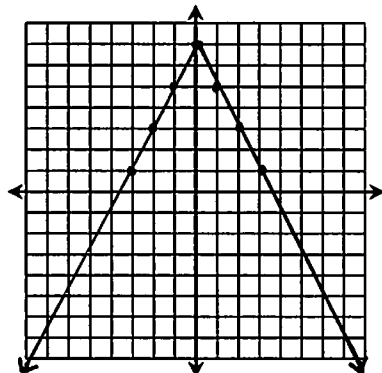


4. $f(x) = -|2x| + 7$

$D = \{ \mathbb{R} \}$

$R = \{ y \mid y \leq 7 \}$

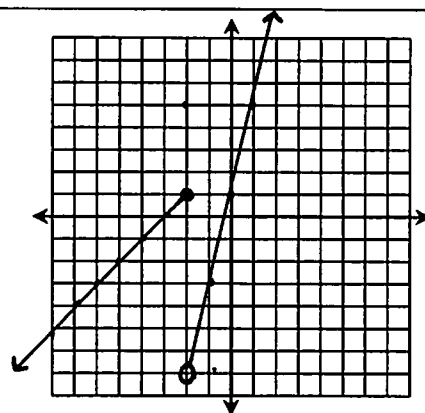
x	y
-3	1
-2	3
-1	5
0	7
1	5
2	3
3	1



$$5. f(x) = \begin{cases} -|x| + 3 & \text{if } x \leq -2 \\ 4x + 1 & \text{if } x > -2 \end{cases}$$

$$D = \underline{\{\mathbb{R}\}}$$

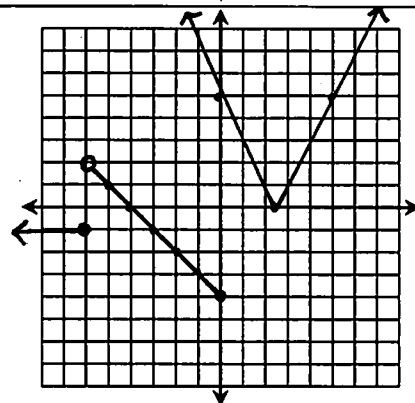
$$R = \underline{\{\mathbb{R}\}}$$



$$6. f(x) = \begin{cases} -1 & \text{if } x \leq -6 \\ -x - 4 & \text{if } -6 < x \leq 0 \\ |2x - 5| & \text{if } x > 0 \end{cases}$$

$$D = \underline{\{\mathbb{R}\}}$$

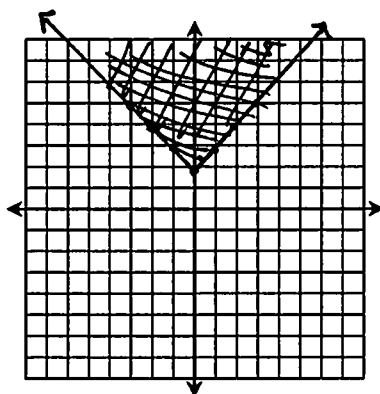
$$R = \underline{\{y \mid y \geq -4\}}$$



Graph each inequality.

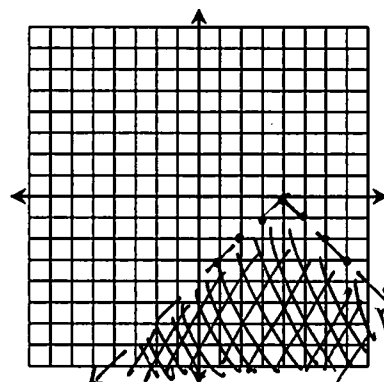
$$7. y \geq |x| + 2$$

x	y
-3	5
-2	4
-1	3
0	2
1	3
2	4
3	5



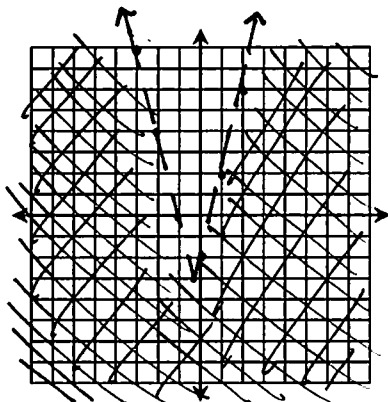
$$8. y < -|x - 4|$$

x	y
1	-3
2	-2
3	-1
4	0
5	-1
6	-2
7	-3



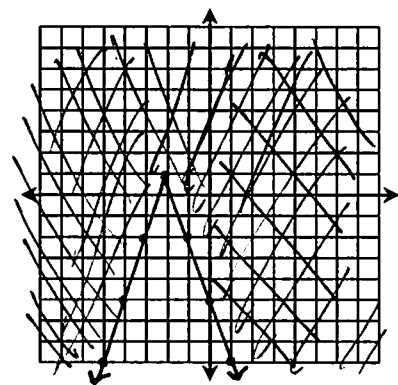
$$9. y < |4x + 1| - 3$$

x	y
-3	8
-2	4
-1	0
-1/4	-3
0	-2
1	2
2	6



$$10. y \geq -|3x + 6| + 1$$

x	y
-5	-8
-4	-5
-3	-2
-2	1
-1	-2
0	-5
1	-8



Name: _____

Algebra 2

Date: _____ Per: _____

Unit 3: Parent Functions & Transformations

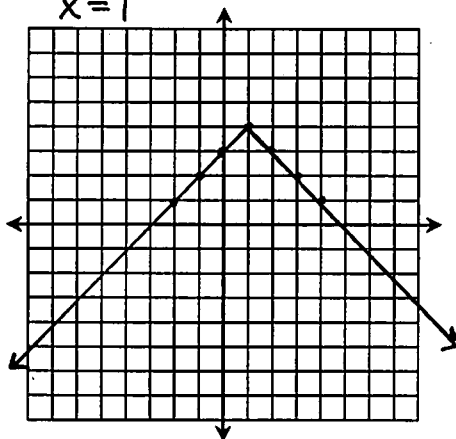
Quiz 3-1: Absolute Value Functions & Piecewise Functions

Use the table to graph each absolute value function. Identify the domain and range.

1. $f(x) = -|x-1| + 4$

$$\begin{aligned} \hookrightarrow x-1 &= 0 \\ x &= 1 \end{aligned}$$

x	$f(x)$
-2	1
-1	2
0	3
1	4
2	3
3	2
4	1



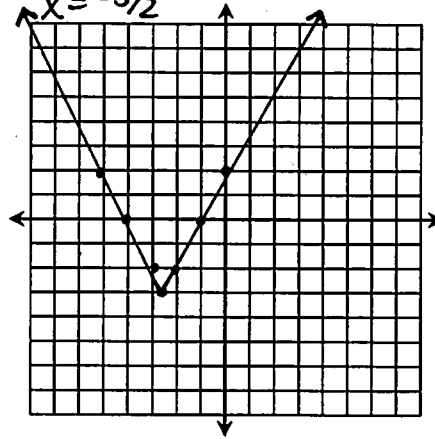
D: $\{ \mathbb{R} \}$

R: $\{ y \mid y \leq 4 \}$

2. $f(x) = |2x+5| - 3$

$$\begin{aligned} \hookrightarrow 2x+5 &= 0 \\ x &= -5/2 \end{aligned}$$

x	$f(x)$
-5	2
-4	0
-3	-2
-5/2	-3
-2	-2
-1	0
0	2



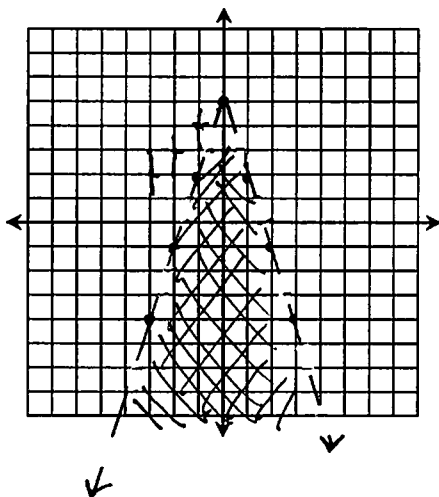
D: $\{ \mathbb{R} \}$

R: $\{ y \mid y \geq -3 \}$

Use the table to graph each absolute value inequality.

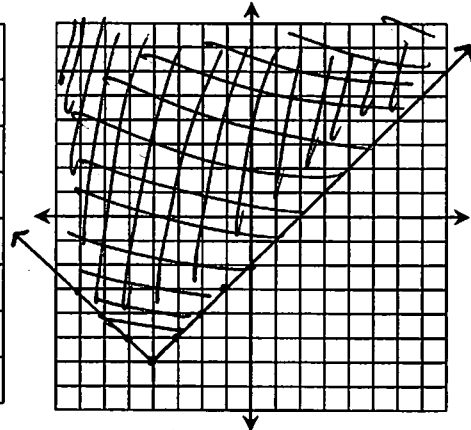
3. $y < -|3x| + 5$

x	y
-3	-4
-2	-1
-1	2
0	5
1	2
2	-1
3	-4



4. $y \geq |x+4| - 6$

x	y
-7	-3
-6	-4
-5	-5
-4	-6
-3	-5
-2	-4
-1	-3



Find each value given the piecewise function below.

$$f(x) = \begin{cases} |2x-1| & \text{if } x < -3 \\ 4-x & \text{if } -3 \leq x < 5 \\ -x^2+7 & \text{if } x \geq 5 \end{cases}$$

$$5. f(-8) = |2(-8) - 1| = |-17| \quad 5. \underline{17}$$

$$6. f(11) = -11^2 + 7 = -121 + 7 \quad 6. \underline{-114}$$

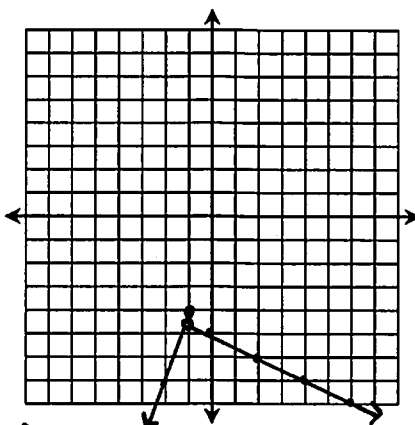
$$7. f(-3) = 4 - (-3) \quad 7. \underline{7}$$

$$8. f\left(-\frac{9}{2}\right) = \left|2\left(-\frac{9}{2}\right) - 1\right| = |-9-1| \quad 8. \underline{10}$$

$$9. f\left(\frac{5}{4}\right) = 4 - \frac{5}{4} = \frac{16}{4} - \frac{5}{4} \quad 9. \underline{\frac{11}{4}} \quad (2.75)$$

Graph each piecewise function. State the domain and range.

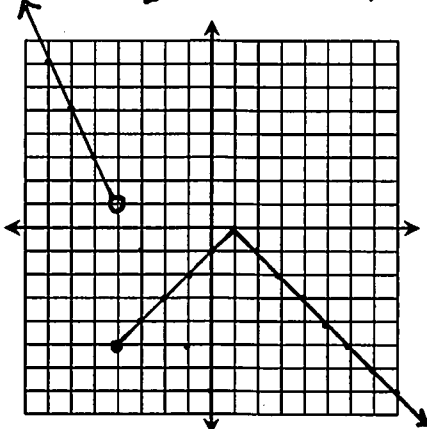
$$10. f(x) = \begin{cases} 3x-1 & \text{if } x \leq -1 \\ -\frac{1}{2}x-5 & \text{if } x > -1 \end{cases}$$



$$10. D: \underline{\{ \mathbb{R} \}}$$

$$R: \underline{\{ y \mid y \leq -4 \}}$$

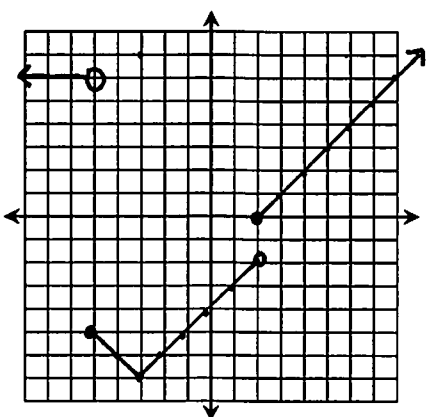
$$11. f(x) = \begin{cases} -2x-7 & \text{if } x < -4 \\ -|x-1| & \text{if } x \geq -4 \end{cases}$$



$$11. D: \underline{\{ \mathbb{R} \}}$$

$$R: \underline{\{ y \mid y > 1 \cup y \leq 0 \}}$$

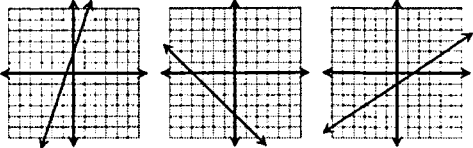
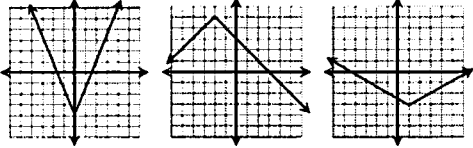
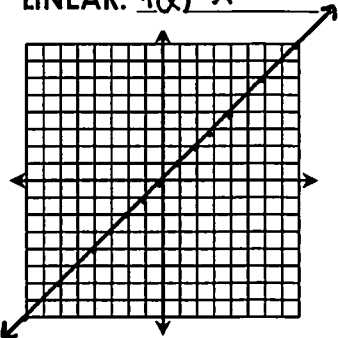
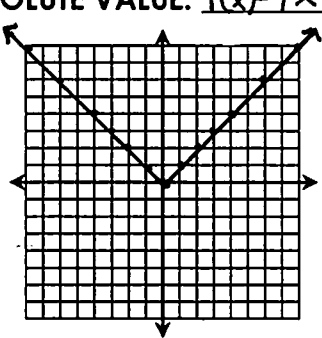
$$12. f(x) = \begin{cases} 6 & \text{if } x < -5 \\ |x+3|-7 & \text{if } -5 \leq x < 2 \\ x-2 & \text{if } x \geq 2 \end{cases}$$



$$12. D: \underline{\{ \mathbb{R} \}}$$

$$R: \underline{\{ y \mid -7 \leq y < -2 \cup y \geq 0 \}}$$

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples
FUNCTION FAMILY	A family of functions (or function family) is a set of functions whose equations and graphs have a similar form and shape. Name the function family for each set of graphs. 1.  Linear Functions 2.  Absolute Value Functions
PARENT FUNCTION	A parent function is the simplest function of the family. Give the parent function for each function family. Then graph the function. LINEAR: $f(x) = x$  ABSOLUTE VALUE: $f(x) = x$ 
TRANSFORMATIONS	A transformation changes the position or size of a figure. Transformations can be applied to parent functions. Let's use the absolute value function as an example. Using your calculator, graph the following functions, then compare to the parent function. Give a brief description. 3. $f(x) = x+2$ Shifted left 2 4. $f(x) = x-2$ Shifted right 2 5. $f(x) = x +2$ Shifted up 2 6. $f(x) = x -2$ Shifted down 2 7. $f(x) = 2 x$ Vertical stretch by 2 8. $f(x) = \frac{1}{2} x$ Vertical compression by $\frac{1}{2}$ 9. $f(x) = - x$ Reflected across the x-axis
Types of TRANSFORMATIONS	• <u>Translation</u> slides (or shifts) a function. • <u>Dilation</u> stretches or compresses a function. • <u>Reflection</u> flips a function.

TRANSFORMATION <i>Rules</i>	We can tell from the function which transformation(s) from the parent function have taken place using the following rules:	
	TRANSLATIONS	$f(x+h)$ Shift left "h" units
		$f(x-h)$ Shift right "h" units
		$f(x)+k$ Shift up "k" units
		$f(x)-k$ Shift down "k" units
	DILATIONS	$a \cdot f(x)$ (when $ a > 1$) Vertical stretch
		$a \cdot f(x)$ (when $ a < 1$) Vertical compression
IDENTIFYING <i>Transformations</i>	REFLECTIONS	$-f(x)$ reflect across x-axis
	Describe how the following functions compare to the parent function.	
	10. $f(x) = x - 3$ Shift down 3 units	11. $f(x) = x+7 $ Shift left 7 units
	12. $f(x) = - x-4 $ reflect across x-axis, Shift right 4	13. $f(x) = 4 x +1$ Vertical stretch by 4, Shift up 1
	14. $f(x) = \frac{1}{3} x $ Vertical compression by $\frac{1}{3}$	15. $f(x) = x+2 - 5$ Shift left 2 and down 5
	16. $f(x) = -2 x-1 $ Reflect across the x-axis, Vertical stretch by 2, shift right 1	17. $f(x) = \frac{4}{3} x +2$ Vertical stretch by $\frac{4}{3}$, Shift up 2
WRITING <i>Functions</i>	Transformations to the absolute value parent function are described. Write the new function.	
	18. Translated 6 units right.	$f(x) = x-6 $
	19. Reflected in the x-axis, then translated 2 units up.	$f(x) = - x +2$
	20. Vertically stretched by a factor of 5, then translated 1 unit left.	$f(x) = 5 x+1 $
	21. Translated 3 units right and 4 units down.	$f(x) = x-3 -4$
	22. Vertically compressed by a factor of $\frac{2}{3}$, then reflected in the x-axis.	$f(x) = -\frac{2}{3} x $
	23. Vertically stretched by a factor of 2, then translated 9 units down.	$f(x) = 2 x -9$
	24. Vertically compressed by a factor of $\frac{1}{4}$, then shifted 1 unit left.	$f(x) = \frac{1}{4} x+1 $

Name: _____

Unit 3: Parent Functions & Transformations

Date: _____ Per: _____

Homework 3: Identifying Transformations & Writing Functions



Give the parent function of each function family.	
1. Linear Functions $f(x) = x$	2. Absolute Value Functions $f(x) = x $
Describe the transformation of each function compared to its parent function.	
3. $f(x) = x - 7$ Shifted down 7	4. $f(x) = \frac{1}{5} x $ Vertical Compression by $\frac{1}{5}$
5. $f(x) = - x $ Reflected across x-axis	6. $f(x) = x + 6 $ Shifted left 6
7. $f(x) = 3 x $ Vertical stretch by 3	8. $f(x) = x - 4 + 6$ Shifted right 4 and up 6
9. $f(x) = - x + 9 $ Reflected across x-axis and shifted left 9	10. $f(x) = 4 x - 1 $ Vertical stretch by 4, Shifted right 1
11. $f(x) = 2 x + 5 - 4$ Vertical stretch by 2, Shifted left 5 and down 4	12. $f(x) = -\frac{5}{3} x + 2$ Vertical stretch by $\frac{5}{3}$, Reflected across x-axis, shift up 2
Transformations to the absolute value parent function are described. Write the new function.	
13. Translated 9 units right. $f(x) = x - 9 $	14. Reflected in the x-axis. $f(x) = - x $
15. Vertically compressed by a factor of $\frac{3}{4}$. $f(x) = \frac{3}{4} x $	16. Shifted 4 units up. $f(x) = x + 4$
17. Reflected in the x-axis, then translated 2 units left. $f(x) = - x + 2 $	18. Vertically stretched by a factor of 5, then translated 3 units right. $f(x) = 5 x - 3 $
19. Translated 7 units down and 4 units left. $f(x) = x + 4 - 7$	20. Vertically compressed by a factor of $\frac{1}{2}$, then reflected in the x-axis. $f(x) = -\frac{1}{2} x $
21. Vertically stretched by a factor of 4, then shifted 2 units right and 1 unit down. $f(x) = 4 x - 2 - 1$	22. Reflected in the x-axis, then translated 8 units left and 5 units up. $f(x) = - x + 8 + 5$

Name:

Date:

Topic:

Class:

Main Ideas/Questions

Notes/Examples

WARM-UP

1. What is the parent function of an absolute equation?

$$f(x) = |x|$$

2. Describe how the following functions compare to the parent function:

a) $f(x) = |x - 5|$

Shift right 5

b) $f(x) = |x| + 9$

Shift up 9

c) $f(x) = -|x|$

Reflect across x-axis

d) $f(x) = 3|x| - 2$

Vertical stretch by 3,
Shifted down 2

e) $f(x) = \frac{1}{4}|x + 6|$

Vertical compression
by $\frac{1}{4}$, shifted left 6

f) $f(x) = -|x + 7| - 2$

reflect across x-axis,
shifted left 7, down 2

The horizontal and vertical shifts describe the location of the vertex!

VERTEX FORM

of an Absolute
Value Function

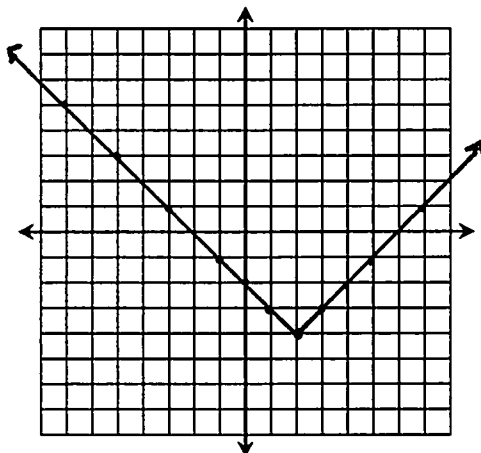
The vertex form of an absolute value function is written as

$$f(x) = a|x - h| + k$$

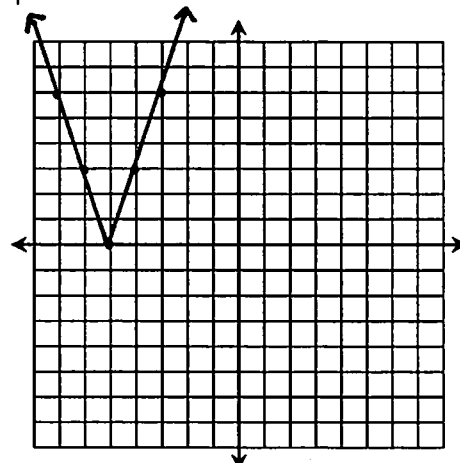
where (h, k) is the vertex, a is the slope on the right side and $-a$ is the slope on the left side

Directions: Give the vertex of each absolute value function. Then graph.

1. $f(x) = |x - 2| - 4$

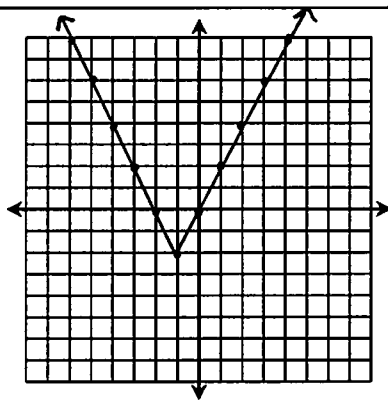
V: $(2, -4)$ 

2. $f(x) = 3|x + 5|$

V: $(-5, 0)$ 

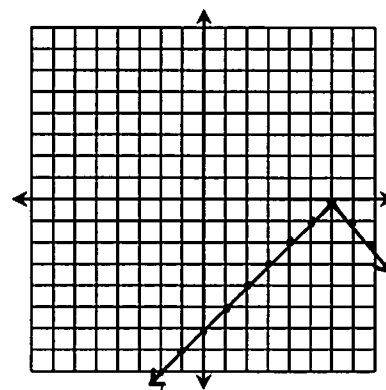
3. $f(x) = 2|x+1| - 2$

$V: (-1, -2)$



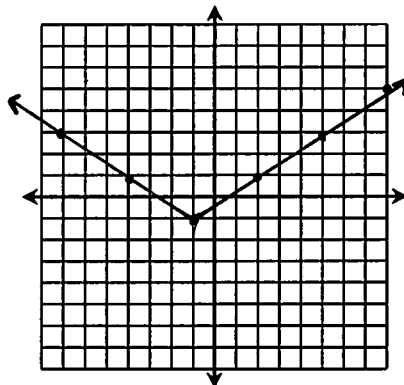
4. $f(x) = -|x-6|$

$V: (6, 0)$



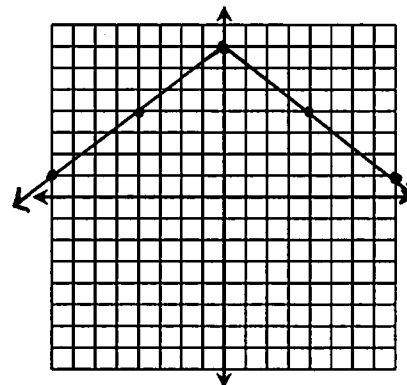
5. $f(x) = \frac{2}{3}|x+1| - 1$

$V: (-1, -1)$



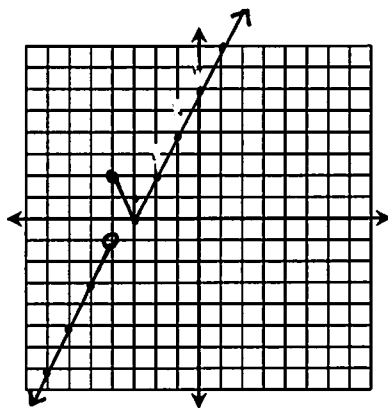
6. $f(x) = -\frac{3}{4}|x| + 7$

$V: (0, 7)$

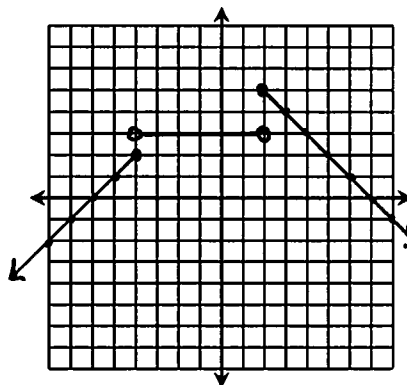


Directions: Graph each piecewise function.

7. $f(x) = \begin{cases} 2x+7 & \text{if } x < -4 \\ 2|x+3| & \text{if } x \geq -4 \end{cases}$



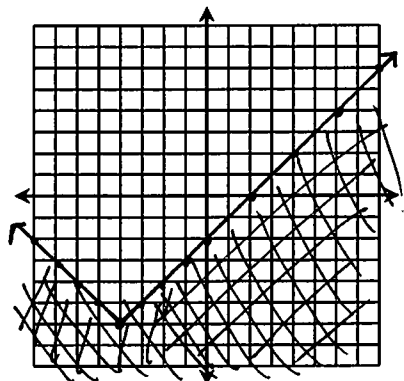
8. $f(x) = \begin{cases} -|x+1|+5 & \text{if } x \leq -4 \\ 3 & \text{if } -4 < x < 2 \\ -x+7 & \text{if } x \geq 2 \end{cases}$



Directions: Graph each absolute value inequality by identifying the vertex.

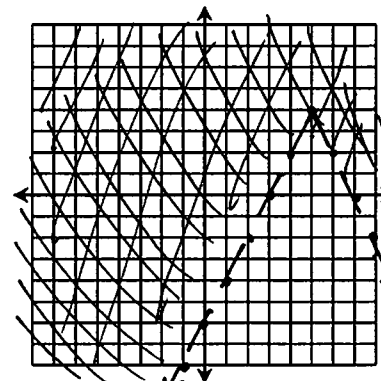
9. $y \leq |x+4| - 6$

$V: (-4, -6)$



10. $y > -2|x-5| + 4$

$V: (5, 4)$



Name: _____

Unit 3: Parent Functions & Transformations

Date: _____ Per: _____

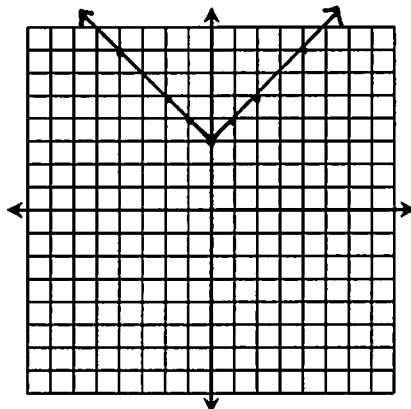
Homework 4: Vertex Form of Absolute Value Functions

**** This is a 2-page document! ****

Give the vertex of each function, then graph. State the domain and range.

1. $f(x) = |x| + 3$

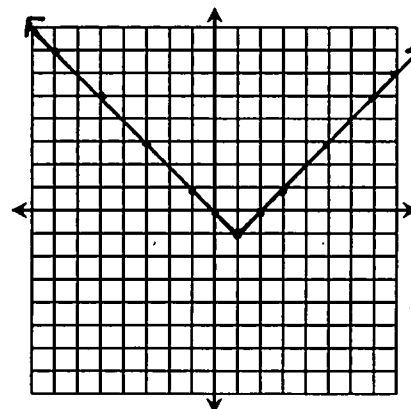
V: (0, 3)



$D = \{ \mathbb{R} \}$ $R = \{ y \mid y \geq 3 \}$

2. $f(x) = |x - 1| - 1$

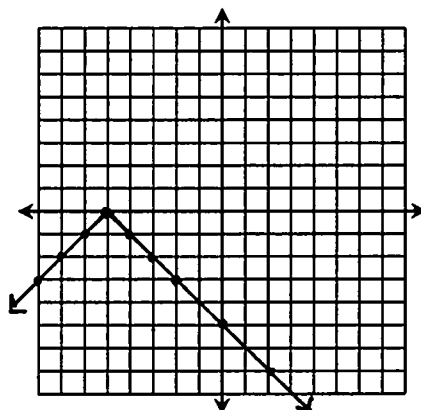
V: (1, -1)



$D = \{ \mathbb{R} \}$ $R = \{ y \mid y \geq -1 \}$

3. $f(x) = -|x + 5|$

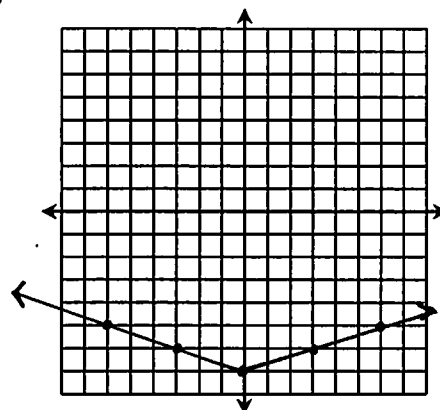
V: (-5, 0)



$D = \{ \mathbb{R} \}$ $R = \{ y \mid y \leq 0 \}$

4. $f(x) = \frac{1}{3}|x| - 7$

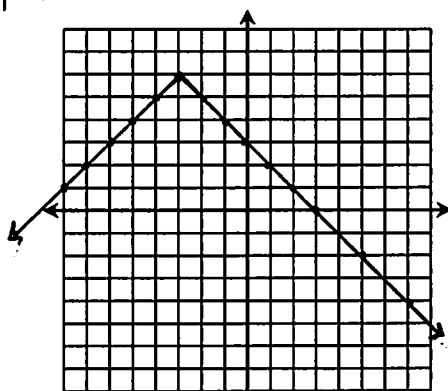
V: (0, -7)



$D = \{ \mathbb{R} \}$ $R = \{ y \mid y \geq -7 \}$

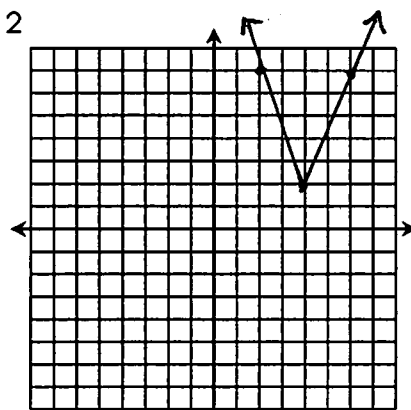
5. $f(x) = -|x + 3| + 6$

V: (-3, 6)



$D = \{ \mathbb{R} \}$ $R = \{ y \mid y \leq 6 \}$

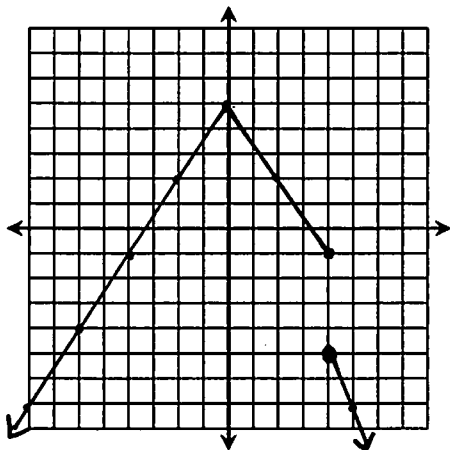
6. $f(x) = \frac{5}{2}|x - 4| + 2$



$D = \{ \mathbb{R} \}$ $R = \{ y \mid y \geq 2 \}$

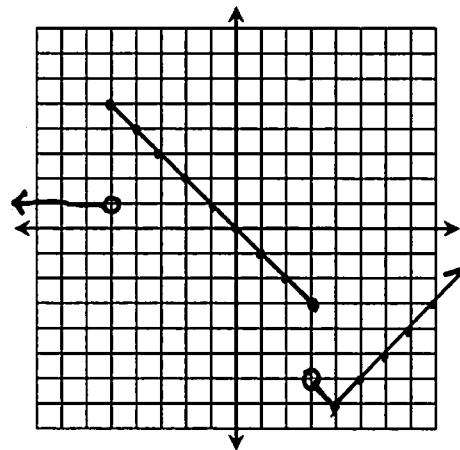
Graph each piecewise function and state the domain and range.

$$7. f(x) = \begin{cases} -\frac{3}{2}|x| + 5 & \text{if } x \leq 4 \\ -2x + 3 & \text{if } x > 4 \end{cases}$$



$$D = \mathbb{R} \quad R = \{y \mid y \leq 5\}$$

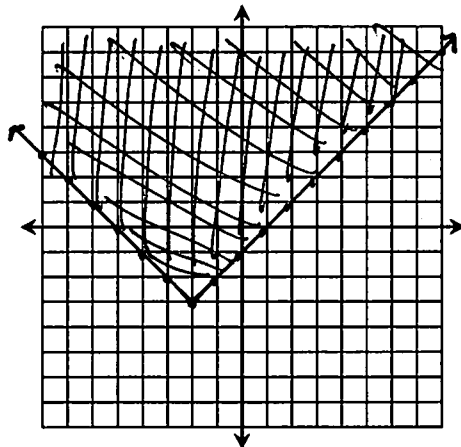
$$8. f(x) = \begin{cases} 1 & \text{if } x < -5 \\ -x & \text{if } -5 \leq x \leq 3 \\ |x - 4| - 7 & \text{if } x > 3 \end{cases}$$



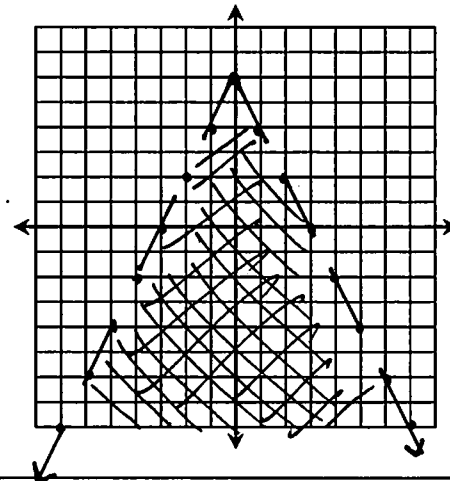
$$D = \mathbb{R} \quad R = \{y \mid y \geq -7\}$$

Graph each absolute value inequality.

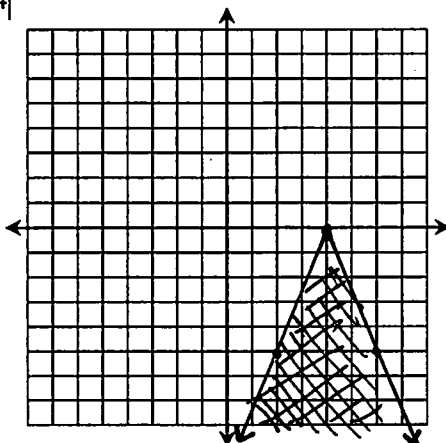
$$9. y \geq |x + 2| - 3$$



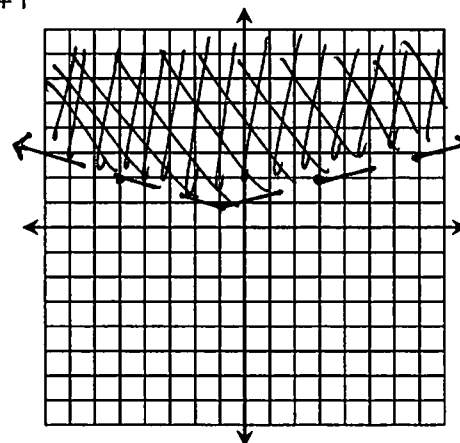
$$10. y < -2|x| + 6$$



$$11. y \leq -\frac{5}{2}|x - 4|$$



$$12. y > \frac{1}{4}|x + 1| + 1$$



Name: _____ Algebra 2

Date: _____ Per: _____ Unit 3: Parent Functions & Transformations

Quiz 3-2: Transformations & Vertex Form of Absolute Value Functions

Identify the parent function of each function family.

1. Linear

1. $f(x) = x$

2. Absolute Value

2. $f(x) = |x|$

Describe the transformations that were applied to the parent function.

3. $f(x) = -|x| - 5$

3. Reflected across x-axis, shifted down 5

4. $f(x) = 3|x + 8|$

4. Vertical stretch by 3, shifted left 8

5. $f(x) = \frac{1}{2}|x - 1| - 1$

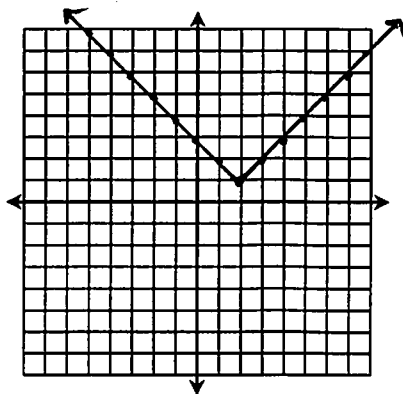
5. Vertical compression by $\frac{1}{2}$, shifted right 1, shifted down 1

6. Dale graphed the absolute value parent function. Then, he reflected the graph over the x-axis, shifted it four units to the right and three units up. Give the new equation.

6. $f(x) = -|x - 4| + 3$

Identify the vertex, then graph. State the domain and range.

7. $f(x) = |x - 2| + 1$

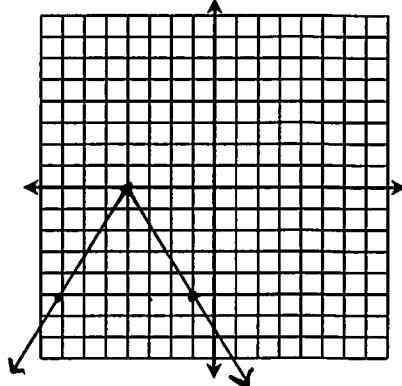


7. Vertex: $(2, 1)$

D: $\{ \mathbb{R} \}$

R: $\{ y \mid y \geq 1 \}$

8. $f(x) = -\frac{5}{3}|x + 4|$



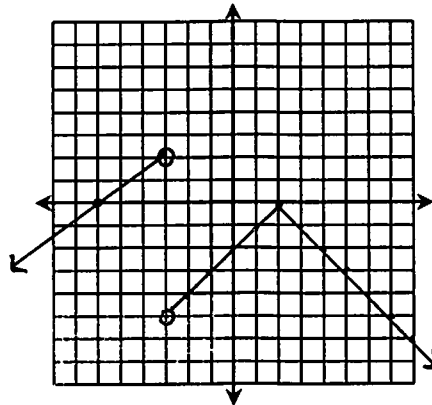
8. Vertex: $(-4, 0)$

D: $\{ \mathbb{R} \}$

R: $\{ y \mid y \leq 0 \}$

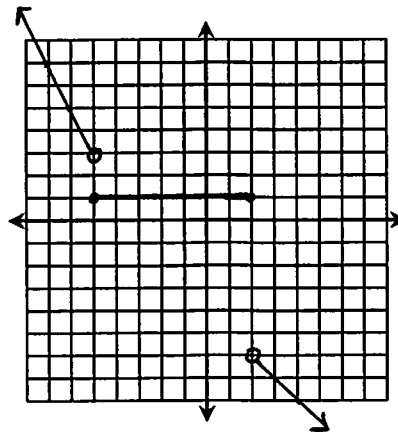
Graph each piecewise function. State the domain and range.

9. $f(x) = \begin{cases} \frac{2}{3}x + 4 & \text{if } x < -3 \\ -|\bar{x} - 2| & \text{if } x > -3 \end{cases}$



9. D: $\{x \mid x \neq -3\}$
R: $\{y \mid y < 2\}$

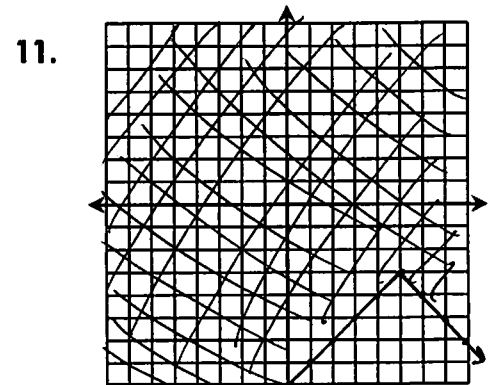
10. $f(x) = \begin{cases} 2|x| - 7 & \text{if } x < -5 \\ 1 & \text{if } -5 \leq x \leq 2 \\ -x - 4 & \text{if } x > 2 \end{cases}$



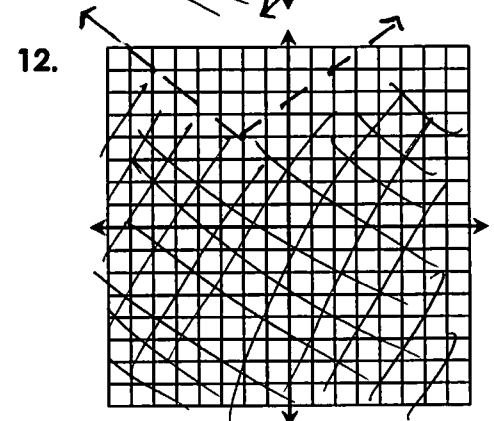
10. D: $\{\mathbb{R}\}$
R: $\{y \mid y > 3 \cup y = 1 \cup y < -6\}$

Graph each inequality.

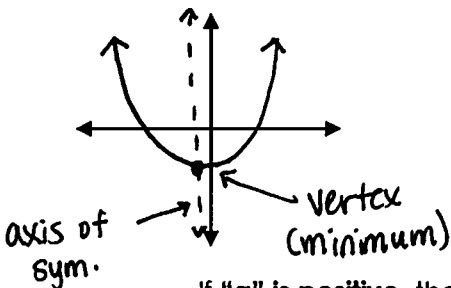
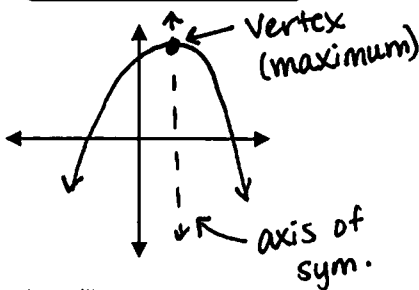
11. $y \geq -|x - 5| - 3$



12. $y < \frac{3}{4}|x + 2| + 4$



Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples
QUADRATIC FUNCTION	- Standard Form of a Quadratic Equation: $f(x) = ax^2 + bx + c$ - A quadratic equation creates a U-shaped curve called a parabola.
Examples	Using your graphing calculator, sketch the following functions: $f(x) = x^2 + 2x - 5$  If "a" is positive, then the parabola will open up. $f(x) = -x^2 + 3x + 7$  If "a" is negative, then the parabola will open down.
Parts of a PARABOLA	- The vertical line that divides the parabola into two equal parts is called the axis of symmetry. - Axis of Symmetry Formula: $x = \frac{-b}{2a}$ - The turning point of the parabola is called the vertex. When the vertex is the lowest point, it's called a minimum, when it's the highest point, it's called a maximum. - Label these parts on the graphs above.
Directions: Determine whether each function has a minimum or maximum. Then find the axis of symmetry and vertex.	
1. $f(x) = 4x^2$ $x = \frac{-0}{2(4)} = 0$ Minimum / Maximum Axis of Symmetry: $x = 0$ Vertex: $(0, 0)$	2. $f(x) = -x^2 + 6x$ $x = \frac{-6}{2(-1)} = 3$ Minimum / Maximum Axis of Symmetry: $x = 3$ Vertex: $(3, 9)$
3. $f(x) = -2x^2 + 7$ $x = \frac{-0}{2(-2)} = 0$ Minimum / Maximum Axis of Symmetry: $x = 0$ Vertex: $(0, 7)$	4. $f(x) = x^2 - 8x + 9$ $x = \frac{-(-8)}{2(1)} = 4$ Minimum / Maximum Axis of Symmetry: $x = 4$ Vertex: $(4, -7)$

GRAPHING

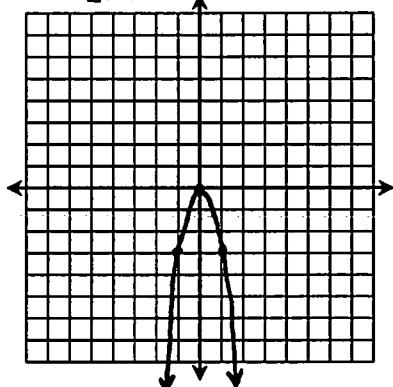
by Table

- ① Find the **AXIS OF SYMMETRY** and **VERTEX**.
- ② **PLACE THE VERTEX AS THE MIDDLE ROW.**
Number up and down, then complete the table.
- ③ **GRAPH!**

Directions: Graph each function. State the domain and range.

5. $f(x) = -3x^2$ $x = \frac{-0}{2(-3)} = 0$

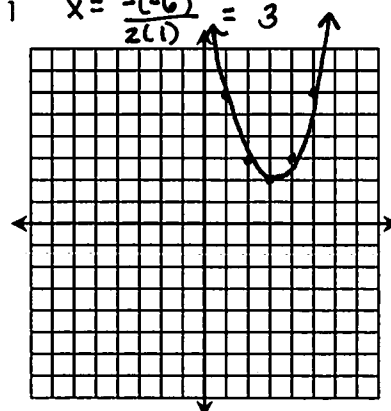
x	f(x)
-3	-27
-2	-12
-1	-3
0	0
1	-3
2	-12
3	-27



$D = \{ \mathbb{R} \}$ $R = \{ y \mid y \leq 0 \}$

6. $f(x) = x^2 - 6x + 11$ $x = \frac{-(-6)}{2(1)} = 3$

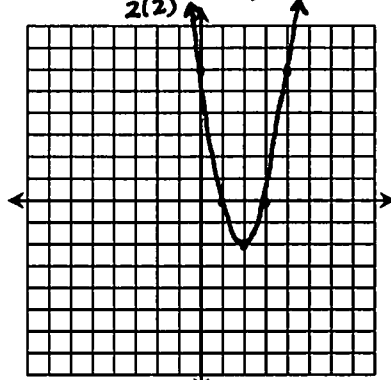
x	f(x)
0	11
1	6
2	3
3	2
4	3
5	6
6	11



$D = \{ \mathbb{R} \}$ $R = \{ y \mid y \geq 2 \}$

7. $f(x) = 2x^2 - 8x + 6$ $x = \frac{-(-8)}{2(2)} = 2$

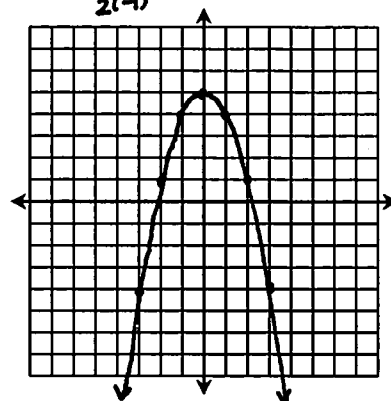
x	f(x)
-1	16
0	6
1	0
2	-2
3	0
4	6
5	16



$D = \{ \mathbb{R} \}$ $R = \{ y \mid y \geq -2 \}$

8. $f(x) = -x^2 + 5$ $x = \frac{-0}{2(-1)} = 0$

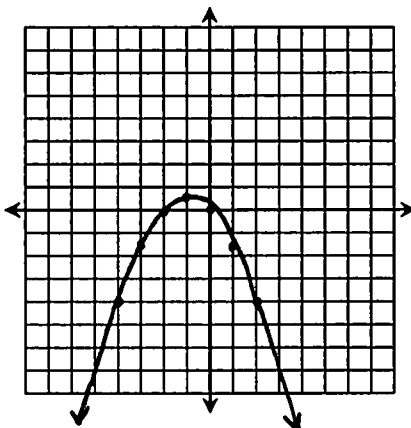
x	f(x)
-3	-4
-2	1
-1	4
0	5
1	4
2	1
3	-4



$D = \{ \mathbb{R} \}$ $R = \{ y \mid y \leq 5 \}$

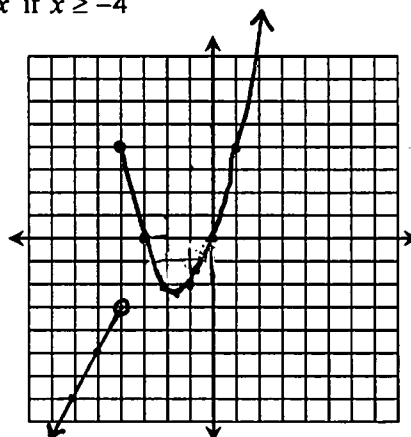
9. $f(x) = -\frac{1}{2}x^2 - x$ $x = \frac{-(-1)}{2(-1/2)} = -1$

x	f(x)
-4	-4
-3	-1.5
-2	0
-1	0.5
0	0
1	-1.5
2	-4



$D = \{ \mathbb{R} \}$ $R = \{ y \mid y \leq 1/2 \}$

10. $f(x) = \begin{cases} 2x+5 & \text{if } x < -4 \\ x^2+3x & \text{if } x \geq -4 \end{cases}$



$D = \{ \mathbb{R} \}$ $R = \{ y \mid y < -3 \cup y \geq -2.25 \}$

Name:

Date:

Topic:

Class:

Main Ideas/Questions

Notes/Examples

GRAPHING QUADRATIC

inequalities

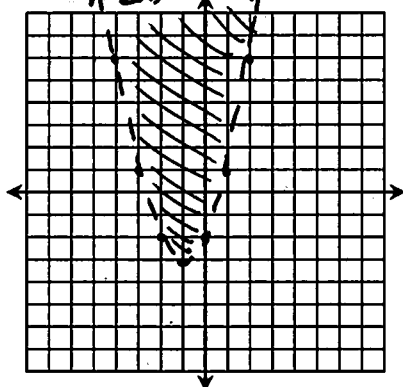
- ① Create a **TABLE OF VALUES**. Recall that the vertex should be in the middle row.
- ② **GRAPH!** Use a dashed line for $<$ and $>$ symbols and a solid line for $\leq$ and $\geq$ symbols.
- ③ **SHADE!** Assuming y is to the left, shade above the line for $>$ or $\geq$ symbols and below the line for $<$ or $\leq$ symbols.

Directions: Graph each inequality.

1. $y > x^2 + 2x - 2$

$$x = \frac{-2}{2(1)} = -1$$

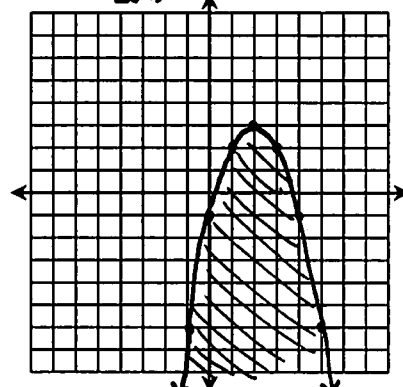
x	y
-4	6
-3	1
-2	-2
-1	-3
0	-2
1	1
2	6



2. $y \leq -x^2 + 4x - 1$

$$x = \frac{-4}{2(-1)} = 2$$

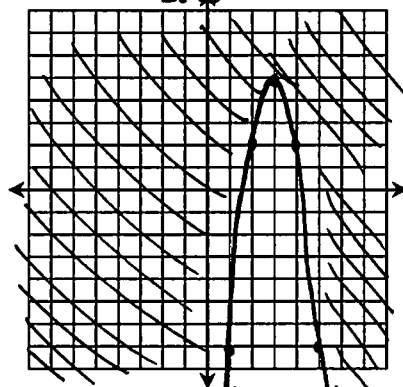
x	y
-1	-6
0	-1
1	2
2	3
3	2
4	-1
5	-6



3. $y \geq -3x^2 + 18x - 22$

$$x = \frac{-18}{2(-3)} = 3$$

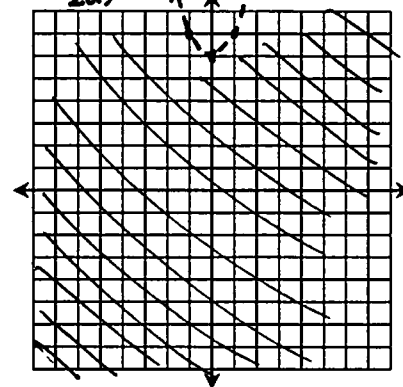
x	y
0	-22
1	-7
2	2
3	5
4	2
5	-7
6	-22



4. $y < x^2 + 6$

$$x = \frac{-0}{2(1)} = 0$$

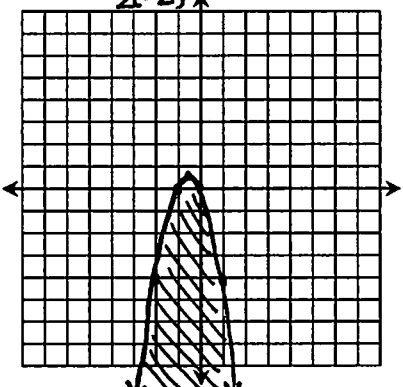
x	y
-3	15
-2	10
-1	7
0	6
1	7
2	10
3	15



5. $y \leq -2x^2 - 2x$

$$x = \frac{-(-2)}{2(-2)} = -\frac{1}{2}$$

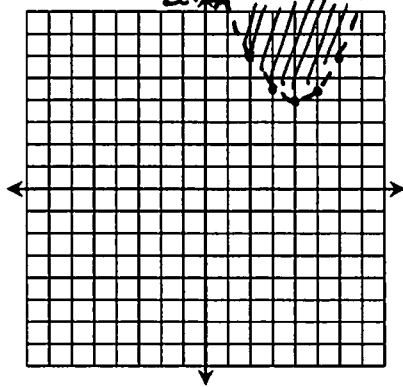
x	y
-3	-12
-2	-4
-1	0
-0.5	0.5
0	0
1	-4
2	-12



6. $y > \frac{1}{2}x^2 - 4x + 12$

$$x = \frac{-(-4)}{2(1/2)} = 4$$

x	y
1	8.5
2	6
3	4.5
4	4
5	4.5
6	6
7	8.5



Name: _____

Unit 3: Parent Functions & Transformations

Date: _____ Per: _____

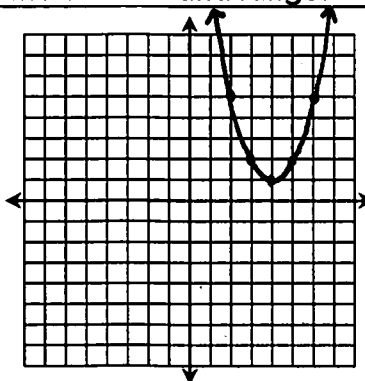
Homework 5: Graphing Quadratic Equations
& Inequalities (Standard Form)**** This is a 2-page document! ******Determine whether the function has a minimum or maximum. Find the axis of symmetry, vertex, then graph using a table of values. Give the domain and range.**

1. $f(x) = x^2 - 8x + 17$

$$x = \frac{-(-8)}{2(1)} = 4$$

$$\begin{aligned} y &= 4^2 - 8(4) + 17 \\ &= 16 - 32 + 17 \\ &= 1 \end{aligned}$$

x	f(x)
1	10
2	5
3	2
4	1
5	2
6	5
7	10

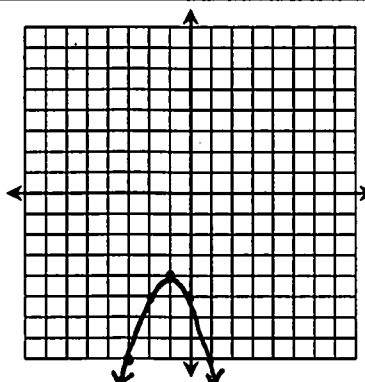
Minimum / MaximumAxis of Symmetry: $x = 4$ Vertex: $(4, 1)$ Domain: $\{ \mathbb{R} \}$ Range: $\{ y \mid y \geq 1 \}$

2. $f(x) = -x^2 - 2x - 5$

$$x = \frac{-(-2)}{2(-1)} = -1$$

$$\begin{aligned} y &= -(-1)^2 - 2(-1) - 5 \\ &= -1 + 2 - 5 \\ &= -4 \end{aligned}$$

x	f(x)
-4	-13
-3	-8
-2	-5
-1	-4
0	-5
1	-8
2	-13

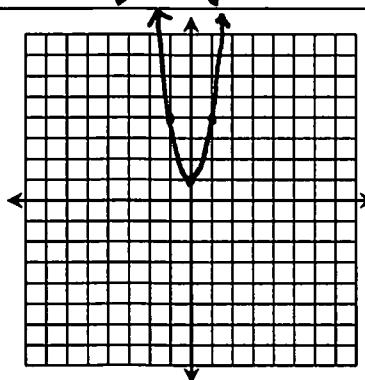
Minimum / MaximumAxis of Symmetry: $x = -1$ Vertex: $(-1, -4)$ Domain: $\{ \mathbb{R} \}$ Range: $\{ y \mid y \leq -4 \}$

3. $f(x) = 3x^2 + 1$

$$x = \frac{-0}{2(3)} = 0$$

$$\begin{aligned} y &= 3(0)^2 + 1 \\ &= 0 + 1 \\ &= 1 \end{aligned}$$

x	f(x)
-3	28
-2	13
-1	4
0	1
1	4
2	13
3	28

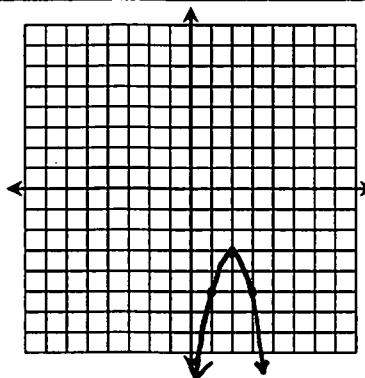
Minimum / MaximumAxis of Symmetry: $x = 0$ Vertex: $(0, 1)$ Domain: $\{ \mathbb{R} \}$ Range: $\{ y \mid y \geq 1 \}$

4. $f(x) = -2x^2 + 8x - 11$

$$x = \frac{-8}{2(-2)} = 2$$

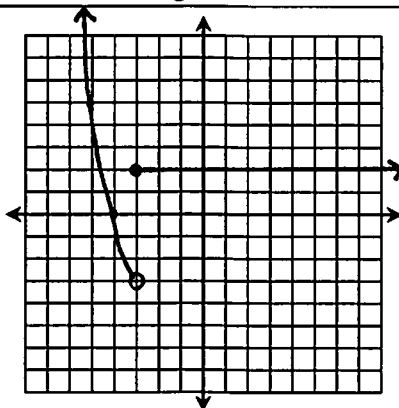
$$\begin{aligned} y &= -2(2)^2 + 8(2) - 11 \\ &= -8 + 16 - 11 \\ &= -3 \end{aligned}$$

x	f(x)
-1	-21
0	-11
1	-5
2	-3
3	-5
4	-11
5	-21

Minimum / MaximumAxis of Symmetry: $x = 2$ Vertex: $(2, -3)$ Domain: $\{ \mathbb{R} \}$ Range: $\{ y \mid y \leq -3 \}$

Graph each function. Give the domain and range.

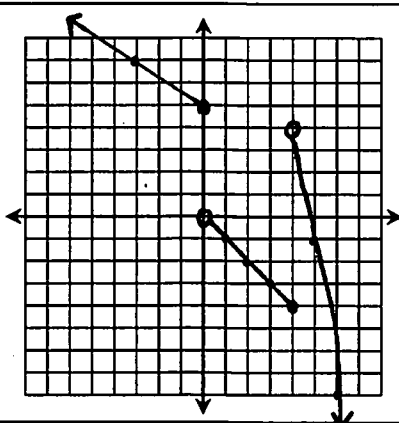
5. $f(x) = \begin{cases} x^2 + 4x & \text{if } x < -3 \\ 2 & \text{if } x \geq -3 \end{cases}$



Domain: $\{\mathbb{R}\}$

Range: $\{y \mid y > -3\}$

6. $f(x) = \begin{cases} -\frac{2}{3}x + 5 & \text{if } x \leq 0 \\ -x & \text{if } 0 < x \leq 4 \\ -x^2 + 4x + 4 & \text{if } x > 4 \end{cases}$



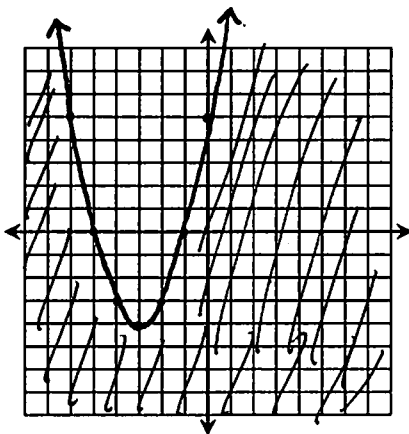
Domain: $\{\mathbb{R}\}$

Range: $\{y \mid y \geq 5 \cup y < 4\}$

Graph each inequality.

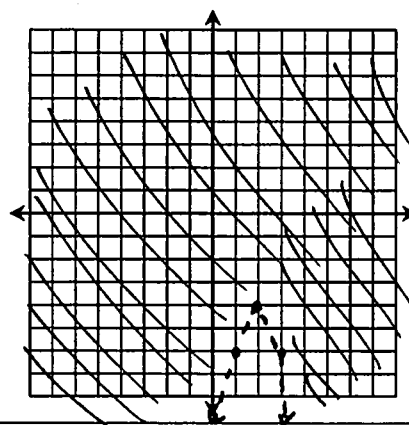
7. $y \leq x^2 + 6x + 5$

x	$f(x)$
-6	5
-5	0
-4	-3
-3	-4
-2	-3
-1	0
0	5



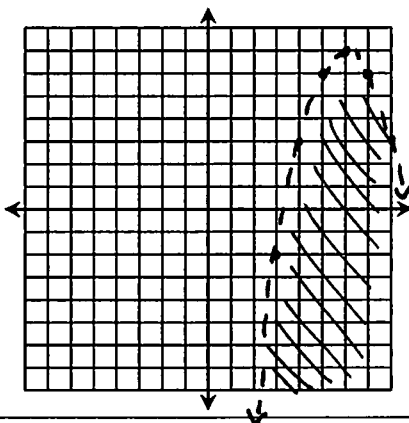
8. $y > -2x^2 + 8x - 12$

x	$f(x)$
-1	-22
0	-12
1	-6
2	-4
3	-6
4	-12
5	-22



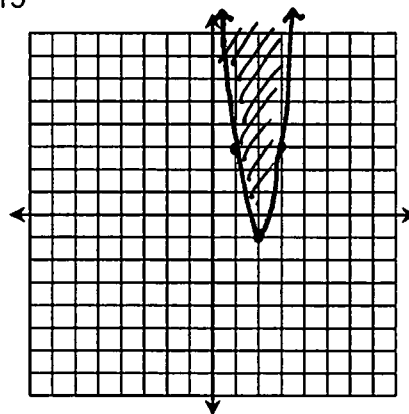
9. $y < -x^2 + 12x - 29$

x	$f(x)$
3	-2
4	3
5	6
6	7
7	6
8	3
9	-2



10. $y \geq 4x^2 - 16x + 15$

x	$f(x)$
-1	35
0	15
1	3
2	-1
3	3
4	15
5	35



Name:

Date:

Topic:

Class:

Main Ideas/Questions	Notes/Examples
VERTEX FORM <i>of a Quadratic Function</i>	- Vertex Form of a Quadratic Equation: $f(x) = a(x-h)^2 + k$ (h, k) is the vertex; $x = h$ is the axis of symmetry a determines the width and direction of the parabola
WHY USE <i>Vertex Form?</i>	Vertex form shows the <u>Vertex</u> on a quadratic equation in respect to its parent function, $f(x) = x^2$.
TRANSFORMATIONS <i>from the Parent Function</i>	Describe how each function compares to the parent function. 1) $f(x) = x^2 + 9$ shift up 9 2) $f(x) = x^2 - 2$ shift down 2 3) $f(x) = -(x+5)^2$ reflected across X-axis, shift left 5 4) $f(x) = (x-4)^2 - 1$ shift right 4, down 1 5) $f(x) = \frac{1}{3}x^2 + 2$ Vert. compression by $\frac{1}{3}$, shift up 2 6) $f(x) = -2(x+4)^2$ Vert. stretch by 2, reflected across X-axis, shift left 4 7) Mason graphed the parent function of a quadratic equation. Then, he reflected the parabola over the x-axis, and translated it seven units up and three units to the right. What is the equation of the new parabola? $f(x) = -(x-3)^2 + 7$

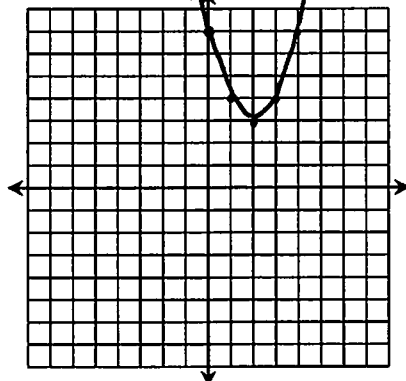
Directions: Give the vertex and axis of symmetry of each quadratic function. Then graph.

8. $f(x) = (x-2)^2 + 3$

V: (2, 3)

A: $x = 2$

x	f(x)
-1	12
0	7
1	4
2	3
3	4
4	7
5	12

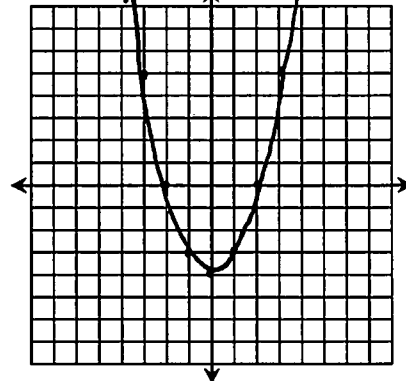


9. $f(x) = x^2 - 4$

V: (0, -4)

A: $x = 0$

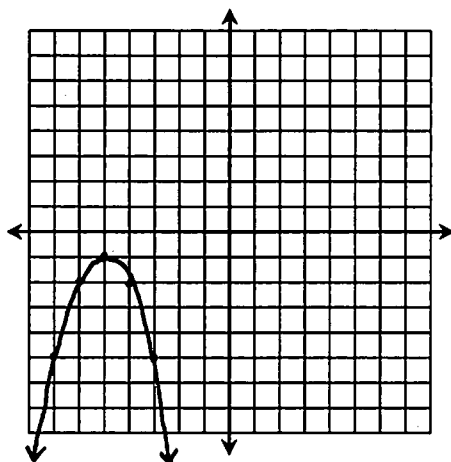
x	f(x)
-3	5
-2	0
-1	-3
0	-4
1	-3
2	0
3	5



10. $f(x) = -(x+5)^2 - 1$

V: $(-5, -1)$
A: $x = -5$

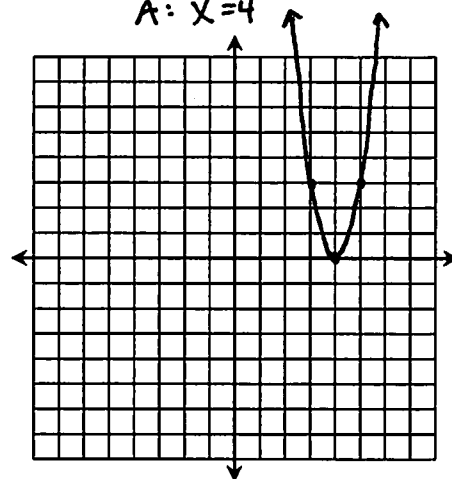
x	f(x)
-8	-10
-7	-5
-6	-2
-5	-1
-4	-2
-3	-5
-2	-10



11. $f(x) = 3(x-4)^2$

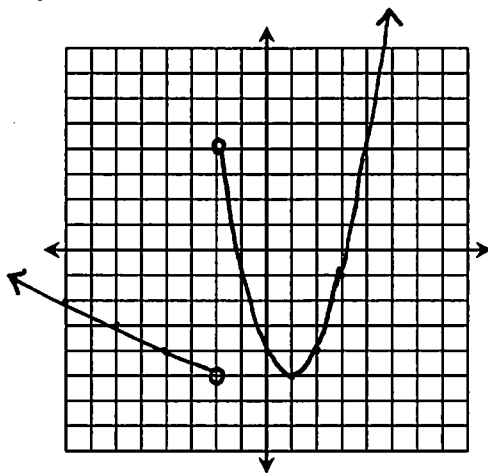
V: $(4, 0)$
A: $x = 4$

x	f(x)
1	27
2	12
3	3
4	0
5	3
6	12
7	27

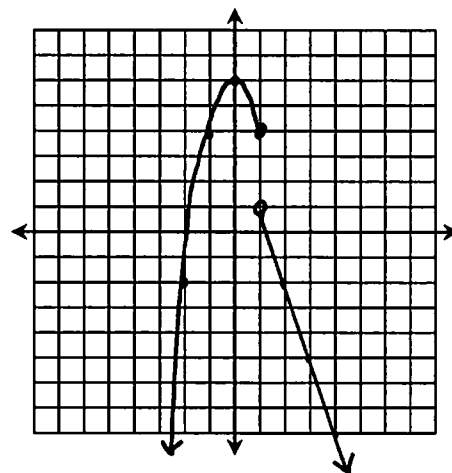


Directions: Graph each piecewise function.

12. $f(x) = \begin{cases} -\frac{1}{2}x - 6 & \text{if } x < -2 \\ (x-1)^2 - 5 & \text{if } x > -2 \end{cases}$



13. $f(x) = \begin{cases} -2x^2 + 6 & \text{if } x \leq 1 \\ -3x + 4 & \text{if } x > 1 \end{cases}$

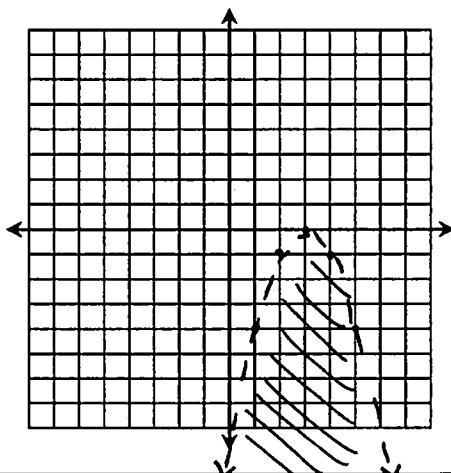


Directions: Give the vertex and axis of symmetry of each quadratic inequality. Then graph.

14. $y < -(x-3)^2$

V: $(3, 0)$
A: $x = 3$

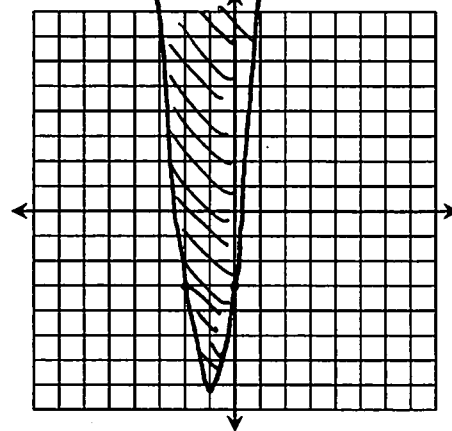
x	f(x)
0	-9
1	-4
2	-1
3	0
4	-1
5	-4
6	-9



15. $y \geq 4(x+1)^2 - 7$

V: $(-1, -7)$
A: $x = -1$

x	f(x)
-4	29
-3	9
-2	-3
-1	-7
0	-3
1	9
2	29



Name: _____

Unit 3: Parent Functions & Transformations

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Homework 6: Vertex Form of a Quadratic Function

**** This is a 2-page document! ******Describe the transformation of each function compared to its parent function.**

1. $f(x) = (x-4)^2 + 9$

right 4, up 9

2. $f(x) = -2x^2 - 3$

Reflect across the x-axis,
Vert. stretch by 2, down 3

3. $f(x) = \frac{1}{4}(x+6)^2$

Vert. compression by $\frac{1}{4}$,
Left 6

4. $f(x) = -(x-7)^2 - 1$

Reflect across the x-axis,
Right 7, down 1

5. $f(x) = \frac{3}{2}x^2$

Vertical stretch by $\frac{3}{2}$

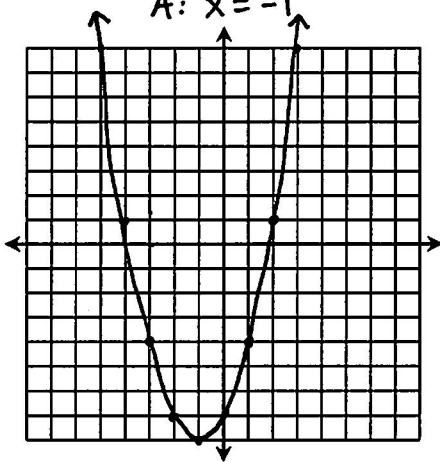
6. $f(x) = -\frac{2}{3}(x-5)^2 + 2$

Reflect across the x-axis,
Vert. compression by $\frac{2}{3}$, right 5, up 2**Give the vertex and axis of symmetry of each quadratic function, then graph.**

7. $f(x) = (x+1)^2 - 8$

V: $(-1, -8)$
A: $x = -1$

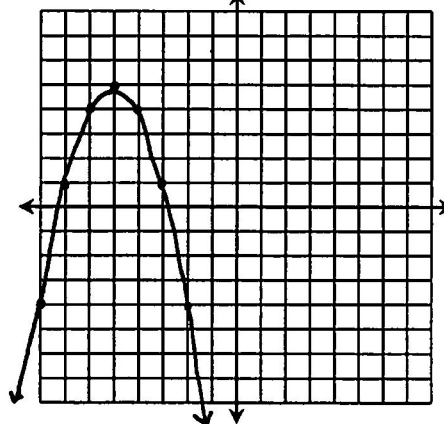
x	f(x)
-4	1
-3	-4
-2	-7
-1	-8
0	-7
1	-4
2	1



8. $f(x) = -(x+5)^2 + 5$

V: $(-5, 5)$
A: $x = -5$

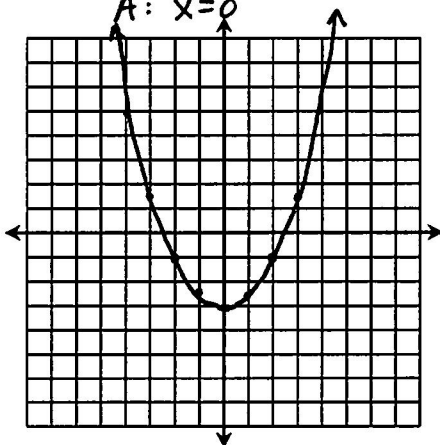
x	f(x)
-8	-4
-7	1
-6	4
-5	5
-4	4
-3	1
-2	-4



9. $f(x) = \frac{1}{2}x^2 - 3$

V: $(0, -3)$
A: $x = 0$

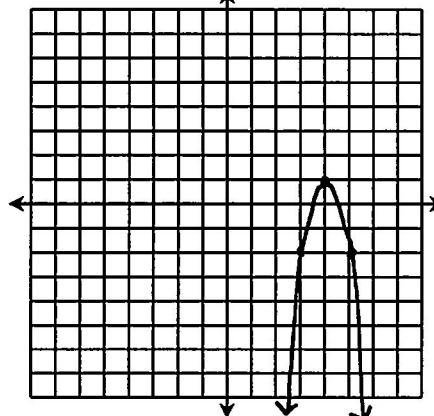
x	f(x)
-3	1.5
-2	-1
-1	-2.5
0	-3
1	-2.5
2	-1
3	1.5



10. $f(x) = -3(x-4)^2 + 1$

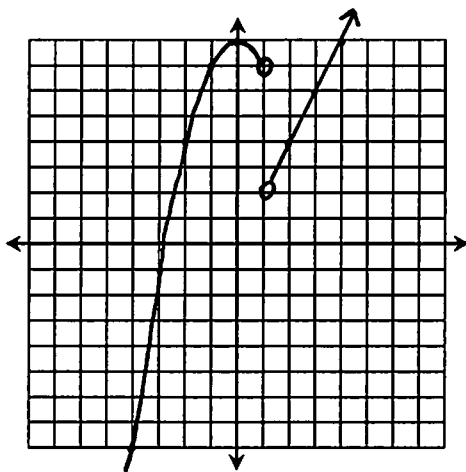
V: $(4, 1)$
A: $x = 4$

x	f(x)
1	-26
2	-11
3	-2
4	1
5	-2
6	-11
7	-26

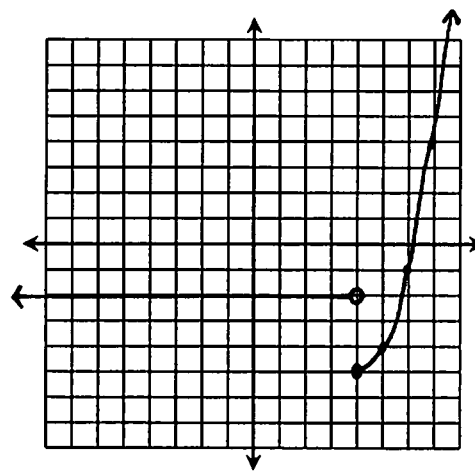


Graph each piecewise function.

11. $f(x) = \begin{cases} -x^2 + 8 & \text{if } x < 1 \\ 2x & \text{if } x > 1 \end{cases}$



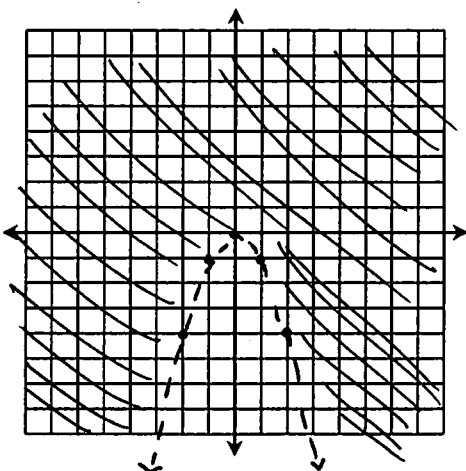
12. $f(x) = \begin{cases} -2 & \text{if } x < 4 \\ (x-4)^2 - 5 & \text{if } x \geq 4 \end{cases}$



Give the vertex and axis of symmetry of each quadratic inequality, then graph.

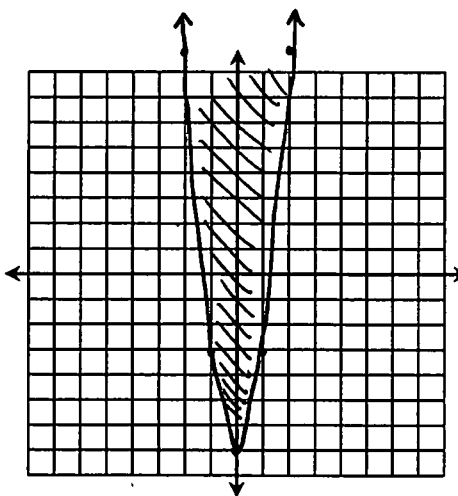
13. $y > -x^2$

x	$f(x)$
-3	-9
-2	-4
-1	-1
0	0
1	-1
2	-4
3	-9



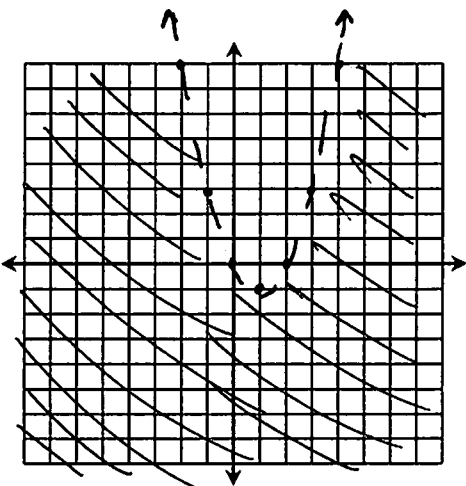
14. $y \geq 4x^2 - 7$

x	$f(x)$
-3	29
-2	9
-1	-3
0	-7
1	-3
2	9
3	29



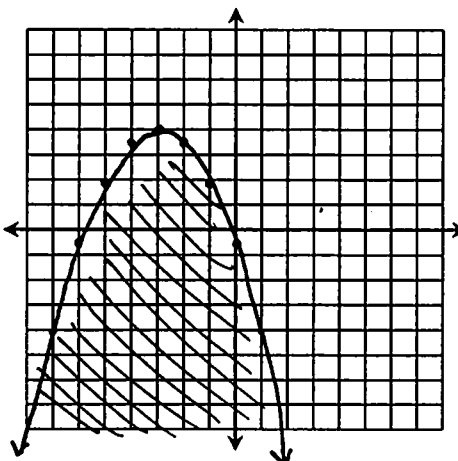
15. $y < (x-1)^2 - 1$

x	$f(x)$
-2	8
-1	3
0	0
1	-1
2	0
3	3
4	8



16. $y \leq -\frac{1}{2}(x+3)^2 + 4$

x	$f(x)$
-6	-6.5
-5	2
-4	3.5
-3	4
-2	3.5
-1	2
0	-0.5



Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples										
ALGEBRA I REVIEW: <i>Perfect Square Trinomials</i>	A trinomial of the form $ax^2 + bx + c$ where $\left(\frac{b}{2}\right)^2 = a \cdot c$. Factor the following trinomials: • $x^2 + 8x + 16 = (x+4)(x+4) = (x+4)^2$ • $m^2 - 18m + 81 = (m-9)^2$ • $a^2 + 2a + 1 = (a+1)(a+1) = (a+1)^2$ • $p^2 - 10p + 25 = (p-5)^2$										
STANDARD FORM $f(x) = ax^2 + bx + c$ ↓ VERTEX FORM $f(x) = a(x-h)^2 + k$	If the standard form equation is a perfect square (like ex. 1), simply factor to convert to vertex form. If not, you can use a process called completing the square to write the equation in vertex form. <table border="1" style="width: 100%; border-collapse: collapse;"> <tr> <td style="width: 5%; text-align: center;">①</td><td>GROUP $(ax^2 + bx)$</td></tr> <tr> <td style="text-align: center;">②</td><td>If $a \neq 1$, FACTOR it outside so it becomes $a(x^2 + bx)$</td></tr> <tr> <td style="text-align: center;">③</td><td>COMPLETE THE SQUARE by taking half of b and square it. This is the new "c". Add this to "$x^2 + bx$". This is the perfect square trinomial.</td></tr> <tr> <td style="text-align: center;">④</td><td>SUBTRACT $a \cdot c$ from the end of the equation.</td></tr> <tr> <td style="text-align: center;">⑤</td><td>FACTOR the trinomial and SIMPLIFY the end of the equation.</td></tr> </table>	①	GROUP $(ax^2 + bx)$	②	If $a \neq 1$, FACTOR it outside so it becomes $a(x^2 + bx)$	③	COMPLETE THE SQUARE by taking half of b and square it. This is the new "c". Add this to " $x^2 + bx$ ". This is the perfect square trinomial.	④	SUBTRACT $a \cdot c$ from the end of the equation.	⑤	FACTOR the trinomial and SIMPLIFY the end of the equation.
①	GROUP $(ax^2 + bx)$										
②	If $a \neq 1$, FACTOR it outside so it becomes $a(x^2 + bx)$										
③	COMPLETE THE SQUARE by taking half of b and square it. This is the new "c". Add this to " $x^2 + bx$ ". This is the perfect square trinomial.										
④	SUBTRACT $a \cdot c$ from the end of the equation.										
⑤	FACTOR the trinomial and SIMPLIFY the end of the equation.										
Directions: Write each function in vertex form. Give the vertex.											
1. $f(x) = x^2 + 4x + 4$ $f(x) = (x^2 + 4x) + 4$ ↓ $\frac{4}{2} = 2 \quad (2)^2 = 4$ $f(x) = (x^2 + 4x + 4) + 4 - 4$ $f(x) = (x+2)^2 + 0$ $f(x) = (x+2)^2$ $V = (-2, 0)$	2. $f(x) = x^2 - 8x + 18$ $f(x) = (x^2 - 8x) + 18$ ↓ $-\frac{8}{2} = -4 \quad (-4)^2 = 16$ $f(x) = (x^2 - 8x + 16) + 18 - 16$ $f(x) = (x-4)^2 + 2$ $V: (4, 2)$										
3. $f(x) = -x^2 + 6x - 5$ $f(x) = (-x^2 + 6x) - 5$ $f(x) = -(x^2 - 6x) - 5$ ↓ $-\frac{6}{2} = -3 \quad (-3)^2 = 9$ $f(x) = -(x^2 - 6x + 9) - 5 + 9$ $f(x) = -(x-3)^2 + 4$ $V: (3, 4)$	4. $f(x) = 2x^2 - 12x + 13$ $f(x) = (2x^2 - 12x) + 13$ $f(x) = 2(x^2 - 6x) + 13$ ↓ $-\frac{6}{2} = -3 \quad (-3)^2 = 9$ $f(x) = 2(x^2 - 6x + 9) + 13 - 18$ $f(x) = 2(x-3)^2 - 5$ $V: (3, -5)$										

$$5. f(x) = x^2 - 3x - 2$$

$$f(x) = (x^2 - 3x) - 2$$

$$\downarrow$$

$$\left(\frac{-3}{2}\right)^2 = \frac{9}{4}$$

$$f(x) = (x^2 - 3x + \frac{9}{4}) - 2 - \frac{9}{4}$$

$$f(x) = \left(x - \frac{3}{2}\right)^2 - \frac{17}{4}$$

$$V: \left(\frac{3}{2}, -\frac{17}{4}\right)$$

$$6. f(x) = -x^2 - 9x - 22$$

$$f(x) = (-x^2 - 9x) - 22$$

$$f(x) = -(x^2 + 9x) - 22$$

$$\downarrow$$

$$\left(\frac{9}{2}\right)^2 = \frac{81}{4}$$

$$f(x) = -(x^2 + 9x + \frac{81}{4}) - 22 + \frac{81}{4}$$

$$f(x) = -\left(x + \frac{9}{2}\right)^2 - \frac{7}{4}$$

$$V: \left(-\frac{9}{2}, -\frac{7}{4}\right)$$

$$7. f(x) = x^2 + 5x$$

$$f(x) = (x^2 + 5x) + 0$$

$$\downarrow$$

$$\left(\frac{5}{2}\right)^2 = \frac{25}{4}$$

$$f(x) = (x^2 + 5x + \frac{25}{4}) - \frac{25}{4}$$

$$f(x) = \left(x + \frac{5}{2}\right)^2 - \frac{25}{4}$$

$$V: \left(-\frac{5}{2}, -\frac{25}{4}\right)$$

$$8. f(x) = 4x^2 - 8x$$

$$f(x) = 4(x^2 - 2x) + 0$$

$$\downarrow$$

$$\frac{-2}{2} = -1 \quad (-1)^2 = 1$$

$$f(x) = 4(x^2 - 2x + 1) - 4$$

$$f(x) = 4(x - 1)^2 - 4$$

$$V: (1, -4)$$

$$9. f(x) = -2x^2 + 20x - 46$$

$$f(x) = -2(x^2 - 10x) - 46$$

$$\downarrow$$

$$\frac{-10}{2} = -5 \quad (-5)^2 = 25$$

$$f(x) = -2(x^2 - 10x + 25) - 46 + 50$$

$$f(x) = -2(x - 5)^2 + 4$$

$$V: (5, 4)$$

$$10. f(x) = -\frac{1}{2}x^2 - 4x - 5$$

$$f(x) = -\frac{1}{2}(x^2 + 8x) - 5$$

$$\downarrow$$

$$\frac{8}{2} = 4 \quad (4)^2 = 16$$

$$f(x) = -\frac{1}{2}(x^2 + 8x + 16) - 5 + 8$$

$$f(x) = -\frac{1}{2}(x + 4)^2 + 3$$

$$V: (-4, 3)$$

Directions: Convert to vertex form. Identify the vertex, axis of symmetry, then graph.

$$11. f(x) = x^2 - 12x + 39$$

$$f(x) = (x^2 - 12x) + 39$$

$$\downarrow$$

$$\frac{-12}{2} = -6 \quad (-6)^2 = 36$$

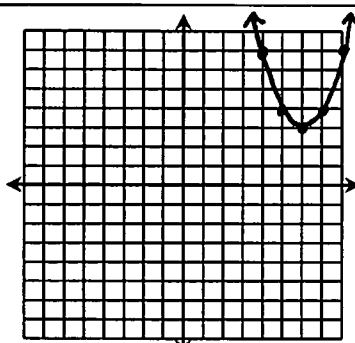
$$f(x) = (x^2 - 12x + 36) + 39 - 36$$

$$f(x) = (x - 6)^2 + 3$$

$$V: (6, 3)$$

$$A: x = 6$$

x	f(x)
3	12
4	7
5	4
6	3
7	4
8	7
9	12



$$12. f(x) = -3x^2 - 12x - 8$$

$$f(x) = -3(x^2 + 4x) - 8$$

$$\downarrow$$

$$\frac{4}{2} = 2 \quad (2)^2 = 4$$

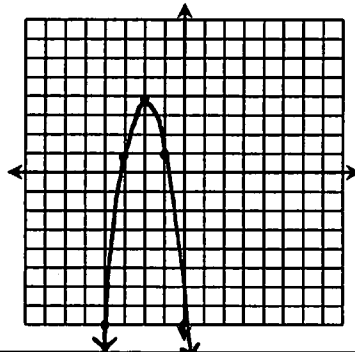
$$f(x) = -3(x^2 + 4x + 4) - 8 + 12$$

$$f(x) = -3(x + 2)^2 + 4$$

$$V: (-2, 4)$$

$$A: x = -2$$

x	f(x)
-5	-23
-4	-8
-3	1
-2	4
-1	1
0	-8
1	-23



Name: _____

Unit 3: Parent Functions & Transformations

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Homework 7: Converting to Vertex Form

**** This is a 2-page document! ******Write each function in vertex form. Give the vertex.**

1. $f(x) = x^2 - 12x + 36$

$$f(x) = (x^2 - 12x) + 36$$

$$\downarrow$$

 $(-6)^2 = 36$

$$f(x) = (x^2 - 12x + 36) + 36 - 36$$

$$f(x) = (x - 6)^2$$

$$V: (6, 0)$$

2. $f(x) = x^2 + 4x - 1$

$$f(x) = (x^2 + 4x) - 1$$

$$\downarrow$$

 $(2)^2 = 4$

$$f(x) = (x^2 + 4x + 4) - 1 - 4$$

$$f(x) = (x + 2)^2 - 5$$

$$V: (-2, -5)$$

3. $f(x) = x^2 - 2x + 5$

$$f(x) = (x^2 - 2x) + 5$$

$$\downarrow$$

 $(-1)^2 = 1$

$$f(x) = (x^2 - 2x + 1) + 5 - 1$$

$$f(x) = (x - 1)^2 + 4$$

$$V: (1, 4)$$

4. $f(x) = 3x^2 + 6x - 4$

$$f(x) = 3(x^2 + 2x) - 4$$

$$\downarrow$$

 $(1)^2 = 1$

$$f(x) = 3(x^2 + 2x + 1) - 4 - 3$$

$$f(x) = 3(x + 1)^2 - 7$$

$$V: (-1, -7)$$

5. $f(x) = x^2 + 7x - 10$

$$f(x) = (x^2 + 7x) - 10$$

$$\downarrow$$

 $(\frac{7}{2})^2 = \frac{49}{4}$

$$f(x) = (x^2 + 7x + \frac{49}{4}) - 10 - \frac{49}{4}$$

$$f(x) = (x + \frac{7}{2})^2 - \frac{89}{4}$$

$$V: (-\frac{7}{2}, -\frac{89}{4})$$

6. $f(x) = -x^2 + 6x - 10$

$$f(x) = -(x^2 - 6x) - 10$$

$$\downarrow$$

 $(-3)^2 = 9$

$$f(x) = -(x^2 - 6x + 9) - 10 + 9$$

$$f(x) = -(x - 3)^2 - 1$$

$$V: (3, -1)$$

7. $f(x) = -6x^2 + 42x$

$$f(x) = -6(x^2 - 7x)$$

$$\downarrow$$

 $(-\frac{7}{2})^2 = \frac{49}{4}$

$$f(x) = -6(x^2 - 7x + \frac{49}{4}) + \frac{147}{2}$$

$$f(x) = -6(x - \frac{7}{2})^2 + \frac{147}{2}$$

$$V: (\frac{7}{2}, \frac{147}{2})$$

8. $f(x) = \frac{1}{2}x^2 - 4x + 7$

$$f(x) = \frac{1}{2}(x^2 - 8x) + 7$$

$$\downarrow$$

 $(-4)^2 = 16$

$$f(x) = \frac{1}{2}(x^2 - 8x + 16) + 7 - 8$$

$$f(x) = \frac{1}{2}(x - 4)^2 - 1$$

$$V: (4, -1)$$

Write each function in vertex form. Identify the vertex and axis of symmetry, then graph.

9. $f(x) = x^2 + 10x + 18$

$$f(x) = (x^2 + 10x) + 18$$

$$\downarrow$$

$$(5)^2 = 25$$

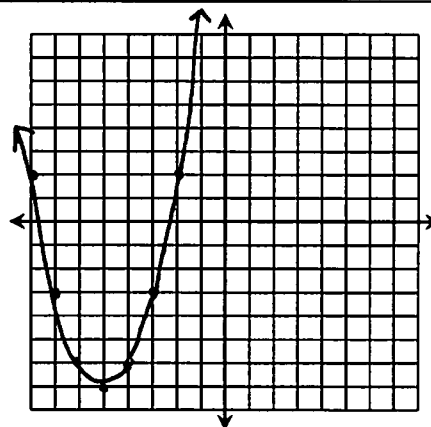
$$f(x) = (x^2 + 10x + 25) + 18 - 25$$

$$f(x) = (x + 5)^2 - 7$$

$$V: (-5, -7)$$

$$A: x = -5$$

x	f(x)
-8	2
-7	-3
-6	-6
-5	-7
-4	-6
-3	-3
-2	2



10. $f(x) = -x^2 + 4x - 6$

$$f(x) = -(x^2 - 4x) - 6$$

$$\downarrow$$

$$(-2)^2 = 4$$

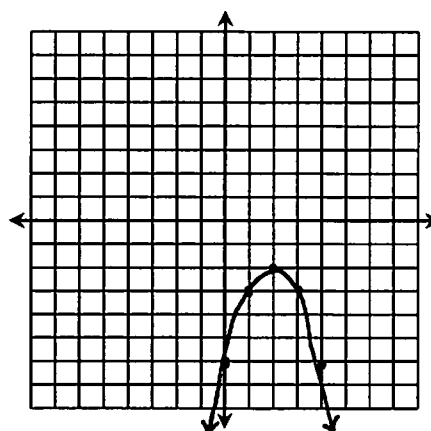
$$f(x) = -(x^2 - 4x + 4) - 6 + 4$$

$$f(x) = -(x - 2)^2 - 2$$

$$V: (2, -2)$$

$$A: x = 2$$

x	f(x)
-1	-11
0	-6
1	-3
2	-2
3	-3
4	-6
5	-11



11. $f(x) = -2x^2 + 8x - 1$

$$f(x) = -2(x^2 - 4x) - 1$$

$$\downarrow$$

$$(-2)^2 = 4$$

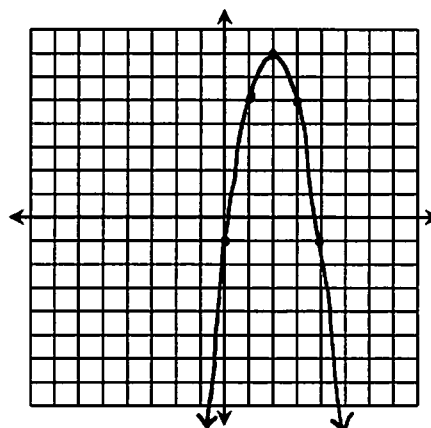
$$f(x) = -2(x^2 - 4x + 4) - 1 + 8$$

$$f(x) = -2(x - 2)^2 + 7$$

$$V: (2, 7)$$

$$A: x = 2$$

x	f(x)
-1	-11
0	-1
1	5
2	7
3	5
4	-1
5	-11



12. $f(x) = 3x^2 + 6x$

$$f(x) = 3(x^2 + 2x) + 0$$

$$\downarrow$$

$$(1)^2 = 1$$

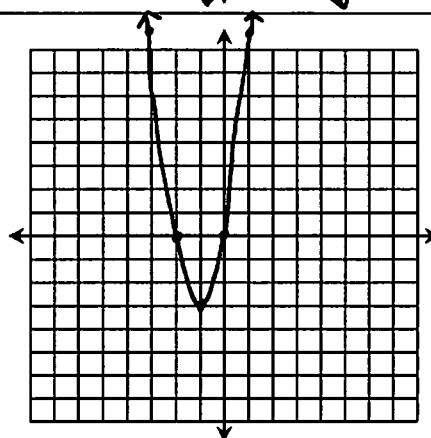
$$f(x) = 3(x^2 + 2x + 1) - 3$$

$$f(x) = 3(x + 1)^2 - 3$$

$$V: (-1, -3)$$

$$A: x = -1$$

x	f(x)
-4	24
-3	9
-2	0
-1	-3
0	0
1	9
2	24



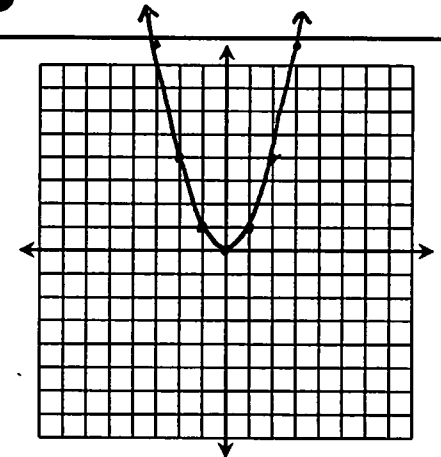
QUADRATIC FUNCTIONS

PARENT FUNCTION:

$$f(x) = x^2$$

- Creates a U-shape curve called a parabola.
- The axis of symmetry is a vertical line that divides the parabola into two equal parts.
- The turning point is called the vertex.

x	f(x)
-3	9
-2	4
-1	1
0	0
1	1
2	4
3	9



STANDARD FORM:

$$f(x) = ax^2 + bx + c$$

Axis of Symmetry: $x = \frac{-b}{2a}$

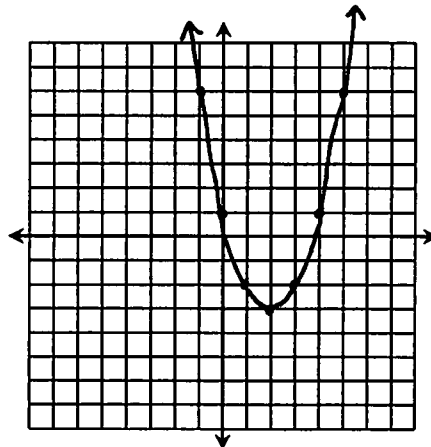
Plug back in to find the vertex!

EXAMPLE: $y = x^2 - 4x + 1$

$$x = \frac{4}{2(1)} = 2$$

$$\begin{aligned} y &= 2^2 - 4(2) + 1 \\ &= 4 - 8 + 1 \\ &= -3 \\ \text{V: } (2, -3) \end{aligned}$$

x	f(x)
-1	6
0	1
1	-2
2	-3
3	-2
4	1
5	6



VERTEX FORM:

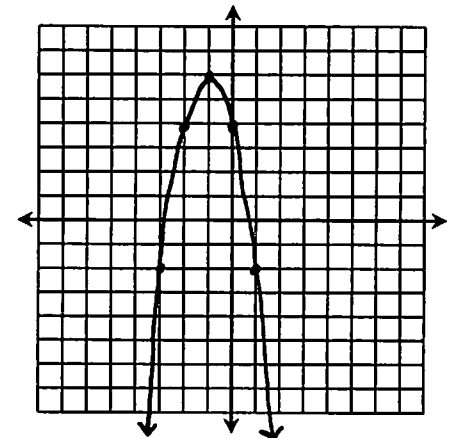
$$f(x) = a(x-h)^2 + k$$

Axis of Symmetry: $x = h$

Vertex: (h, k)

EXAMPLE: $y = -2(x+1)^2 + 6$

x	f(x)
-4	-12
-3	-2
-2	4
-1	6
0	4
1	-2
2	-12



Name: _____ Algebra 2

Date: _____ Per: _____ Unit 3: Parent Functions & Transformations

Quiz 3-3: Graphing Quadratic Functions

Identify the following forms in respect to quadratic functions.

1. Parent Function

1. $f(x) = x^2$

2. Standard Form

2. $f(x) = ax^2 + bx + c$

3. Vertex Form

3. $f(x) = a(x-h)^2 + k$

Give the axis of symmetry, vertex, then graph the function. Give the domain and range.

4. $f(x) = -x^2 + 10x - 17$ $x = \frac{-10}{2(-1)} = 5$

$y = -(5)^2 + 10(5) - 17$
 $y = 8$

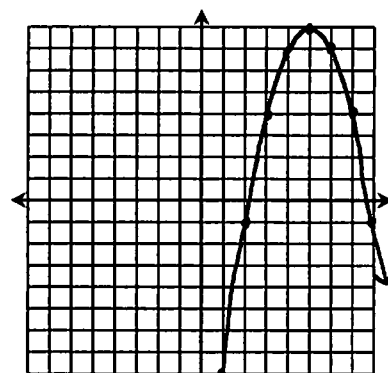
Axis of Symmetry: $x = 5$

Vertex: $(5, 8)$

Domain: $\{ \mathbb{R} \}$

Range: $\{ y \mid y \leq 8 \}$

x	f(x)
2	-1
3	4
4	7
5	8
6	7
7	4
8	-1



5. $f(x) = 2x^2 + 8x + 2$ $x = \frac{-8}{2(2)} = -2$

$y = 2(-2)^2 + 8(-2) + 2$
 $y = -6$

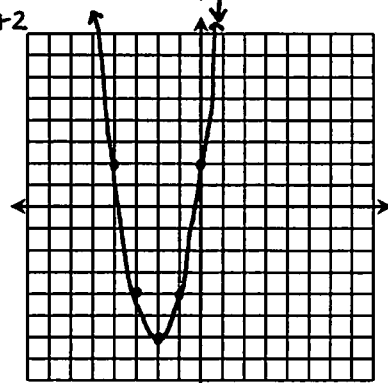
Axis of Symmetry: $x = -2$

Vertex: $(-2, -6)$

Domain: $\{ \mathbb{R} \}$

Range: $\{ y \mid y \geq -6 \}$

x	f(x)
-5	12
-4	2
-3	-4
-2	-6
-1	-4
0	2
1	12



6. $f(x) = x^2 - 6$ $x = \frac{-0}{2(1)} = 0$

$y = 0^2 - 6$
 $y = -6$

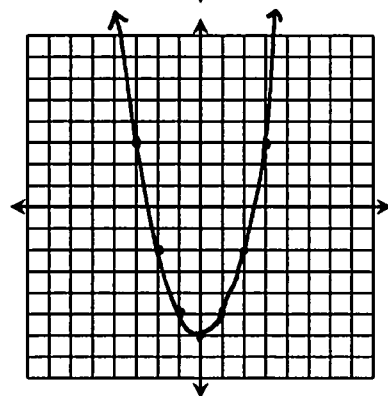
Axis of Symmetry: $x = 0$

Vertex: $(0, -6)$

Domain: $\{ \mathbb{R} \}$

Range: $\{ y \mid y \geq -6 \}$

x	f(x)
-3	3
-2	-2
-1	-5
0	-6
1	-5
2	-2
3	3



7. $f(x) = (x+4)^2 - 5$

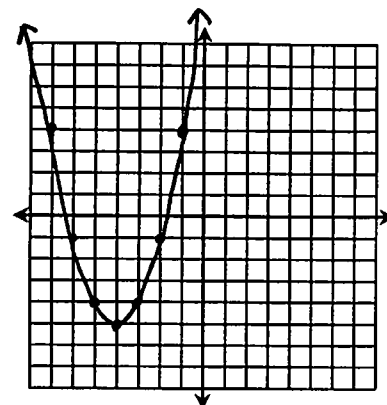
Axis of Symmetry: $x = -4$

Vertex: $(-4, -5)$

Domain: $\{ \mathbb{R} \}$

Range: $\{ y \mid y \geq -5 \}$

x	f(x)
-7	4
-6	-1
-5	-4
-4	-5
-3	-4
-2	-1
-1	4



8. $f(x) = -\frac{1}{2}(x+3)^2$

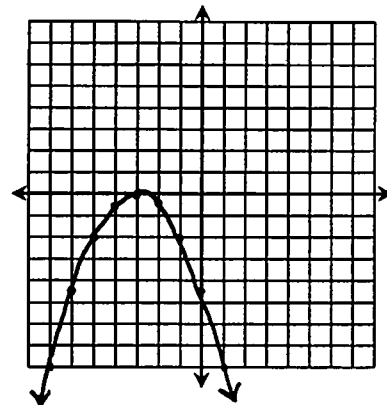
Axis of Symmetry: $x = -3$

Vertex: $(-3, 0)$

Domain: $\{ \mathbb{R} \}$

Range: $\{ y \mid y \leq 0 \}$

x	f(x)
-6	-4.5
-5	-2
-4	-0.5
-3	0
-2	-0.5
-1	-2
0	-4.5



9. The parent function of a quadratic equation is vertically stretched by a factor of 2, then translated 5 units right and 1 unit down. Write the equation of the new function in vertex form.

9. $f(x) = 2(x-5)^2 - 1$

Write each function in vertex form.

10. $f(x) = x^2 - 16x + 64$

$$f(x) = (x^2 - 16x) + 64$$

$$\downarrow$$

$$(-8)^2 = 64$$

$$f(x) = (x^2 - 16x + 64) + 64 - 64$$

$$f(x) = (x-8)^2$$

11. $f(x) = x^2 + 4x - 5$

$$f(x) = (x^2 + 4x) - 5$$

$$\downarrow$$

$$(2)^2 = 4$$

$$f(x) = (x^2 + 4x + 4) - 5 - 4$$

$$f(x) = (x+2)^2 - 9$$

12. $f(x) = -3x^2 + 6x + 4$

$$f(x) = -3(x^2 - 2x) + 4$$

$$\downarrow$$

$$(-1)^2 = 1$$

$$f(x) = -3(x^2 - 2x + 1) + 4 + 3$$

$$f(x) = -3(x-1)^2 + 7$$

10. $f(x) = (x-8)^2$

11. $f(x) = (x+2)^2 - 9$

12. $f(x) = -3(x-1)^2 + 7$

Name:

Date:

Topic:

Class:

Main Ideas/Questions

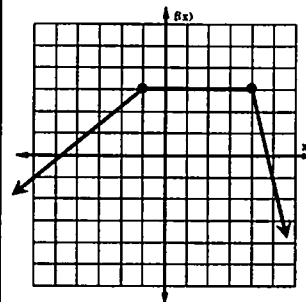
Notes/Examples

INCREASING & DECREASING Intervals

Moving from left to right, a function may increase, decrease, or remain constant on certain intervals.

For example, the function to the right is

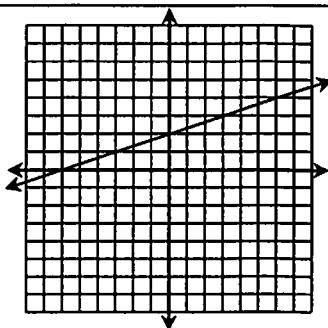
- increasing on the interval $(-\infty, -1)$.
- constant on the interval $(-1, 4)$.
- decreasing on the interval $(4, \infty)$.



EXAMPLES

Identify the intervals on which each function increases or decreases.

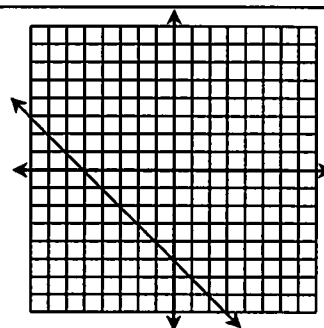
1.



Increasing Interval(s):
 $-\infty < x < \infty$
 $(-\infty, \infty)$

Decreasing Interval(s):
 N/A

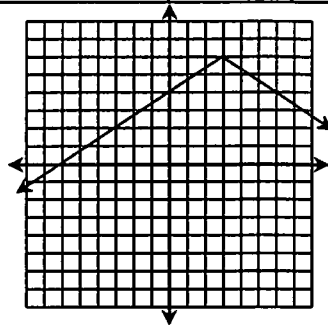
2.



Increasing Interval(s):
 N/A

Decreasing Interval(s):
 $-\infty < x < \infty$
 $(-\infty, \infty)$

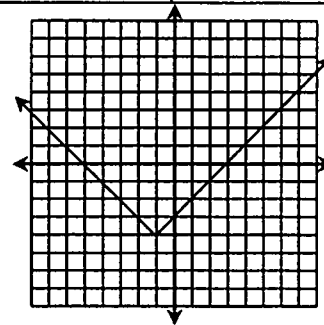
3.



Increasing Interval(s):
 $-\infty < x < 3$
 $(-\infty, 3)$

Decreasing Interval(s):
 $3 < x < \infty$
 $(3, \infty)$

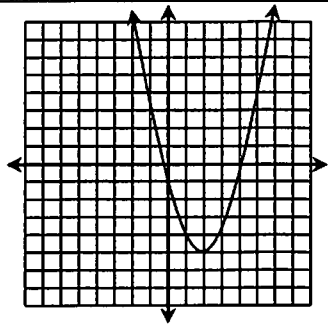
4.



Increasing Interval(s):
 $-1 < x < \infty$
 $(-1, \infty)$

Decreasing Interval(s):
 $-\infty < x < -1$
 $(-\infty, -1)$

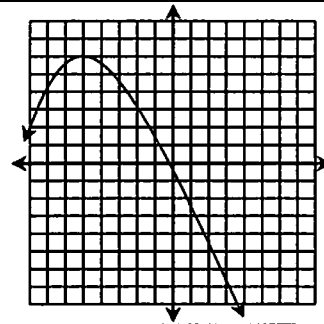
5.



Increasing Interval(s):
 $2 < x < \infty$
 $(2, \infty)$

Decreasing Interval(s):
 $-\infty < x < 2$
 $(-\infty, 2)$

6.



Increasing Interval(s):
 $-\infty < x < -5$
 $(-\infty, -5)$

Decreasing Interval(s):
 $-5 < x < \infty$
 $(-5, \infty)$

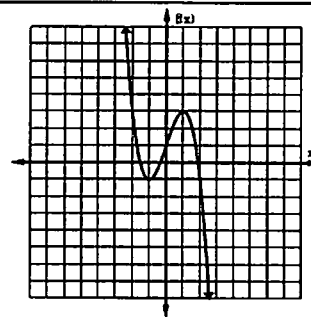
END BEHAVIOR

As x moves towards positive infinity and negative infinity, end behavior describes what is happening to the value of $f(x)$.

Describe the end behavior of the graph to the right:

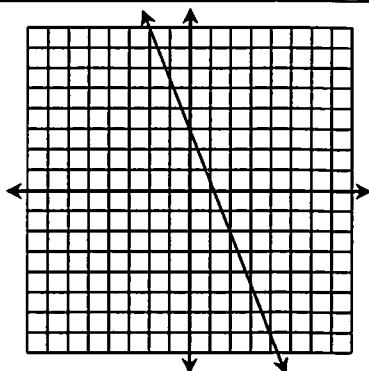
$$\text{As } x \rightarrow \infty, f(x) \rightarrow -\infty$$

$$\text{As } x \rightarrow -\infty, f(x) \rightarrow \infty$$



Describe the end behavior of each function.

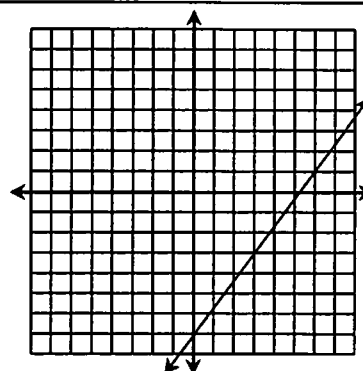
7.



$$\text{As } x \rightarrow \infty, f(x) \rightarrow -\infty$$

$$\text{As } x \rightarrow -\infty, f(x) \rightarrow \infty$$

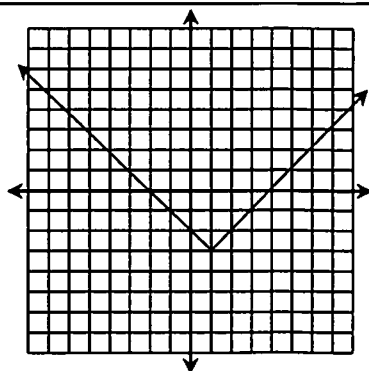
8.



$$\text{As } x \rightarrow \infty, f(x) \rightarrow \infty$$

$$\text{As } x \rightarrow -\infty, f(x) \rightarrow -\infty$$

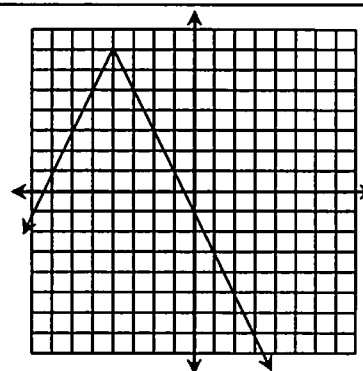
9.



$$\text{As } x \rightarrow \infty, f(x) \rightarrow \infty$$

$$\text{As } x \rightarrow -\infty, f(x) \rightarrow \infty$$

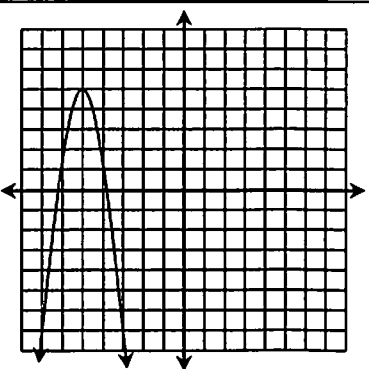
10.



$$\text{As } x \rightarrow \infty, f(x) \rightarrow -\infty$$

$$\text{As } x \rightarrow -\infty, f(x) \rightarrow -\infty$$

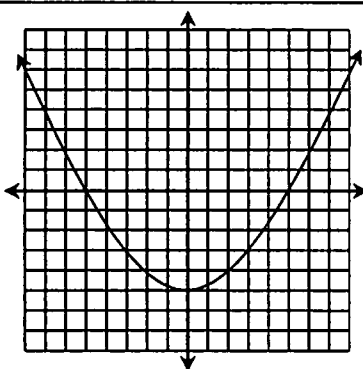
11.



$$\text{As } x \rightarrow \infty, f(x) \rightarrow -\infty$$

$$\text{As } x \rightarrow -\infty, f(x) \rightarrow -\infty$$

12.



$$\text{As } x \rightarrow \infty, f(x) \rightarrow \infty$$

$$\text{As } x \rightarrow -\infty, f(x) \rightarrow \infty$$

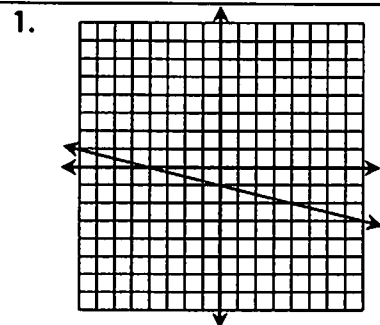
Name: _____

Unit 3: Parent Functions & Transformations

Date: _____ Per: _____

Homework 8: Increasing/Decreasing Intervals
& End Behavior

Identify the increasing and decreasing intervals and the end behavior of each graph.

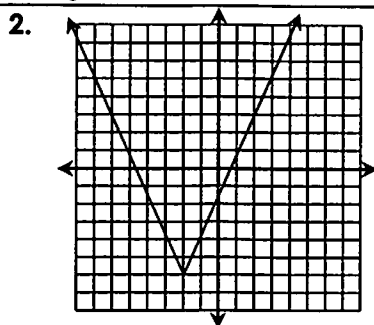


Increasing Int(s): N/A
Decreasing Int(s): $(-\infty, \infty)$

End Behavior:

As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$

As $x \rightarrow -\infty$, $f(x) \rightarrow \infty$

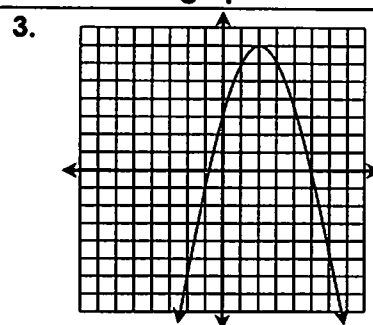


Increasing Int(s): $(-2, \infty)$
Decreasing Int(s): $(-\infty, -2)$

End Behavior:

As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

As $x \rightarrow -\infty$, $f(x) \rightarrow \infty$

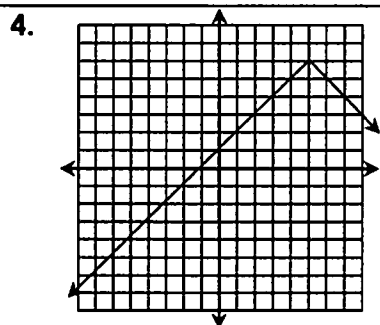


Increasing Int(s): $(-\infty, 2)$
Decreasing Int(s): $(2, \infty)$

End Behavior:

As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$

As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$

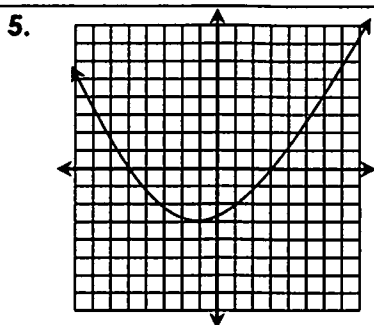


Increasing Int(s): $(-\infty, 5)$
Decreasing Int(s): $(5, \infty)$

End Behavior:

As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$

As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$

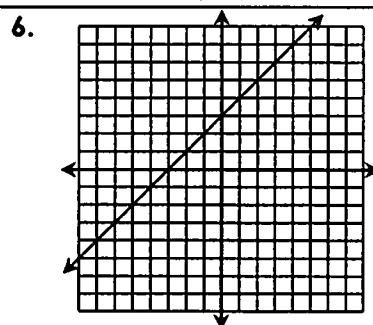


Increasing Int(s): $(-1, \infty)$
Decreasing Int(s): $(-\infty, -1)$

End Behavior:

As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

As $x \rightarrow -\infty$, $f(x) \rightarrow \infty$

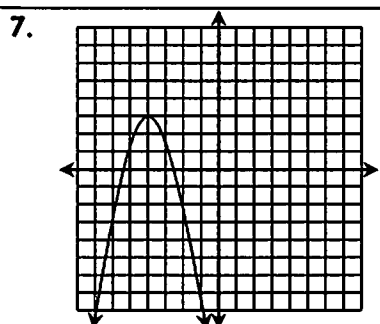


Increasing Int(s): $(-\infty, \infty)$
Decreasing Int(s): N/A

End Behavior:

As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$

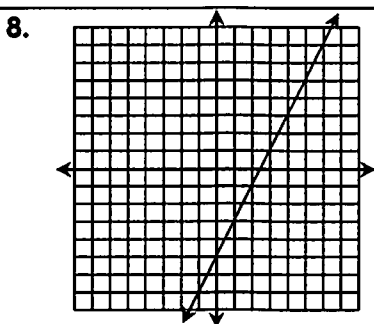


Increasing Int(s): $(-\infty, -4)$
Decreasing Int(s): $(-4, \infty)$

End Behavior:

As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$

As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$

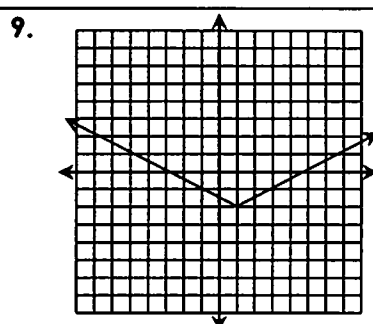


Increasing Int(s): $(-\infty, \infty)$
Decreasing Int(s): N/A

End Behavior:

As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$



Increasing Int(s): $(1, \infty)$
Decreasing Int(s): $(-\infty, 1)$

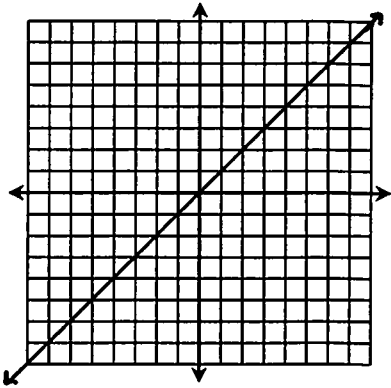
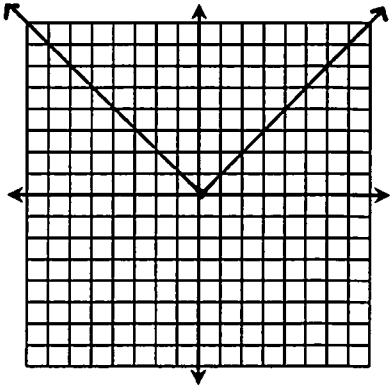
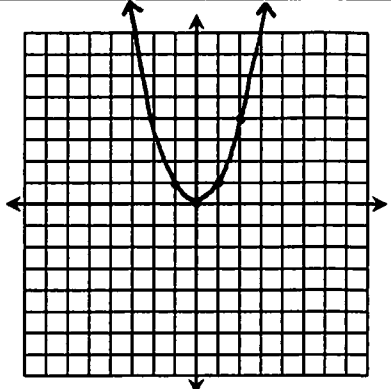
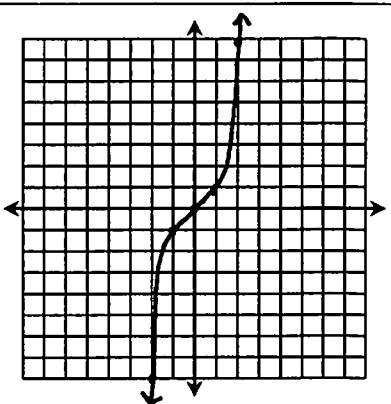
End Behavior:

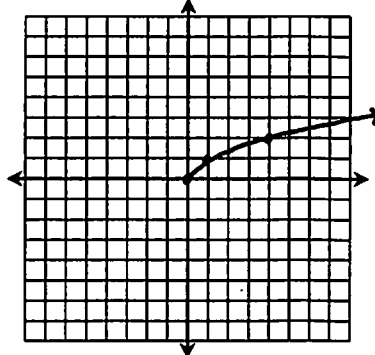
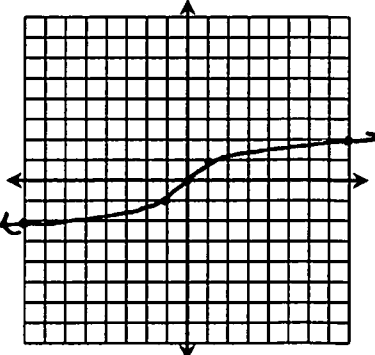
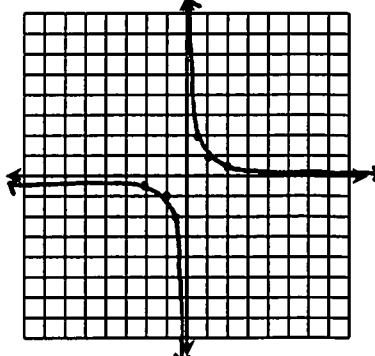
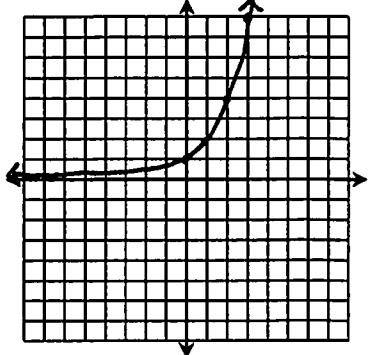
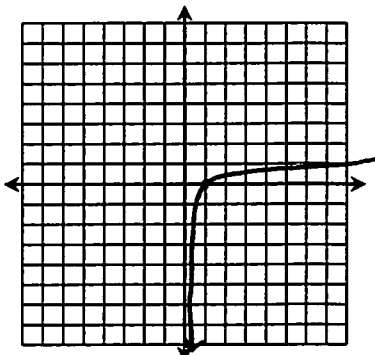
As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

As $x \rightarrow -\infty$, $f(x) \rightarrow \infty$

PARENT FUNCTIONS

A **parent function** is the simplest function of the family.
A **family of graphs** displays similar characteristics.

PARENT FUNCTION	GRAPH	CHARACTERISTICS
LINEAR $f(x) = x$		• D: $\{ \mathbb{R} \}$; R: $\{ \mathbb{R} \}$ • x-Intercept: $(0,0)$; y-Intercept: $(0,0)$ • End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow \infty$ As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$ • Increasing Interval(s): $(-\infty, \infty)$ • Decreasing Interval(s): N/A
ABSOLUTE VALUE $f(x) = x $		• D: $\{ \mathbb{R} \}$; R: $\{ y y \geq 0 \}$ • x-Intercept: $(0,0)$; y-Intercept: $(0,0)$ • End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow \infty$ As $x \rightarrow -\infty$, $f(x) \rightarrow \infty$ • Increasing Interval(s): $(0, \infty)$ • Decreasing Interval(s): $(-\infty, 0)$
QUADRATIC $f(x) = x^2$		• D: $\{ \mathbb{R} \}$; R: $\{ y y \geq 0 \}$ • x-Intercept: $(0,0)$; y-Intercept: $(0,0)$ • End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow \infty$ As $x \rightarrow -\infty$, $f(x) \rightarrow \infty$ • Increasing Interval(s): $(0, \infty)$ • Decreasing Interval(s): $(-\infty, 0)$
CUBIC $f(x) = x^3$		• D: $\{ \mathbb{R} \}$; R: $\{ \mathbb{R} \}$ • x-Intercept: $(0,0)$; y-Intercept: $(0,0)$ • End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow \infty$ As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$ • Increasing Interval(s): $(-\infty, \infty)$ • Decreasing Interval(s): N/A

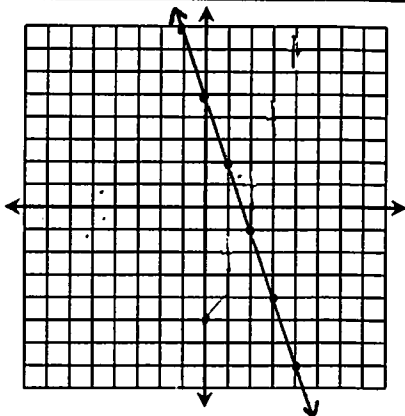
SQUARE ROOT $f(x) = \sqrt{x}$		• D: $\{x x \geq 0\}$; R: $\{y y \geq 0\}$ • x-Intercept: $(0,0)$; y-Intercept: $(0,0)$ • End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow \infty$ As $x \rightarrow 0$, $f(x) \rightarrow 0$ • Increasing Interval(s): $(0, \infty)$ • Decreasing Interval(s): N/A
CUBE ROOT $f(x) = \sqrt[3]{x}$		• D: $\{R\}$; R: $\{R\}$ • x-Intercept: $(0,0)$; y-Intercept: $(0,0)$ • End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow \infty$ As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$ • Increasing Interval(s): $(-\infty, \infty)$ • Decreasing Interval(s): N/A
RECIPROCAL $f(x) = \frac{1}{x}$		• D: $\{x x \neq 0\}$; R: $\{y y \neq 0\}$ • x-Intercept: N/A ; y-Intercept: N/A • End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow 0$ As $x \rightarrow -\infty$, $f(x) \rightarrow 0$ • Increasing Interval(s): N/A • Decreasing Interval(s): $(-\infty, 0), (0, \infty)$ • Asymptote(s): $x=0, y=0$
EXPONENTIAL $f(x) = 2^x$		• D: $\{R\}$; R: $\{y y > 0\}$ • x-Intercept: N/A ; y-Intercept: $(0,1)$ • End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow \infty$ As $x \rightarrow -\infty$, $f(x) \rightarrow 0$ • Increasing Interval(s): $(-\infty, \infty)$ • Decreasing Interval(s): N/A • Asymptote(s): $y=0$
LOGARITHMIC $f(x) = \log x$		• D: $\{x x > 0\}$; R: $\{R\}$ • x-Intercept: $(1,0)$; y-Intercept: N/A • End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow \infty$ As $x \rightarrow 0$, $f(x) \rightarrow -\infty$ • Increasing Interval(s): $(0, \infty)$ • Decreasing Interval(s): N/A • Asymptote(s): $x=0$

FUNCTION FAMILIES REVIEW

(Linear, Absolute Value, and Quadratic)

Graph each function, then identify the characteristics.

1. $f(x) = -3x + 5$



Function Family: Linear

Parent Function: $f(x) = x$

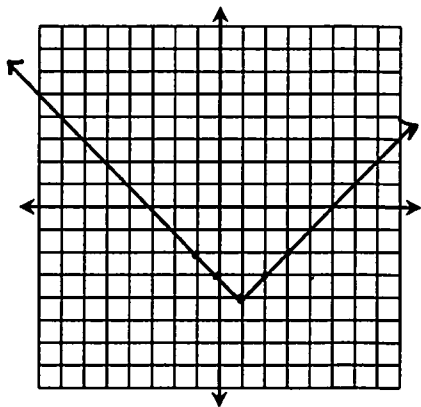
D: $\{\mathbb{R}\}$; R: $\{\mathbb{R}\}$

End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$
As $x \rightarrow -\infty$, $f(x) \rightarrow \infty$

Increasing Interval(s): N/A

Decreasing Interval(s): $(-\infty, \infty)$

2. $f(x) = |x - 1| - 4$



Function Family: Absolute Value

Parent Function: $f(x) = |x|$

D: $\{\mathbb{R}\}$; R: $\{y \mid y \geq -4\}$

End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow \infty$
As $x \rightarrow -\infty$, $f(x) \rightarrow \infty$

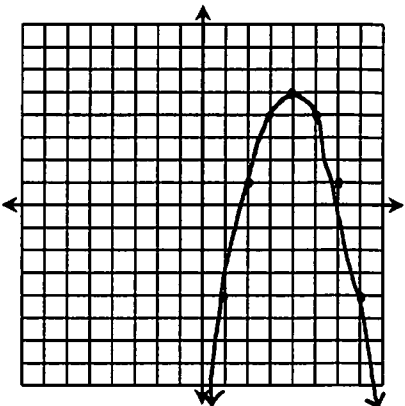
Increasing Interval(s): $(1, \infty)$

Decreasing Interval(s): $(-\infty, 1)$

3. $f(x) = -x^2 + 8x - 11$

$$x = \frac{-(-8)}{2(-1)} = 4$$

X	y
1	-4
2	1
3	4
4	5
5	4
6	1
7	-4



Function Family: Quadratic

Parent Function: $f(x) = x^2$

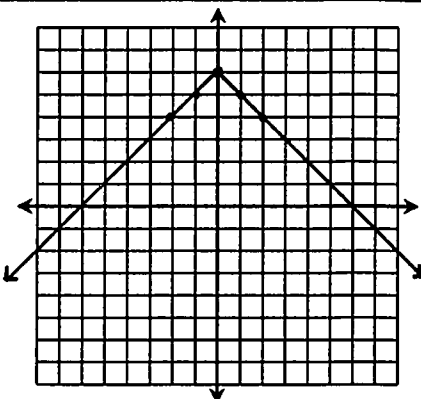
D: $\{\mathbb{R}\}$; R: $\{y \mid y \leq 5\}$

End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$
As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$

Increasing Interval(s): $(-\infty, 4)$

Decreasing Interval(s): $(4, \infty)$

4. $f(x) = -|x| + 6$



Function Family: Absolute Value

Parent Function: $f(x) = |x|$

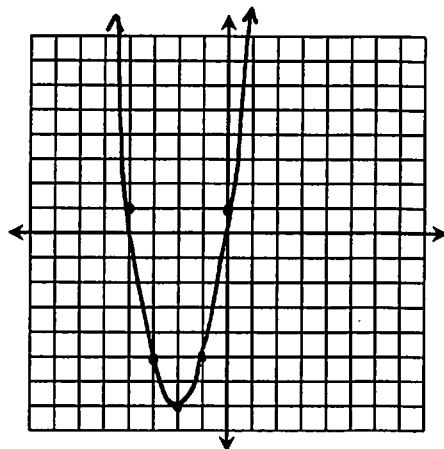
D: $\{\mathbb{R}\}$; R: $\{y \mid y \leq 6\}$

End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$
As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$

Increasing Interval(s): $(-\infty, 0)$

Decreasing Interval(s): $(0, \infty)$

5. $f(x) = 2(x+2)^2 - 7$



Function Family: Quadratic

Parent Function: $f(x) = x^2$

D: $\{ \mathbb{R} \}$; R: $\{ y | y \geq -7 \}$

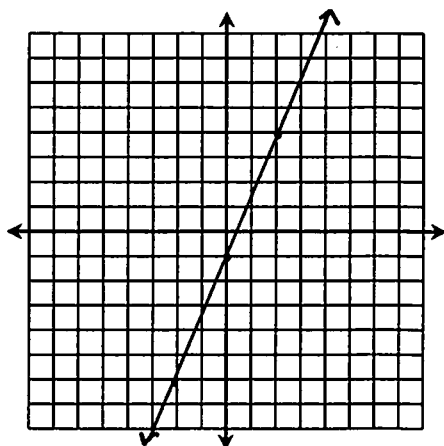
End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

As $x \rightarrow -\infty$, $f(x) \rightarrow \infty$

Increasing Interval(s): $(-2, \infty)$

Decreasing Interval(s): $(-\infty, -2)$

6. $f(x) = \frac{5}{2}x - 1$



Function Family: Linear

Parent Function: $f(x) = x$

D: $\{ \mathbb{R} \}$; R: $\{ \mathbb{R} \}$

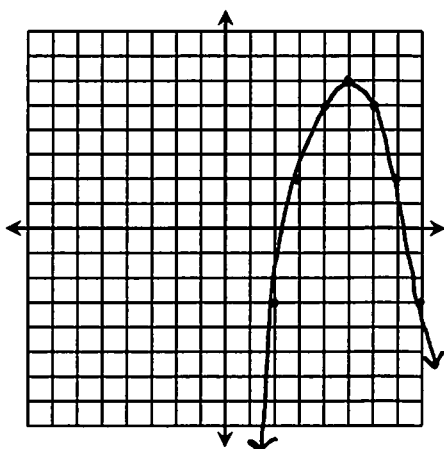
End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$

Increasing Interval(s): $(-\infty, \infty)$

Decreasing Interval(s): N/A

7. $f(x) = -(x-5)^2 + 6$



Function Family: Quadratic

Parent Function: $f(x) = x^2$

D: $\{ \mathbb{R} \}$; R: $\{ y | y \leq 6 \}$

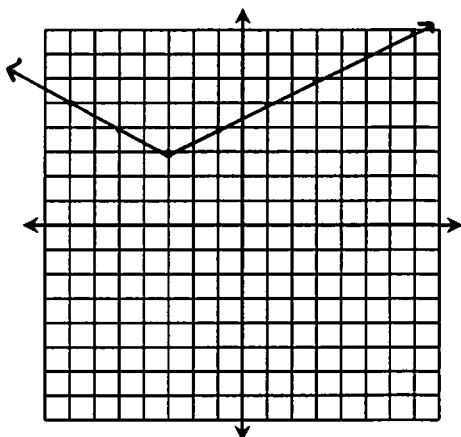
End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$

As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$

Increasing Interval(s): $(-\infty, 5)$

Decreasing Interval(s): $(5, \infty)$

8. $f(x) = \frac{1}{2}|x+3| + 3$



Function Family: Absolute Value

Parent Function: $f(x) = |x|$

D: $\{ \mathbb{R} \}$; R: $\{ y | y \geq 3 \}$

End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

As $x \rightarrow -\infty$, $f(x) \rightarrow \infty$

Increasing Interval(s): $(-3, \infty)$

Decreasing Interval(s): $(-\infty, -3)$

Name: _____

Unit 3: Parent Functions & Transformations

Date: _____ Per: _____

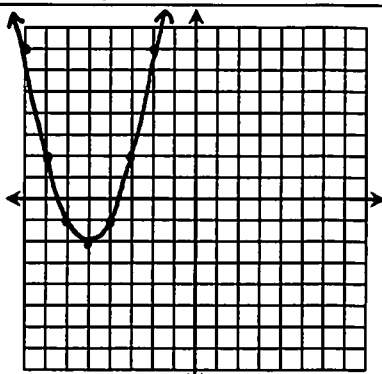
Homework 9: Function Families Review:

Linear, Absolute Value, & Quadratic

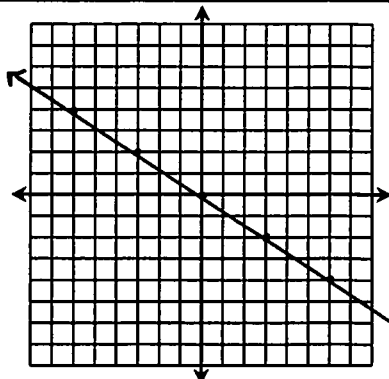
**** This is a 2-page document! ******Graph each function, then identify the characteristics.**

1. $f(x) = (x+5)^2 - 2$

X	y
-8	7
-7	2
-6	-1
-5	-2
-4	-1
-3	2
-2	7

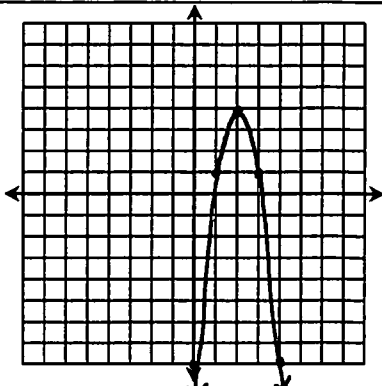
Function Family: QuadraticParent Function: $f(x) = x^2$ D: $\{ \mathbb{R} \}$; R: $\{ y \mid y \geq -2 \}$ End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow \infty$ As $x \rightarrow -\infty$, $f(x) \rightarrow \infty$ Increasing Interval(s): $(-5, \infty)$ Decreasing Interval(s): $(-\infty, -5)$

2. $f(x) = -\frac{2}{3}x$

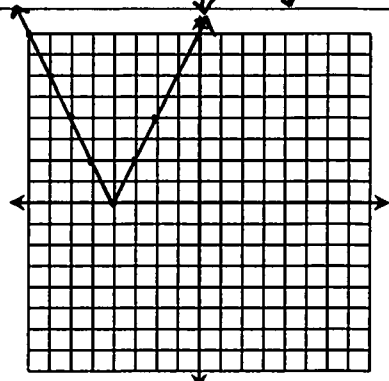
Function Family: LinearParent Function: $f(x) = x$ D: $\{ \mathbb{R} \}$; R: $\{ \mathbb{R} \}$ End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$ As $x \rightarrow -\infty$, $f(x) \rightarrow \infty$ Increasing Interval(s): N/ADecreasing Interval(s): $(-\infty, \infty)$

3. $f(x) = -3x^2 + 12x - 8$

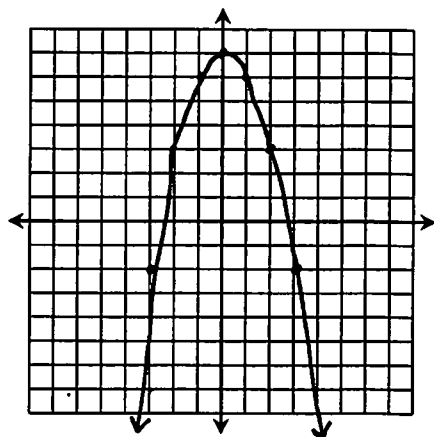
X	y
-1	-23
0	-8
1	1
2	4
3	1
4	-8
5	-23

Function Family: QuadraticParent Function: $f(x) = x^2$ D: $\{ \mathbb{R} \}$; R: $\{ y \mid y \leq 4 \}$ End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$ As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$ Increasing Interval(s): $(-\infty, 2)$ Decreasing Interval(s): $(2, \infty)$

4. $f(x) = 2|x+4|$

Function Family: Absolute ValueParent Function: $f(x) = |x|$ D: $\{ \mathbb{R} \}$; R: $\{ y \mid y \geq 0 \}$ End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow \infty$ As $x \rightarrow -\infty$, $f(x) \rightarrow \infty$ Increasing Interval(s): $(-4, \infty)$ Decreasing Interval(s): $(-\infty, -4)$

5. $f(x) = -x^2 + 7$



Function Family: Quadratic

Parent Function: $f(x) = x^2$

D: $\{ \mathbb{R} \}$; R: $\{ y | y \leq 7 \}$

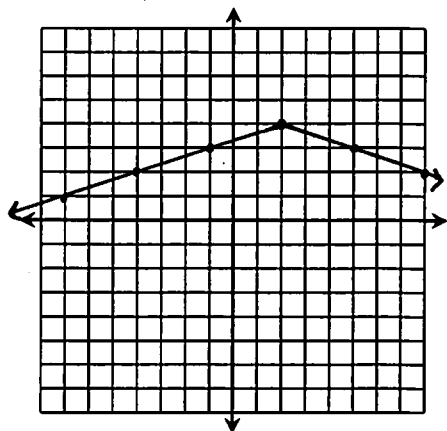
End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$

As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$

Increasing Interval(s): $(-\infty, 0)$

Decreasing Interval(s): $(0, \infty)$

6. $f(x) = -\frac{1}{3}|x-2| + 4$



Function Family: Absolute Value

Parent Function: $f(x) = |x|$

D: $\{ \mathbb{R} \}$; R: $\{ y | y \leq 4 \}$

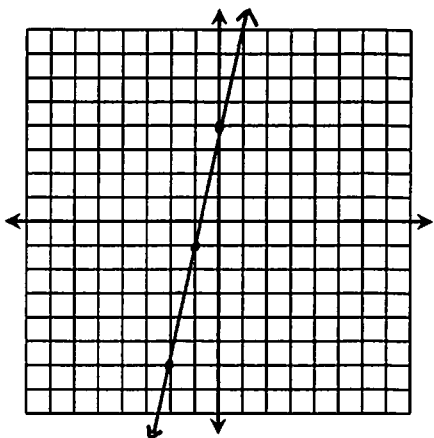
End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$

As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$

Increasing Interval(s): $(-\infty, 2)$

Decreasing Interval(s): $(2, \infty)$

7. $f(x) = 5x + 4$



Function Family: Linear

Parent Function: $f(x) = x$

D: $\{ \mathbb{R} \}$; R: $\{ \mathbb{R} \}$

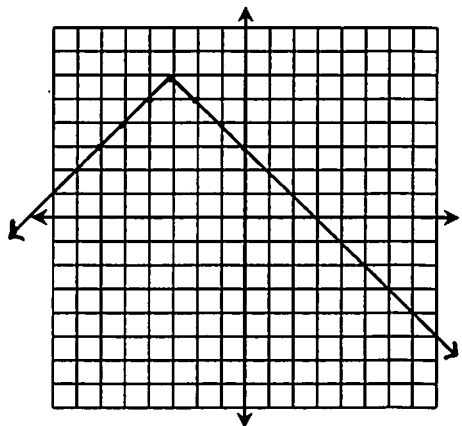
End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$

Increasing Interval(s): $(-\infty, \infty)$

Decreasing Interval(s): N/A

8. $f(x) = -|x+3| + 6$



Function Family: Absolute Value

Parent Function: $f(x) = |x|$

D: $\{ \mathbb{R} \}$; R: $\{ y | y \leq 6 \}$

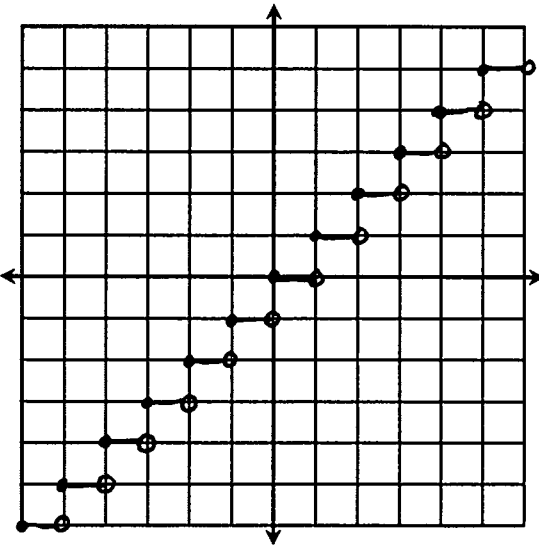
End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$

As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$

Increasing Interval(s): $(-\infty, -3)$

Decreasing Interval(s): $(-3, \infty)$

Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples																				
GREATEST INTEGER Function	- The greatest integer function is written as $f(x) = \lfloor x \rfloor$ - The symbol $\lfloor \rfloor$ means <u>the greatest integer less than or equal to x</u> - In short, this means it will always round the value down to the nearest integer. It is also frequently called a floor function.																				
EVALUATING	1. Given $f(x) = \lfloor x \rfloor$, find each value. a) $f(7) = 7$ b) $f(2.3) = 2$ c) $f(-4.8) = -5$ 2. Given $f(x) = \lfloor x - 5 \rfloor$, find each value. a) $f(9) = 4$ b) $f(0.5) = -5$ c) $f(-8.1) = -14$																				
GREATEST INTEGER Parent Function $f(x) = \lfloor x \rfloor$	Use a table to help generate a pattern, then graph the greatest integer parent function. Identify the domain and range. 3. $f(x) = \lfloor x \rfloor$ <table border="1"> <thead> <tr> <th>x</th><th>y</th></tr> </thead> <tbody> <tr><td>0</td><td>0</td></tr> <tr><td>0.25</td><td>0</td></tr> <tr><td>0.5</td><td>0</td></tr> <tr><td>0.75</td><td>0</td></tr> <tr><td>1</td><td>1</td></tr> <tr><td>1.5</td><td>1</td></tr> <tr><td>2</td><td>2</td></tr> <tr><td>3</td><td>3</td></tr> <tr><td>4</td><td>4</td></tr> </tbody> </table>  $D = \{ \mathbb{R} \}$ $R = \{ \mathbb{Z} \}$	x	y	0	0	0.25	0	0.5	0	0.75	0	1	1	1.5	1	2	2	3	3	4	4
x	y																				
0	0																				
0.25	0																				
0.5	0																				
0.75	0																				
1	1																				
1.5	1																				
2	2																				
3	3																				
4	4																				

x	y
0	-5
0.5	-5
1	-4
1.5	-4
2	-3
3	-2
4	-1

x	y
0	4
0.5	4
1	4
1.5	4
2	5
3	5
4	6

TRANSFORMATIONS

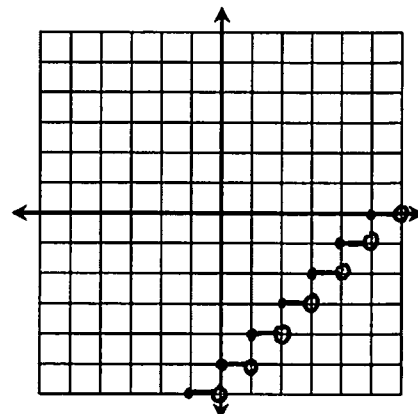
x	y
0	2
0.5	2
1	3
1.5	3
2	4
3	5
4	6

x	y
0	-2
0.5	-2
1	0
1.5	0
2	2
3	4
4	6

Graph each function. Then identify the domain and range.

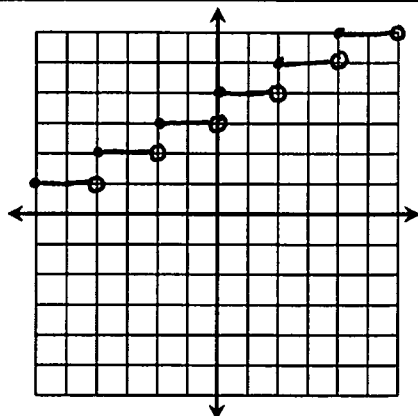
4. $f(x) = \lfloor x \rfloor - 5$

$D = \mathbb{R}$
 $R = \mathbb{Z}$



5. $f(x) = \left\lfloor \frac{1}{2}x + 4 \right\rfloor$

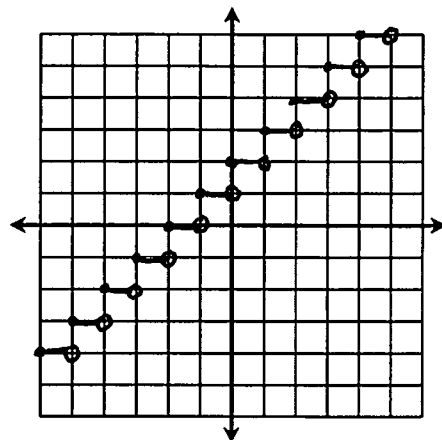
$D = \mathbb{R}$
 $R = \mathbb{Z}$



Transformation(s) from the parent function are described. Write the new function, then graph and identify the domain and range.

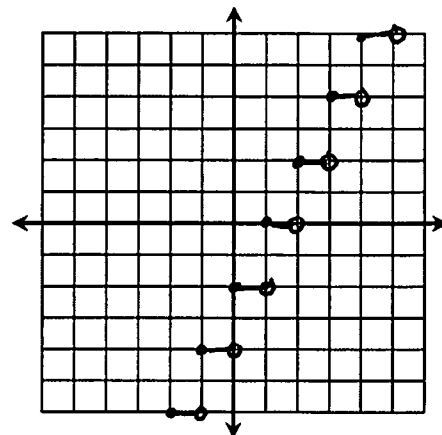
6. Shifted up 2 units

Function: $f(x) = \lfloor x \rfloor + 2$
 $D = \mathbb{R}$
 $R = \mathbb{Z}$



7. Vertical stretch by a factor of 2, then shifted right 1 unit

Function: $f(x) = 2\lfloor x - 1 \rfloor$
 $D = \mathbb{R}$
 $R = \{y \mid y = 2n, n \in \mathbb{Z}\}$



Name: _____

Unit 3: Parent Functions & Transformations

Date: _____ Per: _____

Homework 10: Greatest Integer Function

**** This is a 2-page document! ****

Evaluate the following.

1. $\llbracket 6.8 \rrbracket = 6$

2. $\llbracket 17.21 \rrbracket = 17$

3. $\llbracket 2.99 \rrbracket = 2$

4. $\llbracket -12.7 \rrbracket = -13$

5. $\llbracket -38.15 \rrbracket = -39$

6. $\llbracket 495 \rrbracket = 495$

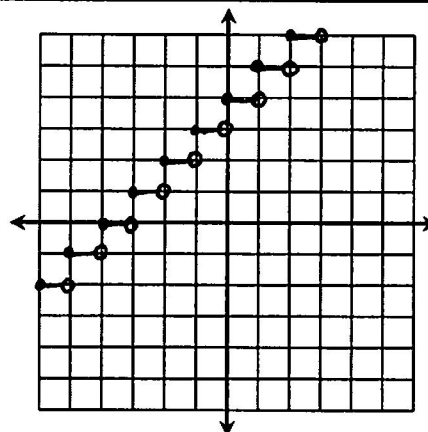
Graph each function, then identify the domain and range.

7. $f(x) = \llbracket x + 4 \rrbracket$

$D = \{ \mathbb{R} \}$

$R = \{ \mathbb{Z} \}$

x	y
0	4
0.5	4
1	5
1.5	5
2	6
3	7
4	8

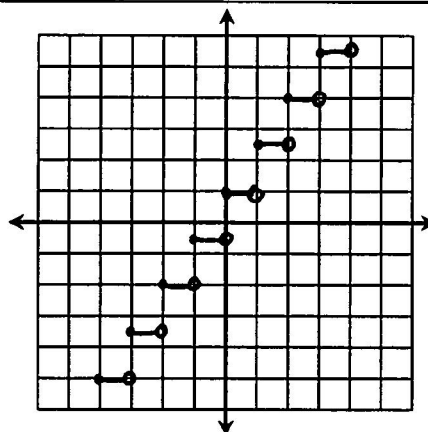


8. $f(x) = \frac{3}{2} \llbracket x \rrbracket + 1$

$D = \{ \mathbb{R} \}$

$R = \{ y \mid y = 1.5n, n \in \mathbb{Z} \}$

x	y
0	1
0.5	1
1	2.5
1.5	2.5
2	4
3	5.5
4	7

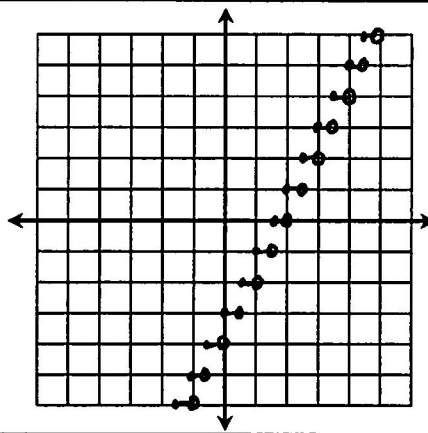


9. $f(x) = \llbracket 2x - 3 \rrbracket$

$D = \{ \mathbb{R} \}$

$R = \{ \mathbb{Z} \}$

x	y
0	-3
0.5	-2
1	-1
1.5	0
2	1
3	3
4	5



Transformation(s) from the greatest integer parent function are described. Write the new function, then graph and identify the domain and range.

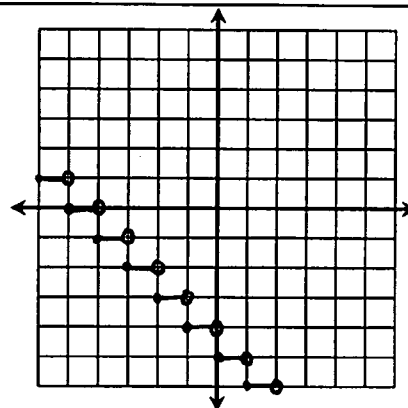
10. Reflected in the x-axis, then shifted down 5 units.

Function: $f(x) = -\lfloor x \rfloor - 5$

D: $\{\mathbb{R}\}$

R: $\{\mathbb{Z}\}$

x	y
-1	-4
-0.5	-4
0	-5
0.5	-5
1	-6
1.5	-6
2	-7
3	-8



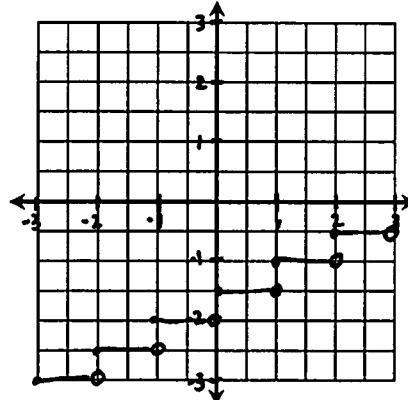
11. Vertically compressed by a factor of $\frac{1}{2}$, then shifted right 3 units.

Function: $f(x) = \frac{1}{2} \lfloor x - 3 \rfloor$

D: $\{\mathbb{R}\}$

R: $\{y \mid y = 0.5n, n \in \mathbb{Z}\}$

x	y
0	-1.5
0.5	-1.5
1	-1
1.5	-1
2	-0.5
3	0
4	0.5
5	1



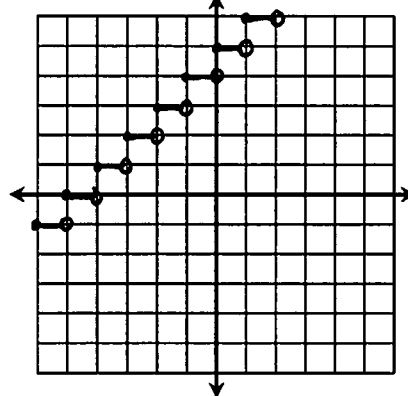
12. Shifted left 1 unit and up 4 units.

Function: $f(x) = \lfloor x + 1 \rfloor + 4$

D: $\{\mathbb{R}\}$

R: $\{\mathbb{Z}\}$

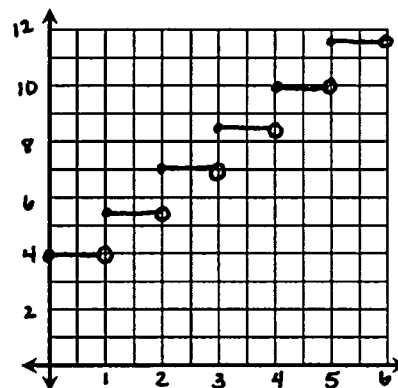
x	y
0	5
0.5	5
1	6
1.5	6
2	7
3	8
4	9
5	10



13. The cost to mail a package is \$4 for up to one pound, then \$1.50 for each additional pound or fraction of a pound. Write a function to represent the cost, C , to mail a package weighing w pounds, then complete the table and graph the function.

Function: $f(x) = 1.5 \lfloor x - 1 \rfloor + 4$

w	C
$0 < w \leq 1$	4.00
$1 < w \leq 2$	5.50
$2 < w \leq 3$	7.00
$3 < w \leq 4$	8.50
$4 < w \leq 5$	10.00
$5 < w \leq 6$	11.50



Unit 3 Test Study Guide

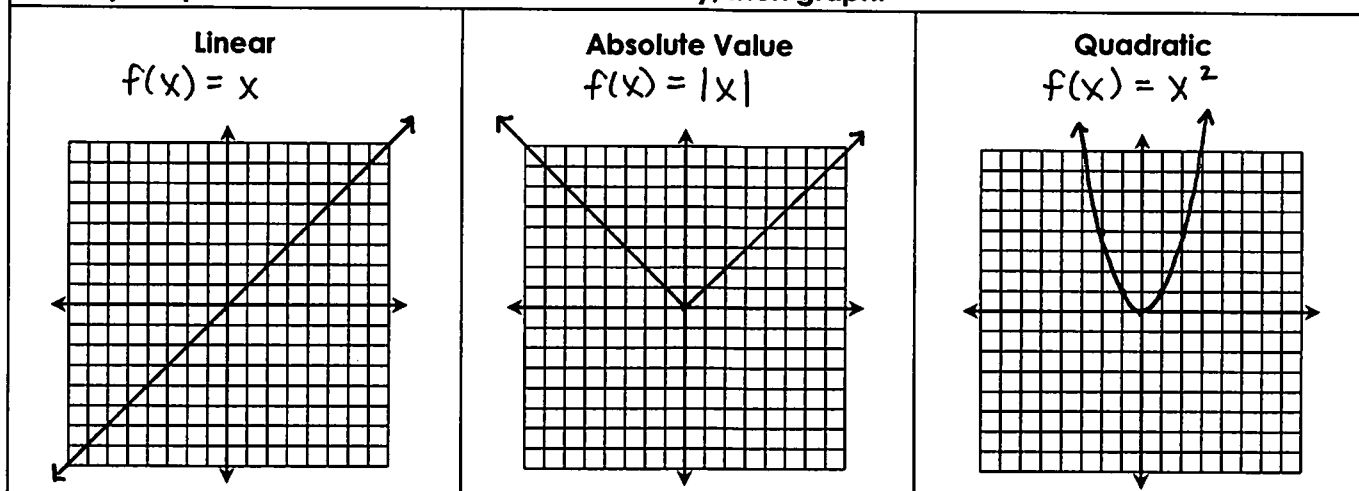
(Parent Functions & Transformations)

Name: _____

Date: _____ Per: _____

Topic 1: Introduction to Parent Functions – Linear, Absolute Value, and Quadratic

Identify the parent function of each function family, then graph:



Topic 2: Transformation on Functions

Describe each transformation rule:

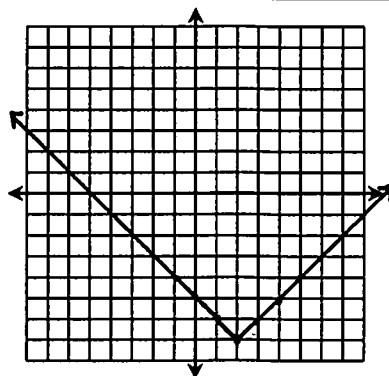
$f(x+h)$	shift left "h" units	$f(x-h)$	shift right "h" units
$f(x)+k$	shift up "k" units	$f(x)-k$	shift down "k" units
$a \cdot f(x)$ when $ a > 1$	vertical stretch	$a \cdot f(x)$ when $ a < 1$	vertical compression
$-f(x)$	reflection across the x-axis		

Topic 3: Absolute Value Functions

Vertex Form of an Absolute Value Function: $f(x) = a|x-h| + k$

Give the vertex, then graph each function. State the domain and range.

1. $f(x) = |x-2| - 7$

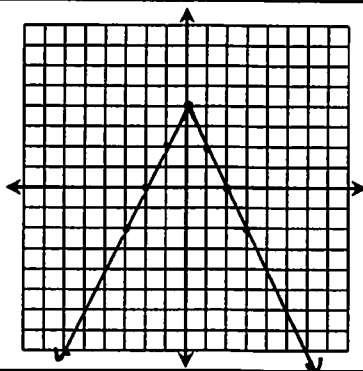


Vertex: $(2, -7)$

Domain: $\{ \mathbb{R} \}$

Range: $\{ y \mid y \geq -7 \}$

2. $f(x) = -2|x| + 4$



Vertex: $(0, 4)$

Domain: $\{ \mathbb{R} \}$

Range: $\{ y \mid y \leq 4 \}$

The parent function of an absolute value equation is transformed as described. Write the new equation in vertex form and identify the vertex.

3. Reflected about the x-axis, then shifted five units left.

$f(x) = -|x+5|$

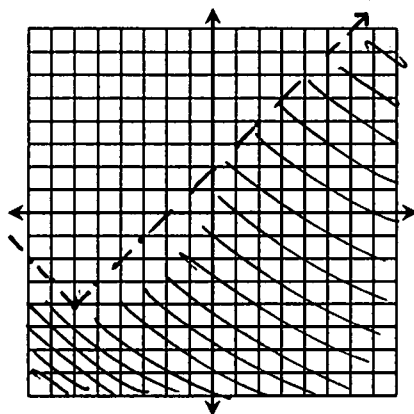
4. Vertically compressed by a factor of one-third, then shifted one unit up and two units right.

$f(x) = \frac{1}{3}|x-2| + 1$

Graph each inequality.

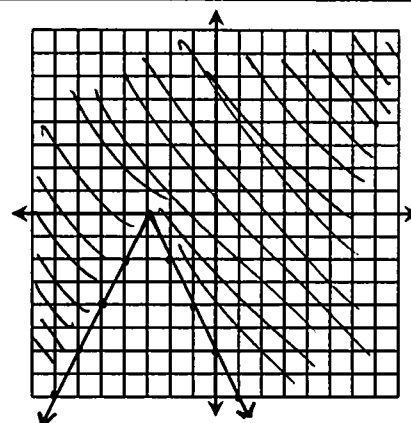
5. $y < |x+6| - 4$

$V: (-6, -4)$



6. $y \geq -2|x+3|$

$V: (-3, 0)$



Topic 4: Quadratic Functions

Standard Form of a Quadratic Function:

$f(x) = ax^2 + bx + c$

Vertex Form of a Quadratic Function:

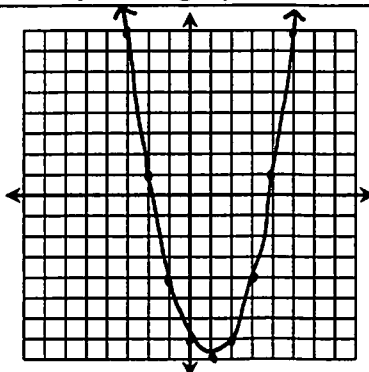
$f(x) = a(x-h)^2 + k$

Give the vertex and axis of symmetry, then graph each function. State the domain and range.

7. $f(x) = x^2 - 2x - 7$

$x = \frac{-(-2)}{2(1)} = 1$

$y = (1)^2 - 2(1) - 7$
 $= 1 - 2 - 7$
 $= -8$



Axis of Symmetry: $x = 1$

Vertex: $(1, -8)$

Domain: $\{ \mathbb{R} \}$

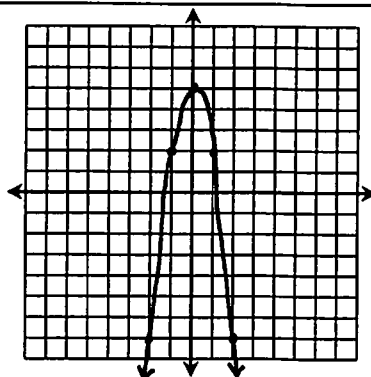
Range: $\{ y \mid y \geq -8 \}$

$$8. f(x) = -3x^2 + 5$$

$$x = \frac{-b}{2a} = 0$$

$$y = -3(0)^2 + 5$$

$$= 5$$



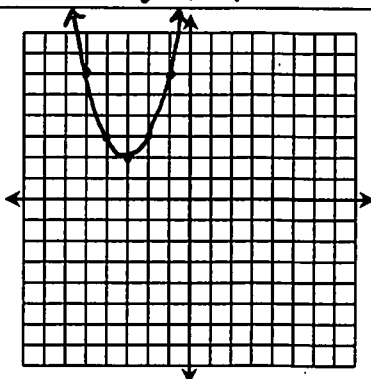
Axis of Symmetry: $x = 0$

Vertex: $(0, 5)$

Domain: $\{ \mathbb{R} \}$

Range: $\{ y \mid y \leq 5 \}$

$$9. f(x) = (x+3)^2 + 2$$



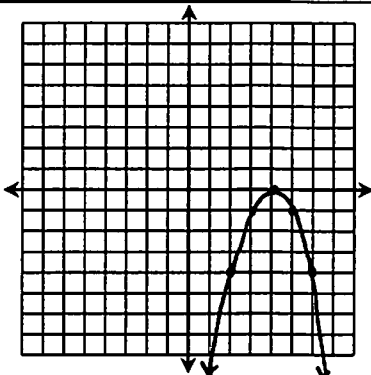
Axis of Symmetry: $x = -3$

Vertex: $(-3, 2)$

Domain: $\{ \mathbb{R} \}$

Range: $\{ y \mid y \geq 2 \}$

$$10. f(x) = -(x-4)^2$$



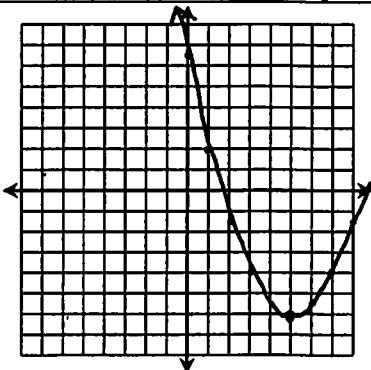
Axis of Symmetry: $x = 4$

Vertex: $(4, 0)$

Domain: $\{ \mathbb{R} \}$

Range: $\{ y \mid y \leq 0 \}$

$$11. f(x) = \frac{1}{2}(x-5)^2 - 6$$



Axis of Symmetry: $x = 5$

Vertex: $(5, -6)$

Domain: $\{ \mathbb{R} \}$

Range: $\{ y \mid y \geq -6 \}$

The parent function of a quadratic equation is transformed as described. Write the new equation in vertex form and identify the vertex.

12. Vertically stretched by a factor of four and shifted seven units right.

$$f(x) = 4(x-7)^2$$

13. Reflected across the x-axis, vertically compressed by a factor of one-half, then shifted four units left and six units down.

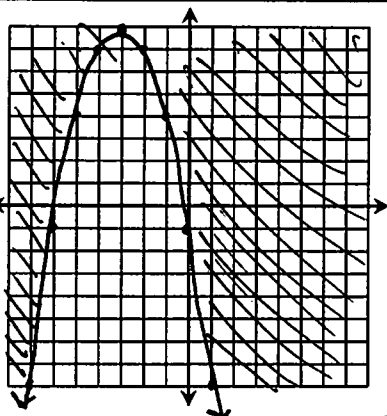
$$f(x) = -\frac{1}{2}(x+4)^2 - 6$$

Graph each inequality.

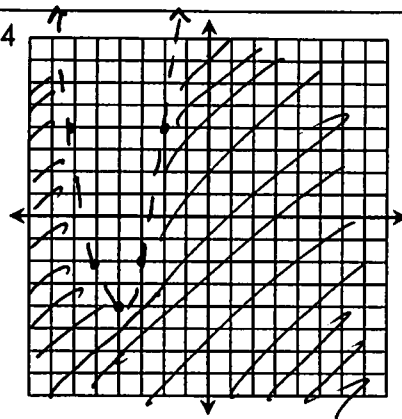
14. $y \geq -x^2 - 6x - 1$

$$x = \frac{-(-6)}{2(-1)} = -3$$

$$\begin{aligned} y &= -(-3)^2 - 6(-3) - 1 \\ &= -9 + 18 - 1 \\ &= 8 \end{aligned}$$



15. $y < 2(x+4)^2 - 4$



Convert each function to vertex form. Then state the vertex.

16. $f(x) = x^2 - 20x + 100$

$$\begin{aligned} f(x) &= (x^2 - 20x) + 100 \\ &\quad \downarrow \\ &\quad (-10)^2 = 100 \end{aligned}$$

$$f(x) = (x^2 - 20x + 100) + 100 - 100$$

$$f(x) = (x - 10)^2 + 0$$

$$f(x) = (x - 10)^2 ; V = (10, 0)$$

17. $f(x) = x^2 + 2x - 8$

$$\begin{aligned} f(x) &= (x^2 + 2x) - 8 \\ &\quad \downarrow \\ &\quad (1)^2 = 1 \end{aligned}$$

$$f(x) = (x^2 + 2x + 1) - 8 - 1$$

$$f(x) = (x + 1)^2 - 9 ; V = (-1, -9)$$

18. $f(x) = -x^2 + 10x - 23$

$$\begin{aligned} f(x) &= -(x^2 - 10x) - 23 \\ &\quad \downarrow \\ &\quad (-5)^2 = 25 \end{aligned}$$

$$f(x) = -(x^2 - 10x + 25) - 23 + 25$$

$$f(x) = -(x - 5)^2 + 2 ; V = (5, 2)$$

19. $f(x) = -2x^2 + 12x - 19$

$$\begin{aligned} f(x) &= -2(x^2 - 6x) - 19 \\ &\quad \downarrow \\ &\quad (-3)^2 = 9 \end{aligned}$$

$$f(x) = -2(x^2 - 6x + 9) - 19 + 18$$

$$f(x) = -2(x - 3)^2 - 1 ; V = (3, -1)$$

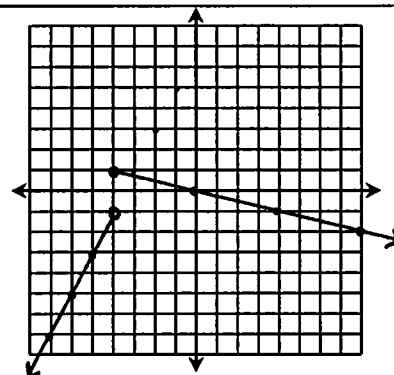
Topic 5: Piecewise Functions

Graph each function. State the domain and range.

20. $f(x) = \begin{cases} 2x + 7 & \text{if } x < -4 \\ -\frac{1}{4}x & \text{if } x \geq -4 \end{cases}$

Domain: $\{ \mathbb{R} \}$

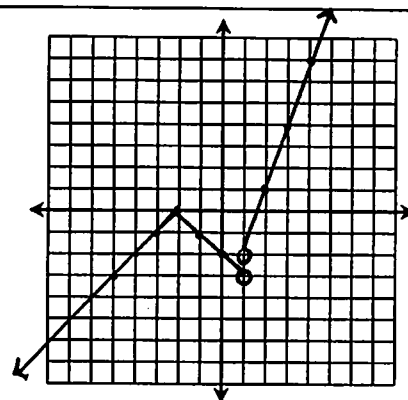
Range: $\{ y \mid y \leq 1 \}$



$$21. g(x) = \begin{cases} -|x+2| & \text{if } x < 1 \\ 3x-5 & \text{if } x > 1 \end{cases}$$

Domain: $\{x | x \neq 1\}$

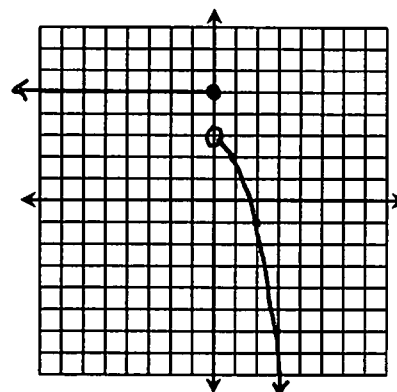
Range: $\{\mathbb{R}\}$



$$22. p(x) = \begin{cases} 5 & \text{if } x \leq 0 \\ -x^2 + 3 & \text{if } x > 0 \end{cases}$$

Domain: $\{\mathbb{R}\}$

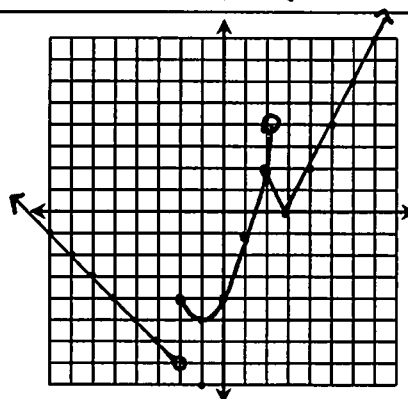
Range: $\{y | y < 3 \cup y = 5\}$



$$23. h(x) = \begin{cases} -x-9 & \text{if } x < -2 \\ (x+1)^2 - 5 & \text{if } -2 \leq x < 2 \\ 2|x-3| & \text{if } x \geq 2 \end{cases}$$

Domain: $\{\mathbb{R}\}$

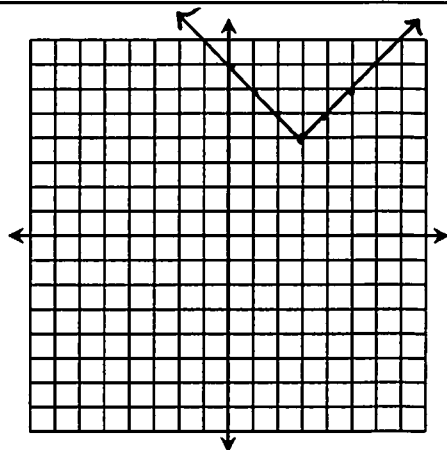
Range: $\{y | y > -7\}$



Topic 6: Mixed Review - Linear, Absolute Value, and Quadratic

Graph each function and identify its characteristics.

$$24. f(x) = |x-3| + 4$$



Function Family: Absolute Value

Parent Function: $f(x) = |x|$

D: $\{\mathbb{R}\}$; R: $\{y | y \geq 4\}$

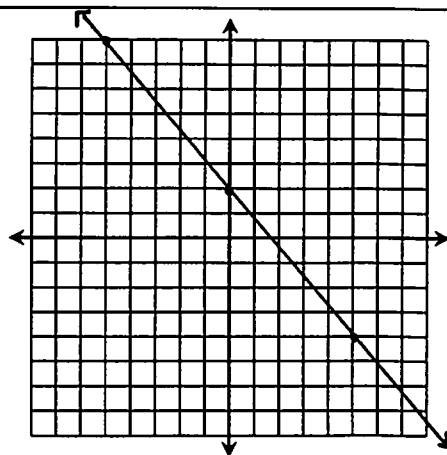
End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

As $x \rightarrow -\infty$, $f(x) \rightarrow \infty$

Increasing Interval: $(3, \infty)$

Decreasing Interval: $(-\infty, 3)$

25. $f(x) = -\frac{6}{5}x + 2$



Function Family: Linear

Parent Function: $f(x) = x$

D: $\{R\}$; R: $\{R\}$

End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$

As $x \rightarrow -\infty$, $f(x) \rightarrow \infty$

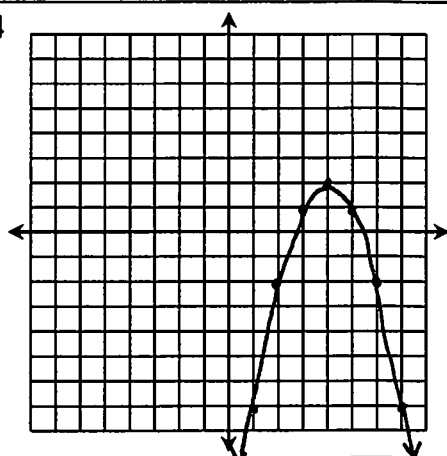
Increasing Interval: N/A

Decreasing Interval: $(-\infty, \infty)$

26. $f(x) = -x^2 + 8x - 14$

$x = \frac{-8}{2(-1)} = 4$

x	y
1	-7
2	-2
3	1
4	2
5	1
6	-2
7	-7



Function Family: Quadratic

Parent Function: $f(x) = x^2$

D: $\{R\}$; R: $\{y | y \leq 2\}$

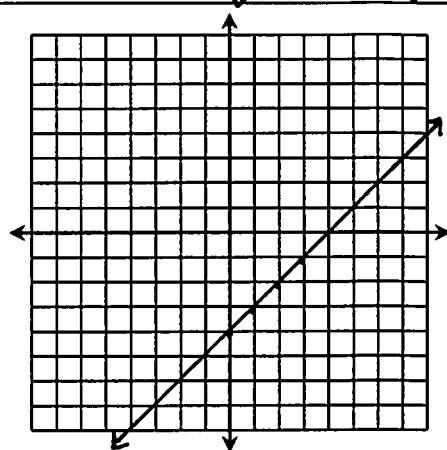
End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$

As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$

Increasing Interval: $(-\infty, 4)$

Decreasing Interval: $(4, \infty)$

27. $f(x) = x - 4$



Function Family: Linear

Parent Function: $f(x) = x$

D: $\{R\}$; R: $\{R\}$

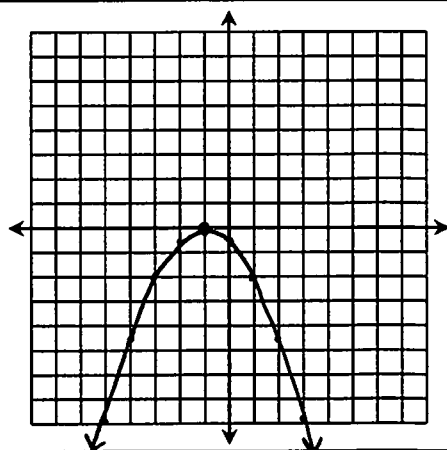
End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow \infty$

As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$

Increasing Interval: $(-\infty, \infty)$

Decreasing Interval: N/A

28. $f(x) = -\frac{1}{2}(x+1)^2$



Function Family: Quadratic

Parent Function: $f(x) = x^2$

D: $\{R\}$; R: $\{y | y \leq 0\}$

End Behavior: As $x \rightarrow \infty$, $f(x) \rightarrow -\infty$

As $x \rightarrow -\infty$, $f(x) \rightarrow -\infty$

Increasing Interval: $(-\infty, -1)$

Decreasing Interval: $(-1, \infty)$

Name: _____

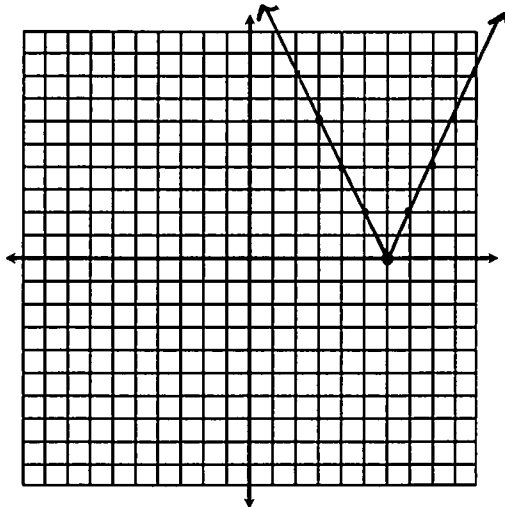
Unit 3 Test

Date: _____ Per: _____

Parent Functions & Transformations
(Linear, Absolute Value, and Quadratic)

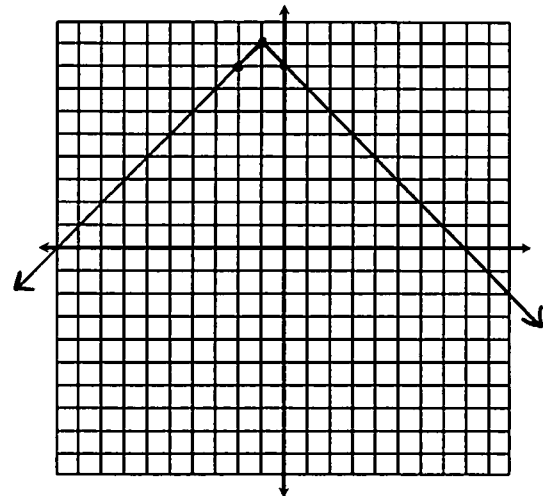
Identify the vertex and graph each function. State the domain and range.

1. $f(x) = 2|x - 6|$ Vertex: (6, 0)



D: $\{ \mathbb{R} \}$ R: $\{ y | y \geq 0 \}$

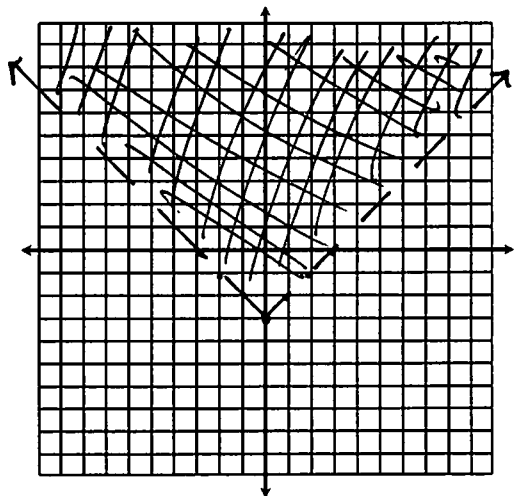
2. $f(x) = -|x + 1| + 9$ Vertex: (-1, 9)



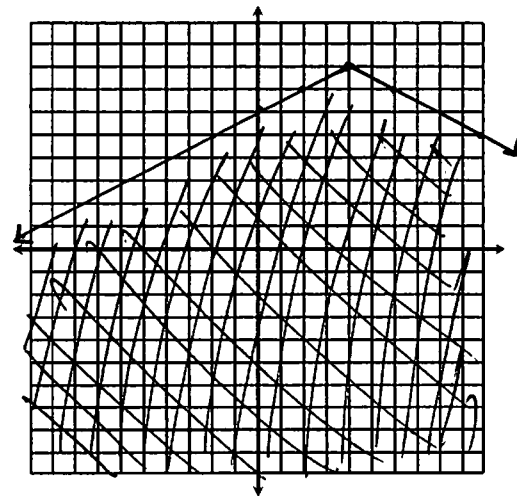
D: $\{ \mathbb{R} \}$ R: $\{ y | y \leq 9 \}$

Identify the vertex, then graph each inequality.

3. $y > |x| - 3$ Vertex: (0, -3)



4. $y \leq -\frac{1}{2}|x - 4| + 8$ Vertex: (4, 8)



5. The graph of the absolute value parent function is vertically stretched by a factor of three, then shifted two units left and five units down. Write the equation of the new function in vertex form.

$$f(x) = 3|x + 2| - 5$$

6. Which function has a vertex at (-1, -3)?

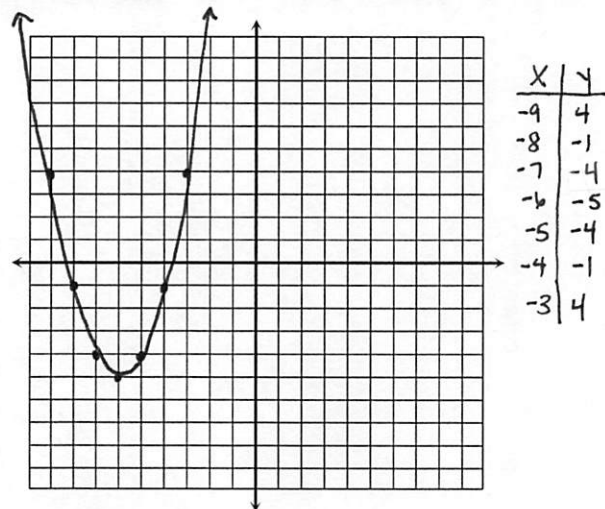
- A. $f(x) = |x + 1| + 3$
- B. $f(x) = |x - 1| + 3$
- C. $f(x) + 3 = |x + 1|$
- D. $f(x) + 3 = |x - 1|$

C

Identify the axis of symmetry and vertex for each function, then graph. State the domain and range.

7. $f(x) = x^2 + 12x + 31$ $x = \frac{-12}{2(1)} = -6$

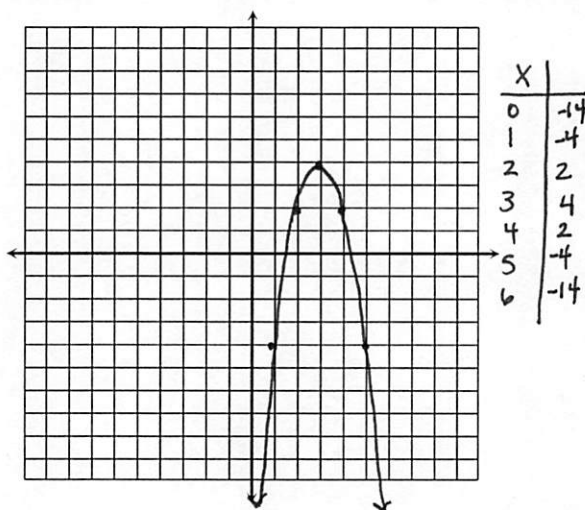
Axis of Symmetry: $x = -6$ Vertex: $(-6, -5)$



D: $\{ \mathbb{R} \}$ R: $\{ y \mid y \geq -5 \}$

8. $f(x) = -2(x - 3)^2 + 4$

Axis of Symmetry: $x = 3$ Vertex: $(3, 4)$

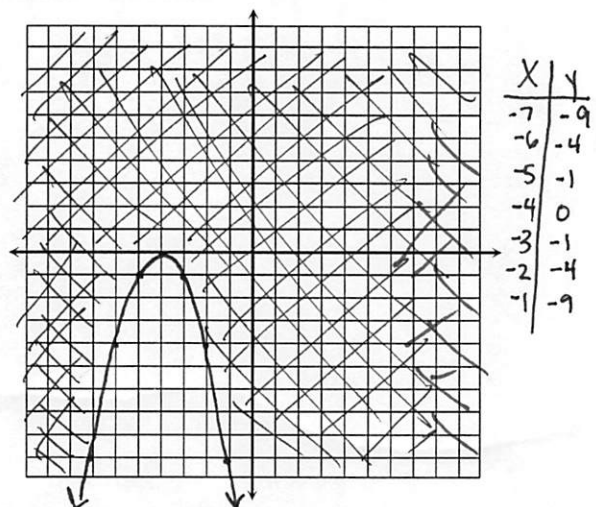


D: $\{ \mathbb{R} \}$ R: $\{ y \mid y \leq 4 \}$

Identify the vertex, axis of symmetry, then graph each inequality.

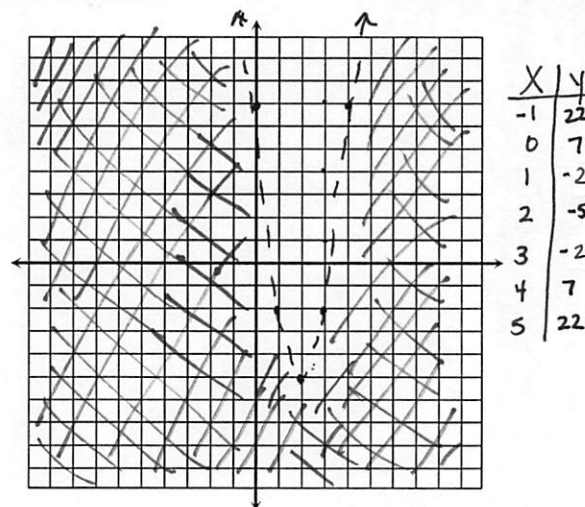
9. $y \geq -(x + 4)^2$

Axis of Symmetry: $x = -4$ Vertex: $(-4, 0)$



10. $y < 3(x - 2)^2 - 5$

Axis of Symmetry: $x = 2$ Vertex: $(2, -5)$



11. The graph of the quadratic parent function is reflected across the x -axis, then shifted five units down and four units right. Write the equation of the new function in vertex form.

$$f(x) = -(x - 4)^2 - 5$$

12. Which function has an axis of symmetry of $x = -2$?

A. $f(x) = x^2 - 4x - 1$

B. $f(x) = -3x^2 - 12x + 4 \rightarrow x = \frac{-(-12)}{2(-3)} = -2$

C. $f(x) = (x - 2)^2 - 8$

D. $f(x) = -2(x + 4)^2 - 3$

B

13. What transformations take place by graphing the function below in respect to its parent function? Check all that apply.

$$f(x) = \frac{3}{2}(x+7)^2 + 2$$

☐ Up 7 units	☒ Up 2 units	☒ Vertical Stretch
☐ Down 7 units	☐ Down 2 units	☐ Vertical Compression
☒ Left 7 units	☐ Left 2 units	☐ Reflection in x-axis
☐ Right 7 units	☐ Right 2 units	☐ Reflection in y-axis

For questions 14-16, convert each function to vertex form. State the vertex.

14. $f(x) = x^2 + 6x + 9$

$$f(x) = (x^2 + 6x) + 9$$

$$\downarrow$$

$$(3)^2 = 9$$

$$f(x) = (x^2 + 6x + 9) + 9 - 9$$

$$f(x) = (x+3)^2 + 0$$

Function:

$$f(x) = (x+3)^2$$

Vertex:

$$(-3, 0)$$

15. $f(x) = x^2 - 10x + 18$

$$f(x) = (x^2 - 10x) + 18$$

$$\downarrow$$

$$(-5)^2 = 25$$

$$f(x) = (x^2 - 10x + 25) + 18 - 25$$

$$f(x) = (x-5)^2 - 7$$

Function:

$$f(x) = (x-5)^2 - 7$$

Vertex:

$$(5, -7)$$

16. $f(x) = -3x^2 + 12x - 7$

$$f(x) = -3(x^2 - 4x) - 7$$

$$\downarrow$$

$$(-2)^2 = 4$$

$$f(x) = -3(x^2 - 4x + 4) - 7 + 12$$

$$f(x) = -3(x-2)^2 + 5$$

Function:

$$f(x) = -3(x-2)^2 + 5$$

Vertex:

$$(2, 5)$$

17. The function $f(x) = -x^2 + 8x - 7$ is translated five units down. Write the new equation in vertex form and state the vertex.

$$f(x) = -(x^2 - 8x) - 7$$

$$\downarrow$$

$$(-4)^2 = 16$$

$$f(x) = -(x^2 - 8x + 16) - 7 + 16$$

$$f(x) = -(x-4)^2 + 9$$

down 5

Function:

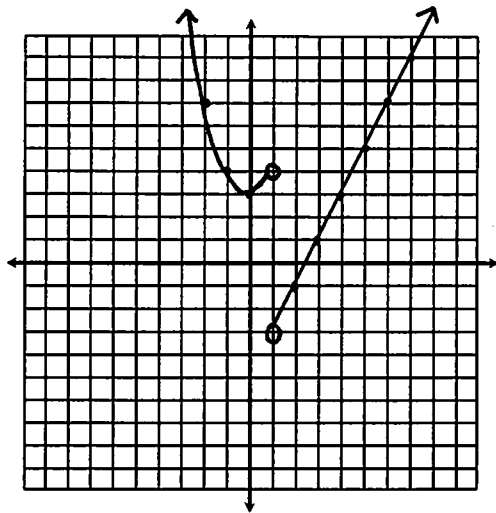
$$f(x) = -(x-4)^2 + 4$$

Vertex:

$$(4, 4)$$

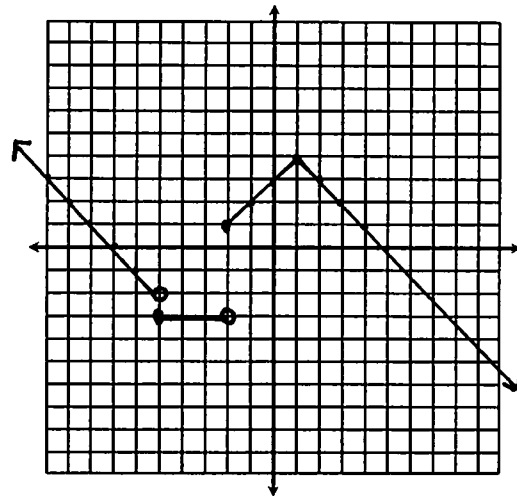
Graph each function. State the domain and range.

18. $f(x) = \begin{cases} x^2 + 3 & \text{if } x < 1 \\ 2x - 5 & \text{if } x > 1 \end{cases}$



D: $\{x \mid x \neq 1\}$ R: $\{y \mid y > -3\}$

19. $f(x) = \begin{cases} -x - 7 & \text{if } x < -5 \\ -3 & \text{if } -5 \leq x < -2 \\ -|x - 1| + 4 & \text{if } x \geq -2 \end{cases}$



D: $\{\mathbb{R}\}$ R: $\{\mathbb{R}\}$

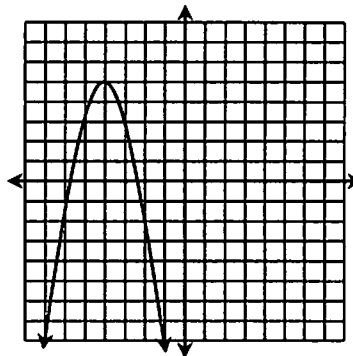
20. Identify the function family and give the equation of the parent function for the set of functions below.

$\{f(x) = -x + 1, g(x) = \frac{1}{4}x + 3, h(x) = 2x - 9\}$

Function Family: Linear

Parent Function: $f(x) = x$

21. The graph of a function is shown below. Identify its function family and the equation of the parent function.



Function Family: Quadratic

Parent Function: $f(x) = x^2$

22. Which parent functions have a domain of all real numbers? Check all that apply.

☒ Linear

☒ Absolute Value

☒ Quadratic

23. Which function has a range of all real numbers greater than or equal to -4?

- A. $f(x) = x - 4$
- B. $f(x) = -4x$
- C. $f(x) = (x + 1)^2 - 4$
- D. $f(x) = -|x - 3| - 4$

C

24. Given the function below, which of the following describes the end behavior as x approaches positive infinity?

$$f(x) = -\frac{4}{3}x + 1$$

- A. $f(x)$ approaches positive infinity
- B. $f(x)$ approaches negative infinity
- C. $f(x)$ approaches 1
- D. $f(x)$ approaches -1

B

25. Given the function below, which of the following describes the end behavior as x approaches negative infinity?

$$f(x) = (x - 6)^2 + 2$$

- A. $f(x)$ approaches positive infinity
- B. $f(x)$ approaches negative infinity
- C. $f(x)$ approaches 6
- D. $f(x)$ approaches -6

A

26. Which of the following describes the behavior of the function below as x approaches positive or negative infinity?

$$f(x) = -|x - h| + k$$

- A. $f(x)$ approaches positive infinity
- B. $f(x)$ approaches negative infinity
- C. $f(x)$ approaches h
- D. $f(x)$ approaches k

B

27. Over what interval is the function below decreasing?

$$f(x) = 2(x + 3)^2 - 5$$

- A. $-\infty < x < -3$
- B. $-\infty < x < 3$
- C. $-3 < x < \infty$
- D. $3 < x < \infty$

A

28. Which of the following describes the function $f(x) = -|x - 5| + 2$?

- A. Increasing over the interval $-\infty < x < -2$; Decreasing over the interval $-2 < x < \infty$
- B. Increasing over the interval $-\infty < x < 2$; Decreasing over the interval $2 < x < \infty$
- C. Increasing over the interval $-\infty < x < -5$; Decreasing over the interval $-5 < x < \infty$
- D. Increasing over the interval $-\infty < x < 5$; Decreasing over the interval $5 < x < \infty$

D

BONUS: Write the function below in vertex form and state the vertex.

$$f(x) = -\frac{1}{2}x^2 + \frac{3}{2}x - 2$$

$$f(x) = -\frac{1}{2}(x^2 - 3x) - 2$$

$$\left(-\frac{3}{2}\right)^2 = \frac{9}{4}$$

$$f(x) = -\frac{1}{2}\left(x^2 - 3x + \frac{9}{4}\right) - 2 + \frac{9}{8}$$

$$f(x) = -\frac{1}{2}\left(x - \frac{3}{2}\right)^2 - \frac{7}{8}$$

Equation:

$$f(x) = -\frac{1}{2}\left(x - \frac{3}{2}\right)^2 - \frac{7}{8}$$

Vertex:

$$\left(\frac{3}{2}, -\frac{7}{8}\right)$$

CREDITS

I use clipart and
fonts in my products by:



Art with Jenny K



Many thanks to these
talented artists!